AWS Certified AI Practitioner

AIF-C01 Exam Preparation

500 Practice Questions with Detailed Explanations
Covering All 5 Exam Domains

Domain	Topic	Questions
Domain 1	AI & ML Fundamentals	Q1 – Q100
Domain 2	Generative AI & Foundation Models	Q101 – Q200
Domain 3	AWS AI & ML Services	Q201 – Q300
Domain 4	Responsible AI	Q301 – Q400
Domain 5	Security, Compliance & Governance	Q401 – Q500

Instructions: Each question has one correct answer. Read all options carefully. The correct answer and a detailed explanation are provided after each question.

Passing Score: 700/1000 | Questions: 65 (+ 15 unscored) | Time: 90 minutes

Domain 1: AI & ML Fundamentals

Q1. What is the primary difference between Artificial Intelligence (AI) and Machine Learning (ML)?

A. AI is a subset of ML
B. ML is a subset of AI that learns from data
C. AI only works with structured data
D. ML does not require any data

✓ Correct Answer: B

Explanation: Machine Learning is a subset of AI. AI is the broader concept of machines performing tasks intelligently, while ML is a specific approach where machines learn patterns from data automatically.

Q2. Which type of machine learning uses labeled training data to learn a mapping from inputs to outputs?

A. Unsupervised learning
B. Reinforcement learning
C. Supervised learning
D. Self-supervised learning

✓ Correct Answer: C

Explanation: Supervised learning uses labeled data where each input has a corresponding correct output label. The model learns to map inputs to outputs based on these examples.

Q3. A company wants to group its customers into segments based on purchasing behavior without any predefined labels. Which ML approach is most appropriate?

A. Supervised learning
B. Reinforcement learning
C. Unsupervised learning
D. Transfer learning

✓ Correct Answer: C

Explanation: Unsupervised learning finds patterns and structure in unlabeled data. Clustering (a type of unsupervised learning) is ideal for customer segmentation without predefined categories.

Q4. What is overfitting in machine learning?

A. When a model performs well on training data but poorly on new data
B. When a model performs equally well on all datasets
C. When a model is too simple to capture patterns
D. When training takes too long

✓ Correct Answer: A

Explanation: Overfitting occurs when a model learns the training data too well, including its noise and outliers, making it fail to generalize to new, unseen data.

Q5. Which technique is used to prevent overfitting by penalizing large model weights?

A. Boosting
B. Regularization
C. Feature engineering
D. Data augmentation

✓ Correct Answer: B

Explanation: Regularization (L1/L2) adds a penalty term to the loss function for large weights, discouraging the model from fitting noise in the training data.

Q6. What is the purpose of a validation dataset in ML training?

A. To train the model
B. To tune hyperparameters and evaluate during training
C. To measure final model performance
D. To preprocess raw data

✓ Correct Answer: B

Explanation: The validation set is used during training to tune hyperparameters and monitor performance, while the test set evaluates final model performance on completely unseen data.

Q7. Which evaluation metric is most appropriate for a highly imbalanced binary classification dataset?

A. Accuracy
B. Mean Squared Error
C. F1 Score
D. R-squared

✓ Correct Answer: C

Explanation: F1 Score balances precision and recall, making it appropriate for imbalanced datasets where accuracy can be misleading (e.g., 95% accuracy by always predicting the majority class).

Q8. In reinforcement learning, what is an 'agent'?

A. The dataset used for training
B. The entity that takes actions in an environment to maximize rewards
C. The reward function
D. The neural network architecture

✓ Correct Answer: B

Explanation: In reinforcement learning, the agent is the learner/decision maker that interacts with an environment, takes actions, and receives rewards to learn an optimal policy.

Q9. What does 'bias' refer to in the context of ML model fairness?

A. The learning rate of the model
B. Systematic errors causing unfair outcomes for certain groups
C. The number of hidden layers
D. The size of the training dataset

✓ Correct Answer: B

Explanation: Bias in ML fairness refers to systematic prejudice in model predictions that results in unfair treatment of certain demographic or protected groups, often originating from biased training data.

Q10. Which of the following best describes a neural network?

A. A decision tree with many branches
B. A system of interconnected nodes that processes data in layers
C. A SQL database for storing model weights
D. A rule-based expert system

✓ Correct Answer: B

Explanation: A neural network consists of interconnected layers of nodes (neurons) that process and transform data, loosely inspired by the structure of biological brains.

Q11. What is 'feature engineering' in machine learning?

A. Training a model on GPU hardware
B. Selecting and transforming raw data into meaningful inputs for a model
C. Evaluating model accuracy
D. Deploying models to production

✓ Correct Answer: B

Explanation: Feature engineering is the process of selecting, transforming, and creating input variables (features) from raw data to improve model performance.

Q12. Which ML algorithm is commonly used for predicting a continuous numerical value?

A. K-means clustering
B. Linear regression
C. Naive Bayes
D. Principal Component Analysis

✓ Correct Answer: B

Explanation: Linear regression predicts a continuous numerical output based on input features. It is a supervised learning algorithm used for regression tasks.

Q13. What is the purpose of cross-validation in model development?

A. To speed up model training
B. To obtain a more reliable estimate of model performance
C. To compress the model size
D. To deploy the model to multiple regions

✓ Correct Answer: B

Explanation: Cross-validation (e.g., k-fold) splits data into multiple training/validation sets to get a more robust and reliable estimate of model generalization performance.

Q14. What is a confusion matrix used for?

A. Visualizing model architecture
B. Evaluating classification model performance across categories
C. Tracking training loss over epochs
D. Comparing two neural networks

✓ Correct Answer: B

Explanation: A confusion matrix shows true positives, true negatives, false positives, and false negatives, allowing detailed analysis of classification model performance across all classes.

Q15. Which of the following is a dimensionality reduction technique?

A. Random Forest
B. Principal Component Analysis (PCA)
C. Gradient Boosting
D. Logistic Regression

✓ Correct Answer: B

Explanation: PCA reduces the number of features while retaining most variance in the data, making it useful for visualization and reducing computational cost.

Q16. What does 'epoch' mean in the context of neural network training?

A. One forward pass on a single sample
B. One complete pass through the entire training dataset
C. One hyperparameter tuning iteration
D. One deployment cycle

✓ Correct Answer: B

Explanation: An epoch is one complete pass through all training samples. Multiple epochs allow the model to learn and adjust weights iteratively.

Q17. Which type of neural network is specifically designed for processing sequential data like text or time series?

A. Convolutional Neural Network (CNN)
B. Recurrent Neural Network (RNN)
C. Generative Adversarial Network (GAN)
D. Autoencoder

✓ Correct Answer: B

Explanation: RNNs are designed to handle sequential data by maintaining a hidden state that captures context from previous time steps.

Q18. A company wants to detect objects in images. Which type of neural network architecture is most appropriate?

A. Recurrent Neural Network
B. Convolutional Neural Network (CNN)
C. Decision Tree
D. Linear Regression

✓ Correct Answer: B

Explanation: CNNs use convolutional layers to automatically learn spatial features from images, making them ideal for image classification, object detection, and recognition tasks.

Q19. What is transfer learning?

A. Moving a model from one cloud region to another
B. Using a pre-trained model as the starting point for a new task

C. Transferring data between S3 buckets

D. Copying model weights between GPUs

✓ Correct Answer: B

Explanation: Transfer learning reuses knowledge from a model trained on a large dataset (e.g., ImageNet) as the starting point for a different but related task, reducing training time and data requirements.

Q20. What is 'batch size' in model training?

A. Total number of epochs

B. Number of training samples processed before updating model weights

C. Number of layers in the network

D. Size of the validation dataset

✓ Correct Answer: B

Explanation: Batch size defines how many training examples are processed in one forward/backward pass before the model's weights are updated via gradient descent.

Q21. Which metric measures the proportion of actual positives correctly identified by a model?

A. Precision

B. Recall (Sensitivity)

C. Specificity

D. AUC-ROC

✓ Correct Answer: B

Explanation: Recall (also called sensitivity) = True Positives / (True Positives + False Negatives). It measures how well the model finds all actual positive cases.

Q22. What is the vanishing gradient problem in deep neural networks?

A. Model weights become too large during training

B. Gradients become extremely small, preventing earlier layers from learning

C. The model trains too fast

D. Training data is insufficient

✓ Correct Answer: B

Explanation: In deep networks, gradients can shrink exponentially as they propagate backward through many layers, causing earlier layers to learn very slowly or not at all.

Q23. Which activation function is most commonly used in hidden layers of modern deep neural networks?

A. Sigmoid

B. Tanh

C. ReLU (Rectified Linear Unit)

D. Softmax

✓ Correct Answer: C

Explanation: ReLU is computationally efficient and helps mitigate the vanishing gradient problem, making it the default choice for hidden layers in deep neural networks.

Q24. What is 'data augmentation' used for in machine learning?

A. Cleaning corrupted data
B. Artificially expanding training data by applying transformations
C. Reducing model size
D. Encrypting training data

✓ Correct Answer: B

Explanation: Data augmentation applies transformations (flips, rotations, crops, noise) to existing training samples to artificially increase dataset size and improve model generalization.

Q25. A model achieves 99% accuracy on training data but only 60% on test data. What problem does this indicate?

A. Underfitting
B. Overfitting
C. Data leakage
D. Class imbalance

✓ Correct Answer: B

Explanation: A large gap between training and test accuracy is the classic sign of overfitting — the model memorized the training data instead of learning generalizable patterns.

Q26. What is the role of the 'learning rate' hyperparameter?

A. Controls how many features are selected
B. Controls how much model weights are updated per training step
C. Determines the number of training epochs
D. Sets the size of each training batch

✓ Correct Answer: B

Explanation: The learning rate determines the step size during gradient descent. A high learning rate may cause divergence; a low rate may cause very slow convergence.

Q27. Which of the following best describes 'underfitting'?

A. Model performs well on training but poorly on test data
B. Model is too complex for the dataset
C. Model fails to capture patterns even in training data
D. Model requires too much memory

✓ Correct Answer: C

Explanation: Underfitting occurs when the model is too simple to capture the underlying patterns in the data, resulting in poor performance on both training and test sets.

Q28. What is 'label encoding' used for in ML preprocessing?

A. Normalizing numerical features
B. Converting categorical text values to numerical integers
C. Removing duplicate records
D. Splitting data into train and test sets

✓ Correct Answer: B

Explanation: Label encoding converts categorical string values (e.g., 'Red', 'Blue', 'Green') into integer values (0, 1, 2) that ML algorithms can process.

Q29. What is the purpose of dropout in neural network training?

A. Speed up inference
B. Reduce overfitting by randomly deactivating neurons during training
C. Initialize weights to zero
D. Increase the learning rate

✓ Correct Answer: B

Explanation: Dropout randomly sets a fraction of neurons to zero during each training pass, preventing co-adaptation of neurons and acting as a regularization technique to reduce overfitting.

Q30. Which of the following is an example of a regression problem?

A. Classifying emails as spam or not spam
B. Predicting tomorrow's stock price
C. Identifying objects in images
D. Grouping customers by behavior

✓ Correct Answer: B

Explanation: Predicting a continuous numerical value (like a stock price) is a regression problem. Classification problems predict discrete categories, while clustering groups unlabeled data.

Q31. What does AUC-ROC measure in classification models?

A. Speed of model inference
B. Model's ability to discriminate between positive and negative classes
C. Total training time
D. Feature importance ranking

✓ Correct Answer: B

Explanation: AUC-ROC (Area Under the ROC Curve) measures a model's ability to distinguish between classes across all classification thresholds. A value of 1.0 is perfect; 0.5 is random.

Q32. What is a hyperparameter in machine learning?

A. A parameter learned from training data
B. A configuration set before training that controls the learning process
C. The output label of a training example
D. A data preprocessing step

✓ Correct Answer: B

Explanation: Hyperparameters (e.g., learning rate, number of layers, batch size) are set manually before training begins, unlike model parameters which are learned from data during training.

Q33. Which algorithm is an ensemble method that combines multiple decision trees?

A. K-nearest neighbors
B. Support Vector Machine
C. Random Forest
D. Logistic Regression

✓ Correct Answer: C

Explanation: Random Forest builds multiple decision trees on random subsets of data and features, then aggregates their predictions, resulting in improved accuracy and reduced overfitting.

Q34. What is 'normalization' in data preprocessing?

A. Removing missing values
B. Scaling features to a standard range (e.g., 0 to 1)
C. Encoding categorical variables
D. Splitting data into folds

✓ Correct Answer: B

Explanation: Normalization scales numerical features to a consistent range, preventing features with large values from dominating the learning process in distance-based or gradient-based algorithms.

Q35. What does NLP stand for in the context of AI?

A. Neural Layer Processing
B. Natural Language Processing
C. Networked Learning Pipeline
D. Normalized Learning Parameter

✓ Correct Answer: B

Explanation: NLP (Natural Language Processing) is the AI field focused on enabling computers to understand, interpret, and generate human language.

Q36. Which of the following tasks is a common NLP application?

A. Image segmentation
B. Sentiment analysis
C. Object detection
D. Anomaly detection in sensor data

✓ Correct Answer: B

Explanation: Sentiment analysis (determining if text is positive, negative, or neutral) is a classic NLP task. Image segmentation and object detection are computer vision tasks.

Q37. A dataset has 90% negative and 10% positive examples. Which approach helps address this imbalance?

A. Reducing the number of epochs
B. Oversampling the minority class or undersampling the majority class
C. Increasing the learning rate
D. Adding more hidden layers

✓ Correct Answer: B

Explanation: Oversampling (e.g., SMOTE) increases minority class samples, while undersampling reduces majority class samples. Both approaches help balance class distribution for better model training.

Q38. What is the purpose of gradient descent in model training?

A. To initialize model weights randomly
B. To iteratively minimize the loss function by adjusting model weights
C. To select the best features from the dataset

D. To evaluate model performance on test data

✓ Correct Answer: B

Explanation: Gradient descent is an optimization algorithm that computes the gradient of the loss function and updates model weights in the direction that minimizes the loss.

Q39. What is 'tokenization' in NLP?

A. Encrypting text data

B. Breaking text into individual units (words, subwords, or characters)

C. Translating text between languages

D. Generating embeddings from text

✓ Correct Answer: B

Explanation: Tokenization splits raw text into smaller units called tokens (words, subwords, or characters) that can be processed and converted into numerical representations by NLP models.

Q40. What does 'precision' measure in a classification model?

A. Proportion of all positives correctly identified

B. Proportion of predicted positives that are actually positive

C. Overall fraction of correct predictions

D. Ability to separate all classes

✓ Correct Answer: B

Explanation: Precision = True Positives / (True Positives + False Positives). It measures how many of the model's positive predictions were actually correct.

Q41. Which of the following is NOT a type of machine learning?

A. Supervised learning

B. Unsupervised learning

C. Reinforcement learning

D. Relational learning

✓ Correct Answer: D

Explanation: The three main types of machine learning are supervised, unsupervised, and reinforcement learning. 'Relational learning' is not a standard ML category.

Q42. What is a 'loss function' in machine learning?

A. The function that selects training data

B. A function measuring the difference between model predictions and actual values

C. The architecture of a neural network

D. A function that increases model complexity

✓ Correct Answer: B

Explanation: The loss function quantifies how wrong the model's predictions are. Minimizing the loss function guides the learning process through optimization algorithms like gradient descent.

Q43. What is the key advantage of ensemble methods like Gradient Boosting?

A. They require less data than single models

B. They combine weak learners to form a stronger predictor

C. They eliminate the need for feature engineering

D. They train faster than single models

✓ Correct Answer: B

Explanation: Ensemble methods combine multiple weak models (often decision trees) sequentially or in parallel, with each new model correcting errors of the previous ones, resulting in a stronger overall predictor.

Q44. What is 'word embedding' in NLP?

A. Storing words in a relational database

B. Representing words as dense numerical vectors capturing semantic meaning

C. Sorting words alphabetically

D. Encrypting text for secure storage

✓ Correct Answer: B

Explanation: Word embeddings (e.g., Word2Vec, GloVe) represent words as dense vectors in a continuous space where semantically similar words are geometrically close to each other.

Q45. What does the term 'inference' mean in ML deployment?

A. Training a new model on new data

B. Using a trained model to make predictions on new data

C. Evaluating model accuracy during training

D. Retraining a model with updated data

✓ Correct Answer: B

Explanation: Inference is the process of using a trained, deployed ML model to make predictions on new, unseen data in production environments.

Q46. Which of the following is a common technique for handling missing data?

A. Oversampling

B. Imputation (replacing missing values with mean/median/mode)

C. Dropout

D. Regularization

✓ Correct Answer: B

Explanation: Imputation fills missing values with statistical estimates (mean, median, mode) or model-based predictions, allowing the dataset to be used without discarding incomplete records.

Q47. What is the Transformer architecture primarily used for?

A. Image segmentation tasks

B. Processing sequences with attention mechanisms, especially in NLP

C. Anomaly detection in tabular data

D. Reinforcement learning environments

✓ Correct Answer: B

Explanation: The Transformer architecture uses self-attention mechanisms to process sequences in parallel, becoming the foundation for modern NLP models like BERT and GPT.

Q48. What is 'model drift' in production ML systems?

A. A model moving between cloud regions

B. Degradation in model performance over time as data distributions change

C. Increasing model accuracy after deployment

D. A model being retrained too frequently

✓ Correct Answer: B

Explanation: Model drift occurs when the statistical properties of input data change over time (data drift) or the relationship between inputs and outputs changes (concept drift), reducing model accuracy.

Q49. What is the purpose of a training/test split in ML?

A. To duplicate data for backup

B. To evaluate model performance on data it has never seen

C. To speed up model training

D. To balance class distribution

✓ Correct Answer: B

Explanation: The test split is held out from training and used only once at the end to evaluate how well the model generalizes to completely new, unseen data.

Q50. Which learning type would you use to train a game-playing AI that learns through trial and error?

A. Supervised learning

B. Unsupervised learning

C. Reinforcement learning

D. Semi-supervised learning

✓ Correct Answer: C

Explanation: Reinforcement learning trains agents through trial and error by rewarding desirable actions and penalizing undesirable ones, making it ideal for game-playing AI.

Q51. What is a 'foundation model' in AI?

A. The first model ever trained on AWS

B. A large model trained on broad data that can be adapted to many downstream tasks

C. A model that only performs basic arithmetic

D. A model stored in AWS Foundations service

✓ Correct Answer: B

Explanation: Foundation models are large AI models trained on vast amounts of data that serve as a base for many different downstream applications through fine-tuning or prompting.

Q52. What does 'fine-tuning' a pre-trained model mean?

A. Deleting unnecessary model layers

B. Further training a pre-trained model on task-specific data

C. Compressing model weights for faster inference

D. Running the model on specialized hardware

✓ Correct Answer: B

Explanation: Fine-tuning takes a pre-trained model and continues training it on a smaller, task-specific dataset to adapt its knowledge to a new domain or task.

Q53. What is the attention mechanism in Transformer models?

A. A memory management technique for large datasets
B. A mechanism allowing the model to focus on relevant parts of the input
C. A regularization technique
D. A data preprocessing step

✓ Correct Answer: B

Explanation: Attention mechanisms allow the model to weigh the importance of different input tokens when generating each output, enabling better handling of long-range dependencies in sequences.

Q54. Which of the following is an example of computer vision?

A. Translating English text to French
B. Detecting tumors in medical images
C. Predicting customer churn
D. Generating product descriptions

✓ Correct Answer: B

Explanation: Computer vision involves processing and analyzing visual data. Medical image analysis for tumor detection is a classic computer vision application.

Q55. What is 'semi-supervised learning'?

A. Learning with only labeled data
B. Learning with a mix of labeled and unlabeled data
C. Learning without any data
D. Learning with only unlabeled data

✓ Correct Answer: B

Explanation: Semi-supervised learning uses a small amount of labeled data combined with a large amount of unlabeled data, leveraging both to improve model performance cost-effectively.

Q56. What does Mean Absolute Error (MAE) measure?

A. Average squared difference between predictions and actual values
B. Average absolute difference between predictions and actual values
C. Proportion of correct classifications
D. Area under the ROC curve

✓ Correct Answer: B

Explanation: MAE calculates the average of absolute differences between predicted and actual values, providing a straightforward measure of prediction error magnitude.

Q57. What is the role of an 'encoder' in an encoder-decoder architecture?

A. Converts model outputs to predictions
B. Compresses input into a compact latent representation
C. Generates outputs from latent representations
D. Preprocesses raw data before training

✓ Correct Answer: B

Explanation: The encoder compresses input data into a compact latent (hidden) representation, capturing the essential features. The decoder then uses this representation to generate the output.

Q58. What is 'explainability' in AI?

A. The speed at which a model makes predictions
B. The ability to understand and interpret how a model makes its decisions
C. The accuracy of a model on test data
D. The process of deploying a model to production

✓ Correct Answer: B

Explanation: Explainability (or interpretability) refers to the degree to which the internal mechanics of an AI model can be understood by humans, which is critical for trust and responsible AI.

Q59. Which of the following describes a 'black box' model?

A. A model stored in an encrypted S3 bucket
B. A model whose internal decision-making process is not easily interpretable
C. A model trained entirely on unlabeled data
D. A model that only processes dark-colored images

✓ Correct Answer: B

Explanation: Black box models (like deep neural networks) make predictions without providing easily interpretable reasoning, contrasting with white box models like decision trees.

Q60. What is the purpose of weight initialization in neural networks?

A. To set the final model weights after training
B. To set starting weight values that enable effective gradient-based learning
C. To encrypt model parameters
D. To determine the number of training epochs

✓ Correct Answer: B

Explanation: Proper weight initialization (e.g., Xavier, He initialization) ensures gradients flow effectively at the start of training, avoiding issues like vanishing or exploding gradients.

Q61. What is 'stochastic gradient descent' (SGD)?

A. Gradient descent using the full dataset per update
B. Gradient descent updating weights using a single or small random batch of samples
C. A method for selecting random hyperparameters
D. A technique for randomly initializing weights

✓ Correct Answer: B

Explanation: SGD updates model weights after each individual sample or mini-batch rather than the full dataset, making training faster and introducing beneficial noise that can help escape local minima.

Q62. Which of the following is NOT a common evaluation metric for regression models?

A. Mean Squared Error (MSE)
B. Root Mean Squared Error (RMSE)
C. Mean Absolute Error (MAE)
D. F1 Score

✓ **Correct Answer: D**

Explanation: F1 Score is a classification metric (combining precision and recall). MSE, RMSE, and MAE are all used to evaluate regression model performance.

Q63. What does 'ETL' stand for in the context of ML data pipelines?

A. Extract, Transform, Load

B. Evaluate, Train, Log

C. Encode, Tokenize, Label

D. Export, Test, Learn

✓ **Correct Answer: A**

Explanation: ETL (Extract, Transform, Load) is the process of extracting data from sources, transforming it into the required format, and loading it into a destination for ML training.

Q64. What is 'model compression' used for?

A. Increasing model accuracy on test data

B. Reducing model size and inference time for deployment on resource-constrained devices

C. Compressing training data

D. Encrypting model weights for security

✓ **Correct Answer: B**

Explanation: Model compression techniques (pruning, quantization, knowledge distillation) reduce model size and computational requirements, enabling deployment on edge devices or for faster inference.

Domain 2: Generative AI & Foundation Models

Q65. What is generative AI?

A. AI that classifies existing data into categories

B. AI that creates new content such as text, images, audio, or video

C. AI that only detects anomalies in data

D. AI that predicts numerical values

✓ **Correct Answer: B**

Explanation: Generative AI models create new, original content by learning patterns from training data, enabling generation of text, images, code, audio, and more.

Q66. What is a Large Language Model (LLM)?

A. A model that processes only large datasets

B. A neural network trained on vast amounts of text data to understand and generate language

C. A model limited to language translation

D. A database of language rules

✓ **Correct Answer: B**

Explanation: LLMs are neural networks (typically Transformer-based) trained on massive text corpora, enabling them to understand, generate, summarize, and reason about language.

Q67. What is 'prompt engineering'?

A. Training a new model from scratch
B. Designing effective input prompts to guide an AI model's output
C. Engineering the model architecture
D. Compressing prompts for faster processing

✓ Correct Answer: B

Explanation: Prompt engineering involves crafting and optimizing input prompts to achieve desired outputs from AI models without changing model weights.

Q68. What is 'zero-shot prompting'?

A. Providing zero training examples to train a model
B. Asking a model to perform a task with no examples in the prompt
C. Running a model with zero temperature
D. A prompt that produces no output

✓ Correct Answer: B

Explanation: Zero-shot prompting asks the model to perform a task with no examples provided, relying entirely on the model's pre-trained knowledge.

Q69. What is 'few-shot prompting'?

A. Training a model on a few examples
B. Including a small number of examples in the prompt to guide the model
C. Using a model that was trained on limited data
D. Prompting a model that has been fine-tuned

✓ Correct Answer: B

Explanation: Few-shot prompting provides a small number of examples within the prompt to demonstrate the desired task format or behavior to the model.

Q70. What does 'temperature' control in LLM generation?

A. The GPU temperature during inference
B. The randomness/creativity of the model's output
C. The length of generated text
D. The number of tokens processed per second

✓ Correct Answer: B

Explanation: Temperature controls output randomness. High temperature (e.g., 1.0+) produces more diverse/creative outputs. Low temperature (e.g., 0.1) produces more deterministic/focused outputs.

Q71. What is RAG (Retrieval Augmented Generation)?

A. A technique for compressing LLMs
B. Augmenting LLM responses by retrieving relevant external knowledge
C. A method for training LLMs from scratch
D. A way to reduce LLM hallucinations by removing training data

✓ Correct Answer: B

Explanation: RAG retrieves relevant documents from a knowledge base at inference time and provides them as context to the LLM, reducing hallucinations and enabling up-to-date responses.

Q72. What is 'hallucination' in the context of LLMs?

A. When a model generates visual outputs
B. When a model generates plausible-sounding but factually incorrect information
C. When a model experiences overheating
D. When a model repeats the same output

✓ Correct Answer: B

Explanation: Hallucination refers to LLMs confidently generating false or fabricated information that sounds plausible, which is a key challenge in deploying LLMs reliably.

Q73. What is 'top-p' (nucleus sampling) in LLM generation?

A. Maximum number of tokens to generate
B. Sampling from the smallest set of tokens whose cumulative probability exceeds p
C. The percentage of training data used
D. The top-performing model in a benchmark

✓ Correct Answer: B

Explanation: Top-p (nucleus) sampling selects from the smallest set of candidate tokens whose cumulative probability exceeds the threshold p, balancing diversity and quality.

Q74. What is Amazon Bedrock primarily used for?

A. Training ML models from scratch
B. Accessing and deploying foundation models via a fully managed API
C. Storing large datasets for ML training
D. Monitoring ML model performance

✓ Correct Answer: B

Explanation: Amazon Bedrock provides fully managed access to foundation models from Amazon and third-party providers via a single API, without needing to manage infrastructure.

Q75. Which of the following is a foundation model available through Amazon Bedrock?

A. Amazon Comprehend
B. Anthropic Claude
C. Amazon Rekognition
D. AWS Glue

✓ Correct Answer: B

Explanation: Amazon Bedrock provides access to models including Anthropic's Claude, Amazon Titan, Meta Llama, Mistral AI, and others through a unified API.

Q76. What is 'context window' in an LLM?

A. The UI window showing model outputs
B. The maximum amount of text (tokens) the model can process at once
C. The time period for model training
D. The number of training examples used

✓ **Correct Answer: B**

Explanation: The context window defines the maximum number of tokens (input + output) an LLM can process in a single interaction. Larger context windows enable handling longer documents.

Q77. What is 'RLHF' in the context of LLM training?

A. Reinforced Learning with Human Feedback

B. Reinforcement Learning from Human Feedback

C. Restricted Language Handling Framework

D. Reliable Language and Hallucination Filtering

✓ **Correct Answer: B**

Explanation: RLHF (Reinforcement Learning from Human Feedback) trains a reward model using human preferences, then uses it to fine-tune the LLM to generate more helpful, harmless, and honest responses.

Q78. What is a 'token' in the context of LLMs?

A. A security credential for API access

B. A unit of text (word, subword, or character) that the model processes

C. A billing unit in AWS

D. An activation in a neural network layer

✓ **Correct Answer: B**

Explanation: Tokens are the basic units LLMs process. A token can be a word, part of a word, or punctuation. For example, 'unhappiness' might be tokenized as 'un', 'happiness'.

Q79. What is 'chain-of-thought' prompting?

A. Linking multiple API calls sequentially

B. Prompting the model to reason step-by-step before giving a final answer

C. Chaining multiple models together

D. A prompt that includes a chain of examples

✓ **Correct Answer: B**

Explanation: Chain-of-thought prompting instructs the model to show its reasoning steps, improving accuracy on complex reasoning tasks by breaking problems into intermediate steps.

Q80. What does 'grounding' mean in the context of generative AI?

A. Preventing the model from generating any output

B. Connecting model outputs to verifiable, factual sources

C. Training the model on domain-specific data

D. Setting temperature to zero

✓ **Correct Answer: B**

Explanation: Grounding connects an AI's responses to verified real-world data or sources, reducing hallucinations and improving factual accuracy of generated outputs.

Q81. What is Amazon Titan Text in Amazon Bedrock?

A. A data processing service

B. Amazon's own family of foundation models for text generation and embedding

C. A text monitoring tool

D. A translation service

✓ **Correct Answer: B**

Explanation: Amazon Titan is Amazon's proprietary family of foundation models available in Bedrock, including text generation (Titan Text) and embedding models (Titan Embeddings).

Q82. What is 'model customization' in Amazon Bedrock?

A. Changing the model's API endpoint

B. Adapting a foundation model using your own data via fine-tuning or continued pre-training

C. Modifying the Bedrock UI

D. Selecting a different AWS region

✓ **Correct Answer: B**

Explanation: Amazon Bedrock model customization allows you to fine-tune foundation models with your proprietary data to improve performance on domain-specific tasks.

Q83. What is a 'knowledge base' in the context of Amazon Bedrock?

A. An AWS documentation library

B. A managed RAG feature that stores and retrieves your data for LLM context

C. A database of foundation model weights

D. A repository of ML training datasets

✓ **Correct Answer: B**

Explanation: Amazon Bedrock Knowledge Bases allow you to connect your private data sources, creating a managed RAG pipeline so LLMs can retrieve relevant context when answering queries.

Q84. What is a 'Bedrock Agent'?

A. An AWS support representative

B. An AI agent that can use tools and APIs to complete multi-step tasks autonomously

C. A monitoring agent for Bedrock infrastructure

D. A data ingestion pipeline

✓ **Correct Answer: B**

Explanation: Amazon Bedrock Agents orchestrate foundation models with action groups (APIs, Lambda functions) and knowledge bases to autonomously complete complex, multi-step tasks.

Q85. What is 'prompt injection' in the context of LLM security?

A. Sending prompts to multiple models simultaneously

B. Malicious inputs designed to override system instructions or manipulate model behavior

C. Injecting additional context into the training process

D. A method to improve prompt quality

✓ **Correct Answer: B**

Explanation: Prompt injection attacks embed malicious instructions in user inputs that can override system prompts, causing the LLM to perform unintended or harmful actions.

Q86. What does 'Guardrails for Amazon Bedrock' provide?

A. Physical security for AWS data centers

B. Safety controls to filter harmful content and implement responsible AI policies

C. Load balancing for foundation model inference

D. Cost controls for Bedrock API usage

✓ Correct Answer: B

Explanation: Guardrails for Amazon Bedrock enables customizable safety policies including content filtering, topic denial, PII redaction, and grounding checks for responsible AI deployment.

Q87. What is 'text embedding' used for in generative AI applications?

A. Displaying text in a web interface

B. Converting text into numerical vectors for semantic search and similarity comparison

C. Formatting model output text

D. Tokenizing text for LLM input

✓ Correct Answer: B

Explanation: Text embeddings convert text into dense numerical vectors where semantically similar texts are geometrically close, enabling semantic search, clustering, and RAG applications.

Q88. What is a 'vector database' used for in AI applications?

A. Storing relational data for ML training

B. Efficiently storing and searching text embeddings for semantic retrieval

C. Managing model deployment configurations

D. Tracking model training metrics

✓ Correct Answer: B

Explanation: Vector databases (e.g., pgvector, OpenSearch, Pinecone) store embeddings and enable fast approximate nearest neighbor search, making them essential for RAG systems.

Q89. What is 'Amazon Bedrock Studio'?

A. A video production service

B. A web-based workspace for building and testing generative AI applications

C. A monitoring dashboard for Bedrock models

D. A code editor for ML models

✓ Correct Answer: B

Explanation: Amazon Bedrock Studio (now part of Amazon SageMaker Studio) provides a visual interface for experimenting with foundation models, building prompts, and developing generative AI apps.

Q90. Which AWS service provides a managed vector store for RAG applications?

A. Amazon RDS

B. Amazon OpenSearch Service with vector search

C. Amazon DynamoDB

D. AWS Glue

✓ Correct Answer: B

Explanation: Amazon OpenSearch Service supports k-NN vector search, making it a popular AWS-managed vector store for RAG applications and semantic search.

Q91. What is 'system prompt' in an LLM application?

A. A prompt generated by the AWS system

B. Instructions provided before the conversation to set model behavior, persona, or constraints

C. The model's internal processing instructions

D. A debug prompt used during model testing

✓ Correct Answer: B

Explanation: A system prompt is a set of instructions given to the LLM at the start of a conversation (before user input) to define its role, tone, constraints, and behavior.

Q92. What is 'Amazon Titan Embeddings'?

A. A video embedding service

B. Amazon's embedding model for converting text into vector representations

C. A feature for embedding Amazon products in websites

D. A compression technique for foundation models

✓ Correct Answer: B

Explanation: Amazon Titan Embeddings is a foundation model in Bedrock that converts text into high-quality vector representations used for semantic search and RAG applications.

Q93. What does 'max tokens' parameter control in LLM API calls?

A. Maximum number of input words

B. Maximum number of tokens the model can generate in a response

C. Maximum number of API calls per minute

D. Maximum model context window size

✓ Correct Answer: B

Explanation: The max_tokens parameter sets an upper limit on how many tokens the model will generate in its response, allowing cost and length control.

Q94. What is 'Stable Diffusion' an example of?

A. A text generation model

B. A diffusion-based image generation model

C. A speech recognition model

D. A video transcription service

✓ Correct Answer: B

Explanation: Stable Diffusion is a latent diffusion model designed to generate high-quality images from text prompts, available through Amazon Bedrock via Stability AI.

Q95. What is 'Amazon Bedrock Playground'?

A. A gaming platform built on Bedrock

B. An interactive environment to test and experiment with foundation models

C. A staging environment for production deployments

D. A benchmarking tool for comparing ML models

✓ Correct Answer: B

Explanation: The Amazon Bedrock Playground (now called the Chat Playground or Text Playground) lets you interactively experiment with different foundation models and parameters without writing code.

Q96. What is 'continued pre-training' in Amazon Bedrock?

A. Re-running the same model training job
B. Further pre-training a foundation model on your domain-specific unlabeled data
C. Training a new model from random weights
D. Continuing training after deployment

✓ Correct Answer: B

Explanation: Continued pre-training in Amazon Bedrock adapts a foundation model to your domain by training on large amounts of domain-specific unlabeled text, updating the model's general knowledge.

Q97. What is 'model evaluation' in Amazon Bedrock?

A. Monitoring model costs
B. Assessing foundation model quality using automatic or human review metrics
C. Evaluating physical hardware requirements
D. Testing API response times

✓ Correct Answer: B

Explanation: Amazon Bedrock Model Evaluation allows you to assess and compare foundation models using built-in metrics (accuracy, robustness, toxicity) or human review to select the best model for your use case.

Q98. What is the primary benefit of using serverless inference in Amazon Bedrock?

A. Lower latency than dedicated endpoints
B. No infrastructure to manage with automatic scaling
C. Ability to train custom models
D. Access to more foundation models

✓ Correct Answer: B

Explanation: Bedrock's serverless inference automatically manages scaling, requires no infrastructure provisioning, and charges only for tokens processed, making it ideal for variable workloads.

Q99. What is 'Amazon Bedrock Flows'?

A. A network traffic management service
B. A visual tool for building multi-step generative AI workflows
C. A data streaming service
D. A CI/CD pipeline for model deployment

✓ Correct Answer: B

Explanation: Amazon Bedrock Flows provides a visual drag-and-drop interface to create multi-step generative AI workflows connecting prompts, models, knowledge bases, and Lambda functions.

Q100. What is 'Constitutional AI' as used by Anthropic?

A. AI governed by government regulations
B. A technique training AI to be helpful, harmless, and honest using a set of principles
C. AI that can write constitutional documents
D. An AI trained exclusively on legal texts

✓ Correct Answer: B

Explanation: Constitutional AI (Anthropic) trains models using a set of principles (a 'constitution') to guide the AI toward helpful, harmless, and honest behavior through self-critique and revision.

Q101. What is 'multimodal AI'?

A. AI that runs on multiple cloud providers
B. AI that processes and generates multiple types of data (text, image, audio)
C. AI that uses multiple optimization algorithms
D. AI deployed across multiple AWS regions

✓ Correct Answer: B

Explanation: Multimodal AI models can process and generate multiple data types (e.g., text and images together), enabling applications like image captioning, visual question answering, and document analysis.

Q102. What is 'Amazon Nova' in the context of Amazon Bedrock?

A. A new AWS storage service
B. Amazon's latest family of state-of-the-art foundation models on Bedrock
C. A monitoring tool for Bedrock
D. A certification program for AI practitioners

✓ Correct Answer: B

Explanation: Amazon Nova is Amazon's latest generation of frontier models on Bedrock, including Nova Micro, Nova Lite, and Nova Pro, offering high performance across text, image, and video tasks.

Q103. What is 'Bedrock Batch Inference'?

A. Sending multiple API calls simultaneously from different users
B. Processing large volumes of inference requests asynchronously at lower cost
C. Batching training data for fine-tuning
D. A method of batch-loading foundation models

✓ Correct Answer: B

Explanation: Bedrock Batch Inference processes large-scale inference jobs asynchronously, enabling cost-effective processing of thousands of prompts without real-time response requirements.

Q104. What does 'Amazon Q' primarily provide?

A. A queue service for AI jobs
B. A generative AI-powered assistant for business and developer productivity
C. A quantization service for model compression
D. A queueing system for Bedrock API requests

✓ Correct Answer: B

Explanation: Amazon Q is a generative AI-powered assistant that helps businesses query their data, answer questions, generate code, and automate tasks using AWS services.

Q105. What is Amazon Q Developer used for?

A. Developing queue-based applications
B. AI-powered code generation, completion, and security scanning for developers
C. Managing developer IAM permissions
D. Deploying Lambda functions

✓ Correct Answer: B

Explanation: Amazon Q Developer (formerly CodeWhisperer) is an AI coding assistant that generates code suggestions, completes code, answers AWS questions, and scans for security vulnerabilities.

Q106. What is 'Amazon Q Business'?

A. A business analytics dashboard
B. A generative AI assistant that answers questions using your enterprise data
C. A CRM system built on AWS
D. A billing management tool

✓ Correct Answer: B

Explanation: Amazon Q Business is a generative AI assistant that connects to enterprise data sources (S3, SharePoint, Salesforce) to answer employee questions using company-specific knowledge.

Q107. What is 'Retrieval Augmented Generation' primarily designed to solve?

A. Slow model inference speed
B. LLM knowledge cutoff and hallucination issues by providing relevant context
C. High cost of training foundation models
D. Security vulnerabilities in model APIs

✓ Correct Answer: B

Explanation: RAG addresses LLM limitations like knowledge cutoffs and hallucinations by retrieving up-to-date, relevant information from external sources and including it in the model's context.

Q108. What is 'Meta Llama' available in Amazon Bedrock?

A. Amazon's internal model family
B. Meta's open-source LLM family accessible through Bedrock
C. A llama image generation model
D. A metadata management service

✓ Correct Answer: B

Explanation: Meta's Llama family of open-source LLMs (including Llama 2 and Llama 3) is available through Amazon Bedrock, providing strong open-source foundation model options.

Q109. What is 'Mistral AI' in the context of Amazon Bedrock?

A. An AWS weather prediction service
B. A European AI company whose models are available on Bedrock
C. A model monitoring service
D. A specialized compliance model

✓ Correct Answer: B

Explanation: Mistral AI is a French AI company providing efficient, high-performance LLMs (Mistral and Mixtral models) available through Amazon Bedrock.

Q110. What is 'Amazon Bedrock Converse API'?

A. A translation service
B. A unified API for multi-turn conversations across different foundation models
C. A social media integration service
D. A customer service routing system

✓ **Correct Answer: B**

Explanation: The Bedrock Converse API provides a consistent interface for multi-turn conversations across all supported foundation models, simplifying development of chat applications.

Q111. What is 'Amazon Bedrock Marketplace'?

A. An e-commerce platform built on Bedrock

B. A catalog for discovering and subscribing to third-party foundation models

C. A marketplace for buying AI training data

D. An AWS service for selling ML applications

✓ **Correct Answer: B**

Explanation: Amazon Bedrock Marketplace expands the model catalog by allowing access to models from additional third-party providers beyond the standard Bedrock offerings.

Q112. What is 'Cohere Command' available through Amazon Bedrock?

A. A CLI command for Bedrock operations

B. Cohere's LLM designed for enterprise text generation tasks

C. A command routing service

D. A security command center

✓ **Correct Answer: B**

Explanation: Cohere's Command models available on Bedrock are enterprise-focused LLMs optimized for text generation, summarization, and RAG applications.

Q113. What are 'inference parameters' in Amazon Bedrock?

A. AWS infrastructure configuration settings

B. Model-specific settings (temperature, top-p, max tokens) controlling generation behavior

C. Network parameters for API connectivity

D. Cost parameters for billing configuration

✓ **Correct Answer: B**

Explanation: Inference parameters control how foundation models generate responses. Key parameters include temperature (randomness), top-p (nucleus sampling), and max_tokens (output length).

Q114. What is 'Amazon Bedrock Provisioned Throughput'?

A. A networking bandwidth allocation

B. Reserved model capacity ensuring consistent performance for high-volume workloads

C. Pre-provisioned S3 storage for training data

D. Dedicated GPU allocation for model training

✓ **Correct Answer: B**

Explanation: Provisioned Throughput in Amazon Bedrock reserves model units (MUs) for consistent, predictable inference performance at high volumes, suitable for production applications.

Q115. What is the primary use of 'Amazon Titan Image Generator'?

A. Analyzing and labeling existing images

B. Generating new images from text prompts

C. Detecting objects in uploaded images

D. Compressing images for storage

✓ **Correct Answer: B**

Explanation: Amazon Titan Image Generator is a foundation model on Bedrock that creates realistic, studio-quality images from text descriptions and supports image editing capabilities.

Q116. What is 'prompt caching' in the context of LLM APIs?

A. Saving prompts to S3

B. Caching processed prompt prefixes to reduce latency and cost for repeated contexts

C. Storing user prompts for compliance

D. Encrypting prompts for security

✓ **Correct Answer: B**

Explanation: Prompt caching stores the KV cache of processed prompt prefixes, allowing reuse when the same context is repeated across requests, reducing both latency and cost.

Q117. What is 'jailbreaking' in the context of LLM safety?

A. Freeing model weights from AWS encryption

B. Crafting inputs to bypass safety filters and make an LLM produce restricted content

C. Running an LLM on personal hardware without license

D. Breaking a model's context limit

✓ **Correct Answer: B**

Explanation: Jailbreaking refers to adversarial prompt techniques designed to circumvent a model's safety guidelines, causing it to produce harmful, offensive, or restricted content.

Domain 3: AWS AI & ML Services

Q118. What is Amazon SageMaker primarily used for?

A. Database management

B. Building, training, and deploying machine learning models at scale

C. Managing EC2 instances

D. Storing large datasets

✓ **Correct Answer: B**

Explanation: Amazon SageMaker is AWS's comprehensive managed ML platform for the full ML lifecycle: data preparation, model building, training, evaluation, deployment, and monitoring.

Q119. What is Amazon Rekognition used for?

A. Speech-to-text conversion

B. Image and video analysis using computer vision

C. Text translation between languages

D. Fraud detection in financial data

✓ **Correct Answer: B**

Explanation: Amazon Rekognition provides computer vision capabilities including object detection, facial recognition, activity detection, content moderation, and text extraction from images/videos.

Q120. What is Amazon Comprehend used for?

A. Image classification
B. Natural language processing tasks like sentiment analysis and entity recognition
C. Speech synthesis
D. Video captioning

✓ Correct Answer: B

Explanation: Amazon Comprehend is a fully managed NLP service for extracting insights from text including sentiment analysis, entity recognition, key phrase extraction, and topic modeling.

Q121. What is Amazon Transcribe used for?

A. Translating text between languages
B. Converting speech to text (automatic speech recognition)
C. Text-to-speech synthesis
D. Generating captions for images

✓ Correct Answer: B

Explanation: Amazon Transcribe is an ASR (Automatic Speech Recognition) service that converts spoken audio to text, supporting multiple languages and custom vocabulary.

Q122. What is Amazon Polly used for?

A. Analyzing sentiment in text
B. Converting text to realistic speech (text-to-speech)
C. Speech-to-text transcription
D. Language translation

✓ Correct Answer: B

Explanation: Amazon Polly is a text-to-speech service that converts text into natural-sounding speech using deep learning, supporting dozens of languages and voices.

Q123. What is Amazon Translate used for?

A. Converting speech to text
B. Neural machine translation between languages
C. Extracting entities from text
D. Generating text from images

✓ Correct Answer: B

Explanation: Amazon Translate is a neural machine translation service providing fast, high-quality translation between a wide range of language pairs.

Q124. What is Amazon Lex used for?

A. Text-to-speech conversion
B. Building conversational interfaces (chatbots) with voice and text
C. Document analysis
D. Image generation

✓ Correct Answer: B

Explanation: Amazon Lex provides ASR and NLU capabilities for building conversational chatbots and voice interfaces, powering Amazon Alexa.

Q125. What is Amazon Kendra used for?

A. Speech recognition
B. Intelligent enterprise search powered by ML
C. Image labeling
D. Time series forecasting

✓ Correct Answer: B

Explanation: Amazon Kendra is an intelligent search service using NLP and ML to provide highly accurate answers from enterprise document repositories.

Q126. What is Amazon Forecast used for?

A. Predicting weather conditions
B. Time series forecasting for business metrics
C. Detecting anomalies in log data
D. Recommending products to customers

✓ Correct Answer: B

Explanation: Amazon Forecast is a fully managed service using ML to deliver highly accurate time series forecasts for business metrics like demand, inventory, and staffing.

Q127. What is Amazon Personalize used for?

A. Personalizing AWS console settings
B. Building real-time personalization and recommendation systems
C. Customizing ML model architecture
D. Personalizing IAM policies

✓ Correct Answer: B

Explanation: Amazon Personalize enables developers to build and deploy real-time personalization and recommendation systems using the same ML technology used by Amazon.com.

Q128. What is Amazon Fraud Detector used for?

A. Detecting security vulnerabilities in code
B. Building ML-powered fraud detection models for online activities
C. Identifying bugs in ML models
D. Monitoring network intrusions

✓ Correct Answer: B

Explanation: Amazon Fraud Detector uses ML and Amazon's fraud detection expertise to build custom fraud detection models for use cases like account takeover and payment fraud.

Q129. What is Amazon Textract used for?

A. Text-to-speech conversion
B. Extracting text and structured data from scanned documents
C. Translating document text
D. Generating text summaries

✓ **Correct Answer: B**

Explanation: Amazon Textract automatically extracts text, forms, tables, and structured data from scanned documents and images using ML, going beyond simple OCR.

Q130. What is Amazon SageMaker Autopilot?

A. A tool for auto-scaling SageMaker instances

B. An AutoML service that automatically trains and tunes ML models

C. A CI/CD pipeline for model deployment

D. A feature for automatic data labeling

✓ **Correct Answer: B**

Explanation: SageMaker Autopilot automatically explores different ML algorithms and hyperparameters to find the best model for tabular datasets, providing full visibility into the process.

Q131. What is Amazon SageMaker Ground Truth used for?

A. Deploying models to production

B. Building high-quality labeled datasets using human annotators

C. Monitoring model performance

D. Tracking model experiments

✓ **Correct Answer: B**

Explanation: SageMaker Ground Truth helps build labeled training datasets using human annotators (AWS Marketplace or private teams) with built-in workflows and active learning.

Q132. What is Amazon SageMaker Feature Store?

A. A marketplace for ML models

B. A fully managed repository for storing, sharing, and reusing ML features

C. A feature flags service

D. A data lake for raw training data

✓ **Correct Answer: B**

Explanation: SageMaker Feature Store is a purpose-built repository for ML features, enabling feature reuse across teams, consistent feature computation, and point-in-time retrieval.

Q133. What is Amazon SageMaker Clarify used for?

A. Clarifying API documentation

B. Detecting bias in training data and explaining model predictions

C. Clarifying data access permissions

D. Cleaning corrupted training data

✓ **Correct Answer: B**

Explanation: SageMaker Clarify detects statistical bias in datasets and model predictions, and provides feature importance explanations (using SHAP values) for model transparency.

Q134. What is Amazon SageMaker Model Monitor used for?

A. Monitoring EC2 instance health

B. Continuously monitoring deployed ML models for data quality and model drift

C. Monitoring training job progress

D. Monitoring S3 data uploads

✓ **Correct Answer: B**

Explanation: SageMaker Model Monitor automatically detects data quality issues and model drift in production, sending alerts when model inputs or predictions deviate from baselines.

Q135. What is Amazon SageMaker Pipelines?

A. Network data pipelines for SageMaker

B. A CI/CD service for automating ML workflows end-to-end

C. An ETL service for data preprocessing

D. A monitoring pipeline for models

✓ **Correct Answer: B**

Explanation: SageMaker Pipelines provides CI/CD for ML, allowing teams to automate, reproduce, and track the entire ML workflow from data preparation to model deployment.

Q136. What is Amazon SageMaker JumpStart?

A. A service for quickly launching EC2 instances

B. A hub for pre-built ML solutions, foundation models, and algorithms

C. A starter kit for new AWS accounts

D. A quick-start guide for SageMaker

✓ **Correct Answer: B**

Explanation: SageMaker JumpStart provides a model hub with hundreds of pre-trained models (including open-source LLMs), built-in algorithms, and ML solution templates deployable with one click.

Q137. What is AWS Glue DataBrew used for in ML workflows?

A. Training ML models

B. Visual data preparation and cleaning without writing code

C. Deploying models to production

D. Monitoring model performance

✓ **Correct Answer: B**

Explanation: AWS Glue DataBrew provides a visual data preparation tool allowing analysts to clean and normalize data using 250+ built-in transformations without writing code.

Q138. What is Amazon SageMaker Studio?

A. A video studio for creating AI demos

B. A web-based integrated development environment for ML

C. A recording studio for Polly voices

D. A design studio for AI applications

✓ **Correct Answer: B**

Explanation: SageMaker Studio is a fully integrated web-based IDE that provides a unified environment for the entire ML workflow: data exploration, model development, training, and deployment.

Q139. What is AWS DeepRacer?

A. A fast data processing service

B. A physical autonomous racing car for learning reinforcement learning

C. A deep learning training accelerator

D. A racing game for AWS employees

✓ Correct Answer: B

Explanation: AWS DeepRacer is a 1/18th scale autonomous racing car that uses reinforcement learning, providing a fun, gamified way to learn RL concepts through racing simulation.

Q140. What is Amazon Lookout for Vision used for?

A. Employee monitoring

B. Detecting visual defects and anomalies in products using ML

C. Analyzing business metrics

D. Security camera monitoring

✓ Correct Answer: B

Explanation: Amazon Lookout for Vision uses computer vision ML to detect manufacturing defects and anomalies in product images, enabling automated quality inspection.

Q141. What is Amazon Lookout for Metrics used for?

A. Monitoring AWS service metrics

B. Automatically detecting anomalies in business metrics

C. Forecasting future metric values

D. Logging ML model metrics

✓ Correct Answer: B

Explanation: Amazon Lookout for Metrics automatically detects anomalies in business metrics (revenue, user engagement, operations) and identifies their root causes.

Q142. What is Amazon Monitron?

A. A security monitoring service

B. An end-to-end equipment monitoring system using ML for predictive maintenance

C. A model performance monitoring tool

D. A network traffic monitor

✓ Correct Answer: B

Explanation: Amazon Monitron is an end-to-end IoT solution using ML to detect abnormal conditions in industrial equipment, enabling predictive maintenance programs.

Q143. What is AWS Panorama used for?

A. Wide-angle photography services

B. Deploying computer vision models to on-premises cameras

C. Panoramic image generation

D. Cloud-based video surveillance

✓ Correct Answer: B

Explanation: AWS Panorama is an ML appliance and SDK that enables deploying computer vision models directly to on-premises cameras for local real-time video analytics.

Q144. What is Amazon Comprehend Medical?

A. A medical diagnosis AI service

B. An NLP service extracting medical information from clinical text

C. A health monitoring service

D. A medical record storage service

✓ Correct Answer: B

Explanation: Amazon Comprehend Medical uses NLP to extract medical information (conditions, medications, dosages, anatomy) from unstructured clinical text and medical records.

Q145. What is Amazon HealthLake?

A. A data lake for general healthcare compliance

B. A HIPAA-eligible service for storing, transforming, and querying health data in FHIR format

C. A lake storage service for medical images

D. A backup service for electronic health records

✓ Correct Answer: B

Explanation: Amazon HealthLake is a HIPAA-eligible service enabling healthcare organizations to store, transform, and query health data in HL7 FHIR format at scale for analytics and ML.

Q146. What is Amazon SageMaker Canvas?

A. A UI design tool for AI apps

B. A no-code ML tool for business analysts to build models without coding

C. A visual model architecture designer

D. A dashboard for monitoring ML costs

✓ Correct Answer: B

Explanation: SageMaker Canvas is a no-code visual interface allowing business analysts to build and deploy ML models for classification, regression, forecasting, and NLP without writing code.

Q147. What is Amazon Augmented AI (A2I) used for?

A. Augmenting training datasets with synthetic data

B. Building human review workflows for ML predictions

C. Augmenting model architectures automatically

D. Increasing model training speed

✓ Correct Answer: B

Explanation: Amazon Augmented AI (A2I) enables building human review workflows to review low-confidence ML predictions, ensuring accuracy for sensitive use cases.

Q148. What is AWS Inferentia?

A. An inference monitoring service

B. A custom AWS chip designed for high-performance, cost-effective ML inference

C. A managed inference endpoint service

D. An inference API for Bedrock models

✓ Correct Answer: B

Explanation: AWS Inferentia is a custom ML inference chip (available via EC2 Inf instances) providing high throughput and low-cost inference for large deep learning models.

Q149. What is AWS Trainium?

A. A training platform for AWS employees
B. A custom AWS chip optimized for high-performance ML model training
C. A managed training service for LLMs
D. A training data management service

✓ Correct Answer: B

Explanation: AWS Trainium is a custom ML chip (available via EC2 Trn instances) optimized for training deep learning models, offering cost savings over GPU-based training.

Q150. What is Amazon SageMaker Experiments?

A. A service for experimental AWS features
B. A tool for tracking, comparing, and managing ML training experiments
C. An A/B testing framework for ML models
D. A sandbox environment for model development

✓ Correct Answer: B

Explanation: SageMaker Experiments enables logging, comparing, and visualizing ML experiments (parameters, metrics, artifacts) to track the evolution of model development.

Q151. What is the purpose of SageMaker Model Registry?

A. Registering AWS accounts for SageMaker
B. Cataloging, versioning, and managing ML models for deployment governance
C. Storing model training data
D. Registering domain names for ML APIs

✓ Correct Answer: B

Explanation: SageMaker Model Registry maintains a catalog of ML model versions with metadata, approval workflows, and deployment tracking for MLOps governance.

Q152. What does Amazon Rekognition Content Moderation do?

A. Moderates access to Rekognition API
B. Detects inappropriate or unsafe content in images and videos
C. Moderates user comments in applications
D. Controls Rekognition service costs

✓ Correct Answer: B

Explanation: Rekognition Content Moderation automatically detects explicit, suggestive, or violent content in images and videos, enabling platforms to enforce content policies at scale.

Q153. What is Amazon Bedrock's 'on-demand' inference pricing model?

A. Paying a fixed monthly fee for unlimited API calls
B. Paying per token (input and output) with no minimum commitment
C. Reserving capacity in advance for discounts
D. Paying per hour of model deployment

✓ Correct Answer: B

Explanation: Bedrock on-demand pricing charges per token processed (input tokens + output tokens), with no upfront costs or minimum commitments, suitable for variable workloads.

Domain 4: Responsible AI

Q154. What are the core principles of responsible AI?

A. Speed, cost, and scalability
B. Fairness, transparency, accountability, privacy, and safety
C. Accuracy, latency, and throughput
D. Automation, optimization, and innovation

✓ Correct Answer: B

Explanation: Responsible AI principles include fairness (unbiased outcomes), transparency (explainability), accountability (human oversight), privacy (data protection), and safety (preventing harm).

Q155. What is 'algorithmic bias'?

A. A technique for improving model accuracy
B. Systematic and unfair discrimination in model outputs toward certain groups
C. The preference for certain algorithms over others
D. Bias introduced by hardware differences

✓ Correct Answer: B

Explanation: Algorithmic bias occurs when ML models produce systematically unfair results that disadvantage certain groups, often stemming from biased training data or model design.

Q156. What is Amazon SageMaker Clarify's primary function regarding fairness?

A. Increasing model training speed
B. Detecting and measuring bias in datasets and model predictions
C. Clarifying model deployment procedures
D. Improving data storage efficiency

✓ Correct Answer: B

Explanation: SageMaker Clarify measures bias metrics in training data and model predictions across demographic groups, helping identify and address fairness issues.

Q157. What does 'model transparency' mean in responsible AI?

A. Making model weights publicly available
B. Being able to understand and explain how a model makes decisions
C. Deploying models with open-source licenses
D. Showing model training costs to users

✓ Correct Answer: B

Explanation: Model transparency means users and stakeholders can understand the factors influencing model decisions, which builds trust and enables identifying and correcting problems.

Q158. What is 'data minimization' in AI privacy practices?

A. Reducing dataset file size
B. Collecting and using only the minimum data necessary for a given purpose
C. Minimizing training time by using less data
D. Reducing duplicate records in datasets

✓ **Correct Answer: B**

Explanation: Data minimization is a privacy principle requiring collection of only the minimum amount of personal data necessary, reducing privacy risks and compliance obligations.

Q159. What is 'PII' in the context of AI data handling?

A. Pre-trained Inference Interface

B. Personally Identifiable Information that can identify an individual

C. Parallel Inference Implementation

D. Privacy Impact Index

✓ **Correct Answer: B**

Explanation: PII (Personally Identifiable Information) includes any data that can identify an individual (name, SSN, email, address). Proper handling and protection of PII is critical in AI systems.

Q160. What does Amazon Macie help with in AI workflows?

A. Training ML models

B. Discovering and protecting sensitive data (PII) in S3 using ML

C. Making model predictions

D. Managing ML pipelines

✓ **Correct Answer: B**

Explanation: Amazon Macie uses ML to automatically discover, classify, and protect sensitive data (PII, financial data) stored in Amazon S3, supporting data privacy compliance.

Q161. What is 'human-in-the-loop' in AI systems?

A. A training technique using human-generated synthetic data

B. Keeping human oversight in AI decision-making for review and correction

C. A method for human-AI collaborative model building

D. Human review of AWS billing

✓ **Correct Answer: B**

Explanation: Human-in-the-loop keeps humans involved in AI decision-making processes, particularly for high-stakes decisions, providing oversight, correction, and accountability.

Q162. What is 'toxicity detection' in generative AI?

A. Detecting toxic chemicals in industrial processes

B. Identifying and filtering harmful, offensive, or dangerous content in AI outputs

C. Monitoring model performance degradation

D. Detecting data poisoning attacks

✓ **Correct Answer: B**

Explanation: Toxicity detection identifies harmful content (hate speech, violence, harassment) in AI-generated outputs, enabling filtering before content reaches users.

Q163. What is the purpose of AWS AI Service Cards?

A. Credit cards for paying for AWS AI services

B. Documentation explaining use cases, limitations, and responsible AI practices for AWS AI services

C. Flash cards for learning AWS AI services

D. Configuration cards for AI model deployment

✓ **Correct Answer: B**

Explanation: AWS AI Service Cards provide transparency by documenting intended use cases, limitations, responsible AI design choices, and performance characteristics of AWS AI services.

Q164. What is 'model cards' in the context of responsible AI?

A. Physical cards used to program AI hardware

B. Structured documents summarizing a model's purpose, performance, limitations, and ethical considerations

C. API authentication cards for model access

D. Billing statements for model usage

✓ **Correct Answer: B**

Explanation: Model cards are short documents accompanying trained ML models that describe their intended use, performance across different groups, limitations, and ethical considerations.

Q165. What is 'differential privacy' in ML?

A. Different privacy levels for different users

B. A technique adding statistical noise to protect individual privacy while enabling aggregate analysis

C. Privacy settings differentiated by data type

D. Encrypting different model layers

✓ **Correct Answer: B**

Explanation: Differential privacy adds carefully calibrated noise to data or model outputs, ensuring that the presence or absence of any single individual's data cannot be determined from the results.

Q166. What is 'federated learning' and why is it relevant to privacy?

A. Learning from data across multiple AWS regions

B. Training models on distributed local data without sending raw data to a central server

C. A federal government AI regulation

D. A technique for sharing model weights securely

✓ **Correct Answer: B**

Explanation: Federated learning trains models locally on devices/institutions and shares only model updates (not raw data), preserving data privacy while enabling collaborative learning.

Q167. What does 'GDPR' stand for and why is it relevant to AI?

A. General Data Processing Regulation

B. General Data Protection Regulation — EU law governing personal data use, including in AI systems

C. Global Database Privacy Requirements

D. Generative Data Protection Rules

✓ **Correct Answer: B**

Explanation: GDPR is the EU's comprehensive data protection law that impacts how AI systems collect, process, and use personal data of EU residents, including rights to explanation of automated decisions.

Q168. What is 'consent management' in AI data practices?

A. Managing IAM consent policies

B. Ensuring individuals provide informed consent before their data is used for AI training
C. Obtaining approval for model deployment
D. Managing access consent for S3 buckets

✓ Correct Answer: B

Explanation: Consent management ensures individuals are informed about how their data will be used for AI purposes and have given explicit permission, fulfilling legal and ethical obligations.

Q169. What is 'Amazon Bedrock Guardrails' content filtering used for?

A. Filtering network traffic to Bedrock
B. Blocking harmful, inappropriate, or unwanted content in LLM interactions
C. Filtering training data quality
D. Controlling API rate limits

✓ Correct Answer: B

Explanation: Bedrock Guardrails content filtering detects and blocks harmful categories (hate speech, violence, sexual content, profanity) in both input prompts and model responses.

Q170. What is 'data provenance' in ML?

A. The geographic origin of training data
B. Tracking the origin, transformation history, and lineage of data used in AI
C. The copyright provenance of model outputs
D. Proving data accuracy

✓ Correct Answer: B

Explanation: Data provenance tracks where training data came from, how it was transformed, and who handled it, supporting accountability, debugging, and compliance in AI systems.

Q171. What is 'content moderation' in AI applications?

A. Moderating model training speed
B. Reviewing and filtering user-generated content to enforce community standards
C. Managing model deployment approvals
D. Monitoring model costs

✓ Correct Answer: B

Explanation: Content moderation uses AI (and humans) to review and filter user-generated or AI-generated content, removing material that violates safety policies or community standards.

Q172. What is 'demographic parity' in AI fairness?

A. Equal sample sizes in training data
B. Equal prediction rates across demographic groups
C. Balanced representation of genders in AI teams
D. Equal compute allocation across regions

✓ Correct Answer: B

Explanation: Demographic parity (statistical parity) requires that a model's positive prediction rate is equal across different demographic groups (e.g., equal loan approval rates regardless of race).

Q173. What is 'explainable AI' (XAI)?

A. AI that provides explicit API documentation
B. AI systems whose decisions can be understood and explained to humans
C. AI that explicitly states its training data
D. AI with explicit cost breakdowns

✓ Correct Answer: B

Explanation: Explainable AI (XAI) refers to methods and techniques that make AI model decisions understandable to humans, crucial for trust, debugging, and regulatory compliance.

Q174. What is 'SHAP' (SHapley Additive exPlanations) used for?

A. A model training optimization technique
B. Explaining individual predictions by calculating each feature's contribution
C. A data augmentation method
D. A hyperparameter tuning algorithm

✓ Correct Answer: B

Explanation: SHAP values explain individual model predictions by computing each feature's contribution to the prediction, providing consistent and locally accurate explanations.

Q175. What is 'watermarking' in generative AI?

A. Adding branding to AI-generated images
B. Embedding imperceptible signals in AI-generated content to identify its origin
C. Protecting model weights from copying
D. Adding timestamps to model outputs

✓ Correct Answer: B

Explanation: AI watermarking embeds imperceptible signals in generated content (text, images) to allow detection of AI-generated material, supporting content authenticity and copyright enforcement.

Q176. What does the 'right to explanation' mean in AI context?

A. The right to access model training code
B. The right for individuals to receive an explanation for automated decisions affecting them
C. The right to explain model architecture to auditors
D. The developer's right to explain model limitations

✓ Correct Answer: B

Explanation: The right to explanation (established in GDPR) allows individuals to request a meaningful explanation of automated decisions that significantly affect them, promoting accountability.

Q177. What is 'adversarial AI'?

A. AI systems competing against each other
B. Attacks designed to manipulate or deceive AI models through malicious inputs
C. AI developed by competitors
D. Reinforcement learning with adversarial agents

✓ Correct Answer: B

Explanation: Adversarial AI involves crafting inputs specifically designed to fool ML models, causing misclassification or harmful behavior, representing a key security and safety concern.

Q178. What is 'model governance' in enterprise AI?

A. The hardware governance for ML infrastructure

B. Policies, processes, and controls for responsible development and deployment of AI models

C. Government regulations about AI

D. Model versioning in a code repository

✓ Correct Answer: B

Explanation: Model governance establishes organizational policies, approval workflows, documentation standards, and monitoring practices to ensure AI models are developed and used responsibly.

Q179. What is 'bias mitigation' in ML?

A. Reducing computational bias toward certain hardware

B. Techniques to detect and reduce unfair bias in models and training data

C. Removing high-variance samples from datasets

D. Reducing overfitting using regularization

✓ Correct Answer: B

Explanation: Bias mitigation encompasses data-level techniques (resampling, reweighting) and model-level techniques (constraint optimization, adversarial debiasing) to reduce unfair model predictions.

Q180. What is 'responsible AI' in the context of AWS?

A. A specific AWS service for AI compliance

B. AWS's framework for developing AI that is fair, explainable, robust, verifiable, privacy-preserving, and secure

C. A cost management approach for AI spending

D. Responsible use of GPU resources

✓ Correct Answer: B

Explanation: AWS Responsible AI encompasses practices and tools (SageMaker Clarify, Bedrock Guardrails, AI Service Cards) supporting ethical AI development aligned with fairness, transparency, and safety.

Q181. What is the purpose of 'red teaming' in AI safety?

A. A performance testing technique for ML pipelines

B. Adversarial testing of AI systems to find safety and security vulnerabilities

C. Color coding ML experiments for tracking

D. A method for parallel model training

✓ Correct Answer: B

Explanation: Red teaming in AI involves adversarial testing where teams attempt to find harmful outputs, safety failures, and security vulnerabilities in AI systems before deployment.

Q182. What does 'accountability' mean in responsible AI?

A. Tracking API call costs

B. Clear ownership and responsibility for AI system behavior and outcomes

C. Tracking model training time

D. Financial accountability for AI projects

✓ Correct Answer: B

Explanation: Accountability in AI means clear assignment of responsibility for AI system behavior to humans/organizations, ensuring someone can be held responsible for harmful outcomes.

Q183. What is 'synthetic data' and its role in responsible AI?

A. Data generated by sensor hardware
B. Artificially generated data preserving statistical properties while protecting privacy
C. Data synthesized from multiple databases
D. Summary statistics used instead of raw data

✓ Correct Answer: B

Explanation: Synthetic data is artificially generated to mimic real data's statistical properties while containing no real individual's information, enabling ML training while preserving privacy.

Domain 5: Security, Compliance & Governance

Q184. What is AWS IAM and how does it apply to ML workloads?

A. An AI model management service
B. Identity and Access Management — controls who can access AWS AI/ML services and resources
C. Infrastructure Automation Management for ML
D. An image analysis module

✓ Correct Answer: B

Explanation: AWS IAM controls access to AWS services including SageMaker and Bedrock through policies. Least-privilege IAM roles should be used to restrict access to ML resources.

Q185. What is the AWS Shared Responsibility Model in the context of AI/ML?

A. AWS and customers share equal responsibility for all aspects
B. AWS secures infrastructure; customers secure their data, models, and application logic
C. Customers are responsible for all security aspects
D. AWS is responsible for model accuracy and fairness

✓ Correct Answer: B

Explanation: Under the Shared Responsibility Model, AWS secures the cloud infrastructure (hardware, network, managed service security), while customers secure their data, models, IAM configurations, and applications.

Q186. What is encryption 'at rest' in the context of ML data?

A. Encrypting data while it's being processed
B. Encrypting data stored in storage services like S3 or EBS
C. Encrypting model API communications
D. Encrypting training logs

✓ Correct Answer: B

Explanation: Encryption at rest protects data stored in storage services (S3, EBS, RDS) from unauthorized access. SageMaker and Bedrock support encryption of training data and model artifacts at rest.

Q187. What is encryption 'in transit' in ML workflows?

A. Encrypting data before uploading to S3
B. Encrypting data as it moves between services using TLS/HTTPS
C. Encrypting model weights during training
D. Encrypting batch inference results

✓ Correct Answer: B

Explanation: Encryption in transit protects data moving between components (e.g., between your application and Bedrock API) using TLS, preventing interception of sensitive data.

Q188. What is AWS Key Management Service (KMS) used for in AI workflows?

A. Managing API keys for Bedrock
B. Creating and managing encryption keys for securing ML data and model artifacts
C. Managing SSH keys for SageMaker instances
D. A key-value store for model parameters

✓ Correct Answer: B

Explanation: AWS KMS manages cryptographic keys used to encrypt training data, model artifacts, and inference results in services like SageMaker and Bedrock.

Q189. What is Amazon VPC and why is it important for ML security?

A. A virtual private cloud for isolating ML resources from public internet access
B. A virtual processing cluster for ML training
C. A VPN service for remote ML teams
D. A virtual pipeline controller for ML workflows

✓ Correct Answer: A

Explanation: Amazon VPC creates an isolated network environment, allowing SageMaker training jobs and endpoints to run without internet exposure, reducing attack surface for ML workloads.

Q190. What is 'data governance' in the context of AI?

A. Government regulations about AI usage
B. Policies and processes managing data availability, quality, security, and usability
C. Governing model training procedures
D. Managing data storage costs

✓ Correct Answer: B

Explanation: Data governance establishes policies for how data is collected, stored, used, and shared, ensuring quality, security, compliance, and responsible use in AI/ML workflows.

Q191. What is AWS CloudTrail used for in AI/ML workflows?

A. Tracking ML model training progress
B. Logging and auditing API calls made to AWS services including SageMaker and Bedrock
C. Tracing distributed ML pipeline performance
D. Managing model deployment trails

✓ Correct Answer: B

Explanation: AWS CloudTrail records all API calls to AWS services, providing an audit trail of who accessed ML resources, what changes were made, and when, supporting security and compliance.

Q192. What is Amazon GuardDuty and its relevance to ML security?

A. A guard service for physical AWS data centers

B. ML-powered threat detection service that monitors for malicious activity in AWS accounts

C. A model guardrails service

D. A service for guarding model intellectual property

✓ Correct Answer: B

Explanation: Amazon GuardDuty uses ML to continuously monitor AWS account activity and detect threats like unauthorized access, data exfiltration, and malicious insider activity affecting ML resources.

Q193. What is HIPAA compliance relevant to in AI?

A. High-performance AI processing acceleration

B. Healthcare data privacy law requiring protection of patient health information in AI applications

C. A performance benchmark for AI models

D. Amazon's internal AI compliance framework

✓ Correct Answer: B

Explanation: HIPAA regulates the use of Protected Health Information (PHI). AI applications processing medical data must comply with HIPAA, using services like Amazon HealthLake that support HIPAA eligibility.

Q194. What is 'least privilege' principle in IAM for AI workloads?

A. Granting the minimum permissions necessary to perform required tasks

B. Using the cheapest IAM configuration

C. Assigning fewer roles to reduce management overhead

D. A simplified IAM policy format

✓ Correct Answer: A

Explanation: Least privilege means granting only the minimum permissions needed (e.g., SageMaker execution roles should only access specific S3 buckets, not all AWS resources).

Q195. What does AWS Config do in the context of AI compliance?

A. Configures SageMaker hyperparameters

B. Continuously monitors and records AWS resource configurations for compliance auditing

C. Configures network routing for ML traffic

D. Manages Bedrock model configurations

✓ Correct Answer: B

Explanation: AWS Config continuously records configuration changes to AWS resources and evaluates them against compliance rules, supporting audit and governance of AI/ML infrastructure.

Q196. What is 'model theft' in AI security?

A. Stealing physical hardware running AI models

B. Extracting a model's functionality or weights through repeated API queries

C. Stealing AI training data

D. Unauthorized use of model API keys

✓ Correct Answer: B

Explanation: Model theft (model extraction) attacks query a deployed model extensively to reconstruct its functionality or approximate its weights, stealing intellectual property.

Q197. What is 'data poisoning' in ML security?

A. Introducing corrupted samples into training data to manipulate model behavior
B. Exposing sensitive data during model training
C. Overfitting on noisy training data
D. Saturating storage with large datasets

✓ Correct Answer: A

Explanation: Data poisoning attacks inject malicious or mislabeled samples into training data to cause the trained model to behave incorrectly or have hidden vulnerabilities.

Q198. What is Amazon Inspector and its relevance to AI/ML?

A. An ML model inspection service
B. An automated vulnerability scanning service for ML application infrastructure
C. A service for inspecting training data quality
D. A code review tool for ML pipelines

✓ Correct Answer: B

Explanation: Amazon Inspector automatically scans EC2 instances, containers, and Lambda functions for software vulnerabilities, helping secure the infrastructure running ML workloads.

Q199. What is AWS Artifact used for?

A. Storing ML model artifacts
B. Accessing AWS compliance reports and security certifications
C. Managing deployment artifacts
D. Storing training data artifacts

✓ Correct Answer: B

Explanation: AWS Artifact provides on-demand access to AWS compliance reports (SOC, PCI, ISO) and agreements, helping customers demonstrate compliance for regulated AI workloads.

Q200. What is 'network isolation' for SageMaker training jobs?

A. Running training jobs in isolated geographic regions
B. Preventing training containers from accessing the internet during training
C. Isolating training jobs by customer account
D. Partitioning network bandwidth for ML workloads

✓ Correct Answer: B

Explanation: SageMaker network isolation mode prevents training containers from making outbound internet connections, reducing the risk of data exfiltration or malicious code downloads.

Q201. What is 'access logging' important for in AI applications?

A. Logging model training progress
B. Recording who accessed AI services and data for audit trails and security investigation
C. Logging model prediction accuracy
D. Tracking billing for AI API usage

✓ **Correct Answer: B**

Explanation: Access logging records all access to AI services, models, and training data, creating audit trails essential for detecting unauthorized access, compliance, and security investigations.

Q202. What does 'SOC 2 compliance' mean for AWS AI services?

A. A security certification for open-source AI models

B. An auditing standard verifying AWS security, availability, and confidentiality controls

C. A software optimization certification

D. A social compliance standard for AI teams

✓ **Correct Answer: B**

Explanation: SOC 2 is an auditing standard that verifies a service provider's security, availability, processing integrity, confidentiality, and privacy controls, important for enterprises using AWS AI services.

Q203. What is 'AWS Organizations' useful for in enterprise AI governance?

A. Organizing ML model categories

B. Centrally managing multiple AWS accounts with policies and guardrails for AI workloads

C. An organizational chart tool for AI teams

D. Managing organization-wide S3 buckets

✓ **Correct Answer: B**

Explanation: AWS Organizations enables central governance of multiple AWS accounts through Service Control Policies (SCPs), allowing enterprise-wide rules about which AI services can be used and how.

Q204. What is 'AWS Security Hub' relevant to AI/ML?

A. A hub for discovering new AI security services

B. A centralized security findings dashboard aggregating security alerts across AWS services

C. A training hub for AWS security certifications

D. A physical security hub for AWS data centers

✓ **Correct Answer: B**

Explanation: AWS Security Hub aggregates security findings from services like GuardDuty, Inspector, and Macie into a central dashboard, enabling comprehensive security monitoring of AI workloads.

Q205. What is 'model versioning' important for in ML governance?

A. Managing software versions of ML frameworks

B. Tracking and managing different versions of deployed models for reproducibility and rollback

C. Versioning training datasets

D. Tracking API version changes

✓ **Correct Answer: B**

Explanation: Model versioning tracks different model versions (training data, hyperparameters, performance metrics), enabling reproducibility, comparison, rollback to previous versions, and audit trails.

Q206. What is 'Amazon Bedrock's data privacy' commitment regarding customer data?

A. AWS uses customer data to improve foundation models

B. Customer data is not used to improve base foundation models without consent

C. Customer prompts are stored permanently for quality review

D. Amazon employees review all Bedrock API calls

✓ Correct Answer: B

Explanation: Amazon Bedrock does not use customer inputs or outputs to train or improve base foundation models by default, protecting enterprise data privacy and intellectual property.

Q207. What is 'AWS PrivateLink' used for in AI workflows?

A. A private social network for AWS employees

B. Accessing AWS services privately without exposing traffic to the public internet

C. Creating private ML model repositories

D. Private billing for AI workloads

✓ Correct Answer: B

Explanation: AWS PrivateLink enables private connectivity from your VPC to AWS services (including Bedrock and SageMaker) without routing traffic through the public internet, enhancing security.

Q208. What is 'compliance as code' in AI governance?

A. Writing AI models to comply with regulations

B. Automating compliance checks through code and infrastructure as code templates

C. Code that automatically passes compliance audits

D. Writing regulations using AI-generated code

✓ Correct Answer: B

Explanation: Compliance as code automates regulatory compliance checks using tools like AWS Config Rules and CloudFormation Guard, ensuring AI infrastructure is consistently compliant.

Q209. What does AWS Well-Architected Framework's Security pillar recommend for ML workloads?

A. Using the cheapest security options available

B. Strong identity, protecting data, detecting events, and automating security responses

C. Prioritizing performance over security

D. Relying entirely on AWS managed security

✓ Correct Answer: B

Explanation: The Security pillar recommends: strong identity foundation (IAM), enabling traceability (CloudTrail), applying security at all layers, automating security, protecting data in transit and at rest.

Q210. What is Amazon SageMaker's 'model package' feature used for in governance?

A. Packaging Python dependencies for training

B. Creating versioned, approved bundles of model artifacts for controlled deployment

C. Packaging data for model training

D. Creating Docker containers for models

✓ Correct Answer: B

Explanation: SageMaker model packages bundle model artifacts with metadata (algorithm, framework, performance metrics) in Model Registry, enabling approval workflows before production deployment.

Q211. What is the purpose of 'tagging' AWS resources in ML governance?

A. Adding labels to training data

B. Attaching metadata to AWS resources for cost allocation, ownership, and compliance tracking

C. Tagging model outputs for review

D. Marking certain models as production-ready

✓ Correct Answer: B

Explanation: Resource tagging attaches key-value metadata to AWS resources (SageMaker jobs, Bedrock endpoints), enabling cost allocation by project/team, ownership identification, and compliance reporting.

Q212. What is 'Amazon Detective' used for?

A. Detecting anomalies in ML model outputs

B. Investigating and analyzing security incidents using log data and ML

C. Detecting data quality issues

D. Detecting compliance violations in training data

✓ Correct Answer: B

Explanation: Amazon Detective uses ML and graph analysis to automatically collect and analyze log data, helping security teams investigate and understand the root cause of security findings.

Q213. What is a key consideration for AI compliance in regulated industries (finance, healthcare)?

A. Using only open-source models

B. Maintaining audit trails, explainability, bias testing, and data lineage documentation

C. Avoiding any use of cloud-based AI

D. Using only models trained on public data

✓ Correct Answer: B

Explanation: Regulated industries require comprehensive audit trails, explainable model decisions, bias/fairness testing, data lineage, and documentation to meet regulatory requirements (SOX, HIPAA, GDPR).

Q214. What does 'Amazon Bedrock invocation logging' enable?

A. Logging model training invocations

B. Recording all API calls and model inputs/outputs for audit and compliance purposes

C. Logging billing for Bedrock API usage

D. Logging user authentication to Bedrock

✓ Correct Answer: B

Explanation: Bedrock invocation logging captures full records of model inputs, outputs, and metadata for all API calls, supporting audit trails, troubleshooting, and regulatory compliance.

Domain 1: AI & ML Fundamentals

Q215. What is the difference between a parametric and non-parametric model?

A. Parametric models have fixed number of parameters; non-parametric grow with data

B. Parametric models require no data; non-parametric require labeled data

C. Parametric models are always more accurate

D. Non-parametric models cannot be deployed

✓ **Correct Answer: A**

Explanation: Parametric models (linear regression, neural networks) have a fixed number of parameters. Non-parametric models (k-NN, decision trees) grow in complexity with the size of training data.

Q216. What is 'model explainability' and why is it important?

A. Documenting model training costs

B. The ability to describe why a model made a specific prediction, important for trust and compliance

C. Explaining the model architecture to engineers

D. Writing model deployment documentation

✓ **Correct Answer: B**

Explanation: Model explainability helps users understand predictions, builds trust, enables debugging, and is often required by regulations (e.g., GDPR's right to explanation) in high-stakes domains.

Q217. What is 'Bayesian optimization' used for in ML?

A. Training Bayesian neural networks

B. Efficiently searching hyperparameter spaces using probabilistic models

C. Adding probabilistic outputs to any model

D. Optimizing database queries

✓ **Correct Answer: B**

Explanation: Bayesian optimization uses a probabilistic surrogate model to intelligently explore hyperparameter spaces, finding good configurations with fewer evaluations than grid or random search.

Q218. Which of the following is an example of a time series forecasting problem?

A. Classifying customer emails

B. Predicting next week's electricity demand

C. Grouping products by category

D. Detecting fraud in transactions

✓ **Correct Answer: B**

Explanation: Time series forecasting predicts future values based on historical temporal patterns. Predicting electricity demand over time is a classic time series problem.

Q219. What is 'knowledge distillation'?

A. Extracting knowledge from enterprise documents

B. Training a small 'student' model to mimic a large 'teacher' model's behavior

C. Distilling training data to its most important features

D. Compressing a model's training dataset

✓ **Correct Answer: B**

Explanation: Knowledge distillation trains a smaller, efficient student model to replicate the outputs of a larger teacher model, achieving similar performance with less computational cost.

Q220. What does 'MLOps' stand for and what does it involve?

A. Machine Learning Options — selecting algorithms

B. Machine Learning Operations — practices for deploying and maintaining ML in production

C. Multi-Layer Operations — managing neural network layers

D. Model Learning Optimization — tuning model performance

✓ **Correct Answer: B**

Explanation: MLOps combines ML and DevOps practices to automate and streamline the ML lifecycle: versioning, CI/CD pipelines, deployment, monitoring, and retraining of ML models in production.

Q221. What is an 'autoencoder' used for?

A. Automatically encoding user passwords

B. Learning compressed data representations and reconstructing data, used for anomaly detection

C. Encoding categorical variables automatically

D. Automatically generating model architecture

✓ **Correct Answer: B**

Explanation: Autoencoders learn to compress data into a lower-dimensional latent space and reconstruct it. They're used for dimensionality reduction, anomaly detection, and generative modeling.

Q222. What is 'active learning' in machine learning?

A. Training models that actively monitor production data

B. A strategy where the model queries for labels on the most informative samples

C. A learning approach that doesn't require any labeled data

D. Real-time model updating during inference

✓ **Correct Answer: B**

Explanation: Active learning reduces labeling costs by having the model identify and request labels only for the most uncertain or informative samples, maximizing learning from limited labels.

Q223. What is a 'recommendation system'?

A. A system recommending which ML model to use

B. An ML system that predicts user preferences to suggest relevant items

C. A rule-based expert system for business decisions

D. A system that recommends AWS services

✓ **Correct Answer: B**

Explanation: Recommendation systems use ML to predict user preferences based on behavior history (collaborative filtering) or item features (content-based) to suggest relevant products, content, or services.

Q224. What is 'collaborative filtering' in recommendation systems?

A. Multiple teams collaborating to label data

B. Making recommendations based on the preferences of similar users

C. Filtering training data collaboratively

D. A method of combining multiple ML models

✓ **Correct Answer: B**

Explanation: Collaborative filtering recommends items to a user based on what similar users (neighbors) liked, without needing explicit item features — used by Netflix and Amazon recommendations.

Q225. What is 'anomaly detection' in ML?

A. Detecting anomalies in model architecture

B. Identifying data points that deviate significantly from expected patterns

C. Detecting anomalies in AWS billing

D. Finding outlier models in an ensemble

✓ Correct Answer: B

Explanation: Anomaly detection identifies unusual data points (outliers) that deviate from normal patterns, used for fraud detection, equipment failure prediction, and network intrusion detection.

Q226. What is 'natural language generation' (NLG)?

A. Generating training labels from text

B. Automatically producing natural language text from structured data or representations

C. Categorizing text into natural language categories

D. Generating language training data

✓ Correct Answer: B

Explanation: NLG automatically converts structured data, summaries, or internal representations into coherent natural language text — the output side of language AI (as opposed to NLU which is the input side).

Q227. What is 'named entity recognition' (NER)?

A. Recognizing the names of ML algorithms

B. Identifying and classifying named entities (people, places, organizations) in text

C. Naming features in a dataset

D. Recognizing voice commands by name

✓ Correct Answer: B

Explanation: NER is an NLP task that locates and classifies named entities in text into categories like person names, organizations, locations, dates, and quantities.

Q228. What is the purpose of a 'test set' in ML?

A. To tune hyperparameters during training

B. To provide a final unbiased evaluation of the trained model on unseen data

C. To augment training data

D. To validate data quality before training

✓ Correct Answer: B

Explanation: The test set is held out entirely until after training and hyperparameter tuning. It provides an unbiased final evaluation of model performance on truly unseen data.

Domain 2: Generative AI & Foundation Models

Q229. What is 'in-context learning' in LLMs?

A. Learning that occurs within a specific organizational context

B. A model's ability to adapt to new tasks using only examples provided in the prompt

C. Learning from data stored in context databases

D. Updating model weights based on user feedback

✓ Correct Answer: B

Explanation: In-context learning is the ability of LLMs to perform new tasks by learning from examples given in the prompt, without any weight updates — a key emergent capability of large models.

Q230. What is 'LoRA' (Low-Rank Adaptation) used for?

A. Low-resource audio transcription
B. Parameter-efficient fine-tuning of large models using low-rank weight updates
C. A low-cost retrieval algorithm
D. Reducing LLM response latency

✓ Correct Answer: B

Explanation: LoRA is a parameter-efficient fine-tuning technique that adds small trainable rank-decomposition matrices to model layers, enabling fine-tuning of large models with far fewer parameters.

Q231. What is the difference between fine-tuning and RAG for adapting LLMs?

A. Fine-tuning updates model weights; RAG provides external context at inference time
B. RAG updates model weights; fine-tuning provides external context
C. They are identical techniques
D. Fine-tuning is only for image models; RAG is for text models

✓ Correct Answer: A

Explanation: Fine-tuning permanently updates model weights for domain adaptation. RAG retrieves external documents at inference time without changing weights, better suited for dynamic or up-to-date knowledge.

Q232. What is 'Amazon Bedrock's model invocation API' used for?

A. Training foundation models
B. Sending prompts to foundation models and receiving generated responses
C. Monitoring model performance
D. Managing model versions

✓ Correct Answer: B

Explanation: The Bedrock InvokeModel API sends a prompt (and inference parameters) to a specified foundation model and returns the model's generated response.

Q233. What is 'Anthropic Claude' optimized for?

A. Image generation tasks
B. Helpful, harmless, and honest text generation with strong reasoning capabilities
C. Real-time speech recognition
D. Time series forecasting

✓ Correct Answer: B

Explanation: Anthropic's Claude models are designed using Constitutional AI to be helpful, harmless, and honest, excelling at complex reasoning, analysis, coding, and long-context document processing.

Q234. What is 'semantic search' and how does it differ from keyword search?

A. Semantic search uses exact word matches; keyword search uses meaning
B. Semantic search understands meaning and intent; keyword search matches exact words
C. Semantic search is faster than keyword search
D. They are the same technique

✓ **Correct Answer: B**

Explanation: Semantic search uses embeddings to understand the meaning and intent of queries, returning relevant results even without exact keyword matches — powered by vector similarity search.

Q235. What is 'Amazon Bedrock's Action Groups' feature in agents?

A. Groups of users with Bedrock access

B. Sets of API actions an agent can call to interact with external services

C. Groups of foundation models for ensemble inference

D. Permission groups for Bedrock resources

✓ **Correct Answer: B**

Explanation: Action Groups in Bedrock Agents define the APIs (via Lambda functions or OpenAPI schemas) that the agent can invoke to take actions, such as querying databases or calling external services.

Q236. What is 'output length' consideration when using LLMs cost-effectively?

A. Longer outputs are always better quality

B. Output tokens cost more per token; concise prompts that elicit focused responses reduce cost

C. Output length does not affect cost

D. Only input tokens are charged in Bedrock

✓ **Correct Answer: B**

Explanation: In Bedrock's token-based pricing, both input and output tokens are charged. Setting appropriate max_tokens and designing prompts for concise responses helps manage inference costs.

Q237. What is a 'GAN' (Generative Adversarial Network)?

A. A gradient-based anomaly network

B. Two neural networks (generator and discriminator) trained adversarially to create realistic data

C. A global attention network for NLP

D. A graph-based analysis network

✓ **Correct Answer: B**

Explanation: GANs consist of a generator (creates fake data) competing against a discriminator (detects fakes). Through adversarial training, the generator learns to produce increasingly realistic data.

Q238. What is 'Amazon Bedrock's Cross-Region Inference'?

A. Running inference across multiple models simultaneously

B. Routing inference requests across AWS regions for better availability and throughput

C. Comparing inference results from different regions

D. Replicating models across regions for disaster recovery

✓ **Correct Answer: B**

Explanation: Cross-Region Inference in Bedrock automatically routes requests to the same model in different AWS regions when the primary region is at capacity, improving availability and throughput.

Domain 3: AWS AI & ML Services

Q239. What is Amazon SageMaker's 'built-in algorithms' feature?

A. Algorithms built into the SageMaker console UI
B. Optimized, pre-built ML algorithms ready to train without writing custom code
C. Algorithms built by third-party vendors
D. Default hyperparameters for training jobs

✓ Correct Answer: B

Explanation: SageMaker offers built-in algorithms (XGBoost, Linear Learner, K-Means, Object Detection, etc.) that are optimized for distributed training on AWS infrastructure.

Q240. What is 'Amazon SageMaker Processing' used for?

A. Processing billing for SageMaker jobs
B. Running data preprocessing, postprocessing, and model evaluation jobs
C. Processing model deployment requests
D. Processing SageMaker Studio notebook outputs

✓ Correct Answer: B

Explanation: SageMaker Processing provides managed infrastructure for running data preprocessing, feature engineering, model evaluation, and postprocessing scripts as containerized jobs.

Q241. What is 'Amazon SageMaker Serverless Inference'?

A. Running SageMaker training without servers
B. A deployment option that automatically scales to zero when idle, ideal for intermittent traffic
C. A serverless database for inference results
D. Inference performed client-side without any server

✓ Correct Answer: B

Explanation: SageMaker Serverless Inference automatically provisions capacity when requests arrive and scales to zero when idle, eliminating costs during quiet periods for infrequently accessed endpoints.

Q242. What is 'SageMaker Real-Time Inference'?

A. Training a model in real-time on streaming data
B. Persistent endpoints for low-latency synchronous predictions
C. Real-time model monitoring
D. Streaming training logs in real-time

✓ Correct Answer: B

Explanation: SageMaker Real-Time Inference deploys persistent HTTPS endpoints that respond synchronously to prediction requests, suitable for applications requiring low-latency responses.

Q243. What is 'SageMaker Batch Transform'?

A. Transforming batch data before storing in S3
B. Running large-scale offline predictions on a full dataset without a persistent endpoint
C. Transforming a model's batch size during training
D. Batching multiple training jobs together

✓ Correct Answer: B

Explanation: SageMaker Batch Transform runs prediction jobs on entire datasets stored in S3, generates predictions, and writes results back to S3 — ideal for offline scoring without maintaining an endpoint.

Q244. What is AWS Lake Formation used for in ML workflows?

A. Forming lake-shaped data visualizations
B. Building, securing, and managing a data lake for ML training data
C. Managing water resources with ML
D. Forming Lambda functions for data processing

✓ Correct Answer: B

Explanation: AWS Lake Formation simplifies building secure data lakes by automating data ingestion, cataloging, and access control, making it easier to manage training data for ML.

Q245. What is 'Amazon Athena' and its relevance to ML?

A. An ML algorithm for time series analysis
B. A serverless SQL query service for analyzing data in S3, useful for ML data exploration
C. A model training service
D. An AI service for automated data labeling

✓ Correct Answer: B

Explanation: Amazon Athena allows running SQL queries directly on data in S3 without loading it into a database, making it easy to explore and prepare training datasets.

Q246. What is Amazon SageMaker's 'managed spot training'?

A. Training models at specific geographic spots
B. Using EC2 Spot Instances for training to reduce costs by up to 90%
C. Identifying the best training spot in a dataset
D. Managing specific training checkpoints

✓ Correct Answer: B

Explanation: SageMaker Managed Spot Training uses lower-cost EC2 Spot Instances with automatic checkpointing, reducing training costs significantly while handling potential interruptions.

Q247. What is 'Amazon Rekognition Face Liveness' used for?

A. Detecting if celebrities are currently alive
B. Verifying that a face in a camera is a real, live person (anti-spoofing)
C. Tracking facial expression changes over time
D. Recognizing faces in historical photos

✓ Correct Answer: B

Explanation: Rekognition Face Liveness detects presentation attacks (spoofing with photos/videos/masks), verifying that a face in a camera feed is a live person for identity verification.

Q248. What is 'Amazon Connect' and how does AI enhance it?

A. A database connection service
B. A cloud contact center service enhanced by AI for sentiment analysis and agent assistance
C. A service connecting ML models to applications
D. A VPN connectivity service

✓ **Correct Answer: B**

Explanation: Amazon Connect is a cloud contact center. AI features like Amazon Q in Connect provide real-time agent assistance, Amazon Transcribe handles call transcription, and Comprehend analyzes sentiment.

Domain 4: Responsible AI

Q249. What is 'impact assessment' in AI deployment?

A. Measuring model training speed
B. Evaluating potential societal, ethical, and legal impacts of an AI system before deployment
C. Assessing cloud infrastructure impact
D. Measuring revenue impact of AI features

✓ **Correct Answer: B**

Explanation: An AI impact assessment systematically evaluates potential harms, fairness issues, privacy risks, and societal implications of deploying an AI system before it goes live.

Q250. What does 'inclusive AI design' mean?

A. Including all ML frameworks in a model
B. Designing AI systems that work equitably across diverse user groups and demographics
C. Including all data sources in training
D. Making AI interfaces visually inclusive

✓ **Correct Answer: B**

Explanation: Inclusive AI design ensures systems work well for diverse users regardless of age, ability, race, gender, or geography, reducing disparate impacts on marginalized groups.

Q251. What is 'counterfactual explanation' in AI?

A. An explanation of what would have happened without AI
B. An explanation showing what minimal change to input would alter the model's decision
C. A factual summary of model training
D. An explanation contradicting model outputs

✓ **Correct Answer: B**

Explanation: Counterfactual explanations show users what minimal input changes would produce a different outcome (e.g., 'You would have been approved if your income were $5,000 higher').

Q252. What is 'AI governance framework'?

A. A technical framework for building AI models
B. A structured set of policies, standards, and processes ensuring responsible AI development and use
C. AWS governance for AI service pricing
D. A government regulatory framework for AI

✓ **Correct Answer: B**

Explanation: An AI governance framework establishes organizational standards, oversight mechanisms, ethics boards, accountability structures, and compliance processes for responsible AI use.

Q253. What is 'training data consent' and why does it matter?

A. Getting model approval to train on new data
B. Ensuring data used to train AI was collected with appropriate permissions from individuals
C. Consenting to AWS terms for training jobs
D. Data quality approval before training

✓ Correct Answer: B

Explanation: Training data consent ensures that personal data used to train AI models was lawfully collected with appropriate consent or legal basis, a key requirement under privacy laws like GDPR.

Q254. What is 'copyright concern' in generative AI?

A. A concern about AI models that are copyrighted
B. The risk that AI may reproduce copyrighted content from training data in its outputs
C. Copyright protection for AI-generated content
D. Concerns about copyright in AI marketing materials

✓ Correct Answer: B

Explanation: Generative AI models trained on copyrighted content may reproduce protected material in outputs, raising legal risks around copyright infringement that organizations must manage.

Q255. What is 'dual-use concern' in AI?

A. Using AI on two devices simultaneously
B. The risk that AI capabilities developed for beneficial purposes could also be used for harm
C. A model deployed in two different regions
D. Using both supervised and unsupervised learning together

✓ Correct Answer: B

Explanation: Dual-use concerns arise when AI systems (like image generation or text synthesis) designed for legitimate purposes can also be misused for harmful activities like deepfakes or disinformation.

Q256. What is 'Amazon Bedrock's topic denial' feature in Guardrails?

A. Denying API requests based on topic
B. Preventing LLMs from discussing user-defined off-limits topics
C. Filtering irrelevant topics from model training
D. Denying user access to certain topics

✓ Correct Answer: B

Explanation: Bedrock Guardrails topic denial allows organizations to define topics they don't want their AI application to discuss (e.g., competitors, legal advice), blocking responses about those topics.

Domain 5: Security, Compliance & Governance

Q257. What is 'SageMaker role-based access control' (RBAC)?

A. Controlling SageMaker UI roles
B. Assigning minimum-necessary IAM permissions to SageMaker execution roles based on job function

C. A role-playing training technique

D. Automatically rotating access credentials

✓ Correct Answer: B

Explanation: RBAC for SageMaker assigns specific IAM roles to different users/jobs (data scientists, MLOps engineers) with only the permissions needed for their function, following least privilege.

Q258. What is 'model lineage tracking' in ML governance?

A. Tracking model versions in a code repository

B. Recording the full history of data, code, and parameters used to produce a model

C. Tracking model performance over time

D. Documenting model deployment locations

✓ Correct Answer: B

Explanation: Model lineage tracking records the complete provenance of a model including training data, preprocessing steps, code versions, hyperparameters, and evaluation results, supporting reproducibility and audit.

Q259. What is 'Amazon SageMaker ML Lineage Tracking'?

A. A service for tracking ML project timelines

B. Automatically recording relationships between datasets, training jobs, models, and endpoints

C. A tool for visualizing model family trees

D. A method for tracking ML costs over time

✓ Correct Answer: B

Explanation: SageMaker ML Lineage Tracking automatically captures and stores the relationships (lineage) between all ML workflow artifacts, enabling governance, audit trails, and reproducibility.

Q260. What is 'Amazon S3 Object Lock' relevant to ML compliance?

A. Locking S3 buckets from all access

B. Preventing deletion or modification of training data for regulatory retention requirements

C. Encrypting objects in S3

D. Locking model weights from modification

✓ Correct Answer: B

Explanation: S3 Object Lock implements WORM (Write Once Read Many) storage, preventing deletion or modification of training data and model artifacts for a specified retention period, meeting compliance requirements.

Q261. What does 'AWS Trusted Advisor' offer for ML workloads?

A. Trusted human advisors for ML projects

B. Automated recommendations for cost optimization, performance, and security of AWS resources

C. AI-powered code review for ML pipelines

D. Compliance advice from AWS legal experts

✓ Correct Answer: B

Explanation: AWS Trusted Advisor analyzes AWS infrastructure and provides automated recommendations across cost optimization, performance, security, fault tolerance, and service limits for ML workloads.

Q262. What is 'infrastructure as code' (IaC) and its value for ML governance?

A. Writing ML algorithms using infrastructure tools
B. Managing and provisioning ML infrastructure through version-controlled configuration files
C. Using ML to generate infrastructure code
D. Code that runs directly on hardware without OS

✓ Correct Answer: B

Explanation: IaC (AWS CloudFormation, CDK) defines ML infrastructure in code, enabling version control, reproducibility, consistent environments across dev/prod, and automated compliance checks.

Q263. What is 'Amazon EventBridge' used for in ML workflows?

A. A bridge service connecting ML models
B. Event-driven automation triggering ML pipeline actions based on events
C. A network bridge for ML traffic
D. An event logging service for model outputs

✓ Correct Answer: B

Explanation: Amazon EventBridge enables event-driven ML workflows by triggering SageMaker pipelines, Lambda functions, or other actions in response to events like S3 uploads or model monitoring alerts.

Q264. What is the 'AWS Well-Architected ML Lens'?

A. A camera lens for ML hardware inspection
B. A framework of best practices for designing well-architected ML workloads on AWS
C. A ML model for architecture diagram analysis
D. An AWS certification for ML architects

✓ Correct Answer: B

Explanation: The AWS Well-Architected ML Lens extends the Well-Architected Framework with ML-specific best practices covering data preparation, model training, deployment, and monitoring.

Q265. What is 'Amazon Bedrock's service quotas' and why do they matter?

A. Budget limits set by the customer
B. Default limits on API requests per minute/second that may need to be increased for production
C. Quotas on the number of foundation models available
D. Training data size limits

✓ Correct Answer: B

Explanation: Bedrock service quotas set default limits on invocations per minute/second per model. Production workloads may require quota increases through AWS Support to handle expected traffic.

Q266. What is 'cost allocation tags' used for in ML projects?

A. Tagging expensive ML algorithms
B. Attributing AWS costs to specific ML projects, teams, or cost centers
C. Tagging data with cost metadata
D. Allocating compute budgets to training jobs

✓ Correct Answer: B

Explanation: Cost allocation tags label AWS resources (SageMaker training jobs, Bedrock calls) with project/team metadata, enabling detailed cost reporting and showback to business units.

Q267. What is 'AWS Budgets' useful for in AI/ML cost governance?

A. Creating ML models for financial forecasting

B. Setting cost and usage alerts to prevent unexpected AI/ML spending

C. Managing payroll budgets using AI

D. Budgeting training data storage

✓ Correct Answer: B

Explanation: AWS Budgets allows setting custom cost and usage thresholds with automatic alerts, helping organizations control AI/ML spending and avoid unexpected bills from high Bedrock or SageMaker usage.

Domain 1: AI & ML Fundamentals

Q268. What is 'concept drift' in deployed ML models?

A. When the UI of a model changes over time

B. When the relationship between input features and target labels changes over time

C. When model weights gradually increase

D. When training data becomes unavailable

✓ Correct Answer: B

Explanation: Concept drift occurs when the statistical relationship between inputs and outputs changes over time (e.g., customer behavior shifts), causing model predictions to degrade in production.

Q269. What is 'class imbalance' and one common way to handle it?

A. When classes have different names; fix by renaming

B. When one class has far more samples than others; fix by oversampling minority or undersampling majority

C. When classes overlap in feature space; fix by adding more features

D. When class labels are wrong; fix by relabeling

✓ Correct Answer: B

Explanation: Class imbalance (e.g., 99% negative, 1% positive) causes models to be biased toward the majority class. Techniques like SMOTE oversampling or class weighting help balance learning.

Domain 2: Generative AI & Foundation Models

Q270. What is 'Stable Diffusion XL' available through Amazon Bedrock?

A. An extended text generation model

B. An enhanced image generation model producing higher quality and larger resolution images

C. A model for video stabilization

D. A diffusion-based speech synthesis model

✓ **Correct Answer: B**

Explanation: Stable Diffusion XL (SDXL) by Stability AI is an advanced diffusion model available on Bedrock that generates higher quality, more detailed images than earlier Stable Diffusion versions.

Q271. What is 'Amazon Bedrock's document chat' pattern?

A. Chatting with AWS support via documents

B. Allowing users to ask questions about uploaded documents using an LLM

C. A document approval workflow

D. Converting documents to chat transcripts

✓ **Correct Answer: B**

Explanation: The document chat pattern uses Bedrock to let users upload documents (PDFs, text) and ask natural language questions, with the LLM extracting and reasoning about the document content.

Domain 3: AWS AI & ML Services

Q272. What is 'Amazon SageMaker Debugger'?

A. A tool for debugging Python code in notebooks

B. A service that captures training metrics and detects training issues automatically

C. A debugger for SageMaker infrastructure

D. A tool for debugging model API calls

✓ **Correct Answer: B**

Explanation: SageMaker Debugger captures real-time training metrics (tensors, gradients, weights) and automatically detects common training problems like vanishing gradients or overfitting.

Q273. What is 'Amazon SageMaker Data Wrangler'?

A. A data storage management service

B. A visual data preparation and feature engineering tool within SageMaker Studio

C. A service for wrangling distributed training jobs

D. A data migration service to SageMaker

✓ **Correct Answer: B**

Explanation: SageMaker Data Wrangler provides a visual interface for importing, exploring, transforming, and exporting ML training data with 300+ built-in transformations and data visualization.

Q274. What is 'Amazon CodeGuru' and its AI capability?

A. A code generation service

B. An AI-powered service for automated code reviews and application performance recommendations

C. A curriculum for learning to code

D. A code compliance checker

✓ **Correct Answer: B**

Explanation: Amazon CodeGuru uses ML to provide automated code reviews (CodeGuru Reviewer) detecting bugs and security vulnerabilities, and performance recommendations (CodeGuru Profiler).

Q275. What is 'Amazon Lookout for Equipment'?

A. A service for finding lost equipment
B. An ML service for detecting abnormal equipment behavior to enable predictive maintenance
C. A supply chain management service
D. A tool for equipment inventory tracking

✓ Correct Answer: B

Explanation: Amazon Lookout for Equipment analyzes sensor data from industrial machinery using ML to detect early signs of equipment abnormalities, preventing costly failures and downtime.

Domain 4: Responsible AI

Q276. What is 'AI accountability' in practice?

A. Tracking AI API costs
B. Ensuring humans and organizations can be held responsible for AI-caused harms
C. Making AI systems self-accountable
D. Auditing cloud spending on AI

✓ Correct Answer: B

Explanation: AI accountability means establishing clear ownership for AI system behavior, documentation of decisions, oversight mechanisms, and processes for addressing harms caused by AI systems.

Q277. What is 'demographic bias' in training data?

A. Using demographic data for model features
B. Underrepresentation or overrepresentation of demographic groups causing unfair model outcomes
C. Demographic differences in model training speed
D. Bias toward specific geographic regions in infrastructure

✓ Correct Answer: B

Explanation: Demographic bias occurs when training data does not proportionally represent demographic groups, causing models to perform worse for underrepresented groups.

Q278. What is the purpose of 'AI ethics review boards'?

A. Reviewing the technical accuracy of AI models
B. Providing oversight of AI projects to ensure alignment with ethical principles and organizational values
C. Reviewing AI vendor contracts
D. Approving AI marketing campaigns

✓ Correct Answer: B

Explanation: AI ethics review boards provide governance oversight by evaluating proposed AI systems for ethical risks, bias, privacy concerns, and alignment with responsible AI principles before deployment.

Domain 5: Security, Compliance & Governance

Q279. What is 'AWS Security Token Service (STS)' used for in ML?

A. Long-term credentials for SageMaker
B. Generating temporary, limited-privilege credentials for secure access to ML resources
C. Storing security tokens for model APIs
D. A token-counting service for LLMs

✓ Correct Answer: B

Explanation: AWS STS generates temporary security credentials with limited permissions and expiration times, enabling secure, short-lived access to ML resources like SageMaker and Bedrock.

Q280. What does 'data residency' mean for AI compliance?

A. Where ML engineers are located
B. The requirement that data must be stored and processed within specific geographic boundaries
C. Residual data remaining after model training
D. The residence time of data in memory during inference

✓ Correct Answer: B

Explanation: Data residency requirements mandate that certain data (especially personal data) must remain within specific countries or regions, affecting which AWS regions can be used for AI workloads.

Q281. What is 'Amazon Bedrock's encryption' capability?

A. Encrypting prompts before sending to models
B. Encrypting model customization data and outputs using AWS KMS keys
C. End-to-end encryption of LLM weights
D. Encrypting Bedrock API credentials

✓ Correct Answer: B

Explanation: Amazon Bedrock encrypts data at rest and in transit. For model customization, you can use your own AWS KMS keys to encrypt training data and customized model artifacts.

Q282. What is a 'service control policy' (SCP) in AWS Organizations for AI governance?

A. A policy controlling how AI services work
B. An organization-level policy restricting which AWS AI services can be used across accounts
C. A service pricing control mechanism
D. A policy for controlling model inference parameters

✓ Correct Answer: B

Explanation: SCPs allow central administrators to restrict which AWS services (including Bedrock or SageMaker features) can be used in any account within an organization, enforcing AI governance guardrails.

Domain 2: Generative AI & Foundation Models

Q283. What is 'Amazon Bedrock's Responsible AI' detection capability?

A. Monitoring Bedrock infrastructure health
B. Detecting and filtering harmful, biased, or inappropriate content using built-in classifiers

C. Detecting responsible AI frameworks used

D. Monitoring responsible use of AWS credits

✓ Correct Answer: B

Explanation: Bedrock includes built-in responsible AI capabilities including content classifiers for detecting harmful content categories and Guardrails for custom policy enforcement.

Domain 1: AI & ML Fundamentals

Q284. What is 'Monte Carlo simulation' used for in ML contexts?

A. A casino-themed training dataset

B. Using random sampling to estimate complex distributions or uncertain outcomes

C. A reinforcement learning environment

D. A model training acceleration technique

✓ Correct Answer: B

Explanation: Monte Carlo methods use random sampling to approximate complex mathematical problems. In ML, they're used for uncertainty estimation, hyperparameter search, and approximating intractable distributions.

Domain 3: AWS AI & ML Services

Q285. What is 'Amazon Titan Text Embeddings V2' optimized for?

A. Generating long-form text content

B. Creating high-dimensional embeddings optimized for semantic search and RAG applications

C. Translating text embeddings between languages

D. Summarizing embedded documents

✓ Correct Answer: B

Explanation: Amazon Titan Text Embeddings V2 generates 1024-dimensional embeddings optimized for semantic similarity, search, and RAG applications, with improved performance over V1.

Domain 4: Responsible AI

Q286. What is 'algorithmic auditing'?

A. An audit of AWS AI billing

B. Systematic examination of an AI system's behavior for fairness, accuracy, and compliance

C. Auditing the algorithm selection process

D. A financial audit of AI projects

✓ Correct Answer: B

Explanation: Algorithmic auditing involves independent examination of AI systems to verify they behave as intended, treat people fairly, comply with regulations, and don't produce harmful outcomes.

Domain 5: Security, Compliance & Governance

Q287. What is 'AWS Control Tower' used for in AI governance?

A. Controlling individual ML model parameters
B. Setting up and governing a secure multi-account AWS environment with pre-configured guardrails
C. A control panel for AI service configuration
D. Managing model deployment controls

✓ Correct Answer: B

Explanation: AWS Control Tower automates setup of a well-architected multi-account AWS environment with built-in governance guardrails, helping organizations standardize AI workload governance across accounts.

Domain 1: AI & ML Fundamentals

Q288. What is 'BERT' and what was its significance in NLP?

A. A basic encoding and retrieval technique
B. A bidirectional Transformer model that revolutionized NLP by understanding context from both directions
C. A benchmark evaluation for retrieval tasks
D. A binary encoding representation technique

✓ Correct Answer: B

Explanation: BERT (Bidirectional Encoder Representations from Transformers) processes text in both directions simultaneously, capturing richer context and achieving state-of-the-art performance on many NLP tasks when released.

Domain 2: Generative AI & Foundation Models

Q289. What does 'model card' documentation for Amazon Bedrock models include?

A. Credit card information for billing
B. Intended use cases, limitations, performance benchmarks, and responsible AI considerations
C. Technical installation instructions
D. API authentication credentials

✓ Correct Answer: B

Explanation: Model cards for Bedrock foundation models document intended use cases, evaluation results, limitations, safety considerations, and responsible AI practices to help customers make informed choices.

Domain 3: AWS AI & ML Services

Q290. What is 'Amazon SageMaker HyperPod'?

A. A hyperparameter optimization tool
B. A managed cluster for training large foundation models with resiliency and efficiency features
C. A high-performance model hosting service
D. A pod orchestration service for inference

✓ Correct Answer: B

Explanation: SageMaker HyperPod provides purpose-built infrastructure for training large foundation models, with automatic failure recovery, performance optimizations, and distributed training support.

Domain 5: Security, Compliance & Governance

Q291. What is 'Amazon SageMaker's inter-container encryption'?

A. Encrypting data between SageMaker notebooks
B. Encrypting traffic between training instances in distributed training jobs
C. Container image encryption at rest
D. Encrypting data between SageMaker and S3

✓ Correct Answer: B

Explanation: SageMaker's inter-container traffic encryption protects data transmitted between training instances during distributed training jobs, preventing potential interception within the network.

Domain 4: Responsible AI

Q292. What is the 'right to be forgotten' and its AI implications?

A. The right to delete your AWS account
B. The right for individuals to have their personal data deleted, which may require model retraining
C. The right for models to forget training data after deployment
D. A data backup deletion policy

✓ Correct Answer: B

Explanation: The right to be forgotten (GDPR Article 17) allows individuals to request deletion of their personal data. For AI, this may require removing data from training sets and potentially retraining affected models.

Domain 2: Generative AI & Foundation Models

Q293. What is 'Amazon Bedrock's Converse API' streaming capability?

A. Streaming video through Bedrock
B. Receiving model responses token-by-token in real-time for a better user experience
C. Streaming training data into the model
D. Broadcasting model outputs to multiple users

✓ Correct Answer: B

Explanation: Bedrock's Converse API streaming returns generated tokens as they are produced rather than waiting for the full response, enabling real-time typing-indicator effects in chat applications.

Domain 3: AWS AI & ML Services

Q294. What is 'Amazon Q in QuickSight' used for?

A. Queuing QuickSight dashboard requests
B. Natural language querying and AI-powered business intelligence insights in QuickSight
C. A quota management service
D. A quick-start template for dashboards

✓ Correct Answer: B

Explanation: Amazon Q in QuickSight allows users to ask business intelligence questions in natural language, generating visualizations and narratives from data automatically using generative AI.

Domain 1: AI & ML Fundamentals

Q295. What is 'multi-task learning'?

A. Training multiple separate models in parallel
B. Training a single model on multiple related tasks simultaneously to improve generalization
C. Assigning ML tasks to multiple team members
D. Running multiple training jobs on the same GPU

✓ Correct Answer: B

Explanation: Multi-task learning trains one model on multiple related tasks simultaneously, allowing shared representations to improve generalization and performance on all tasks compared to training separately.

Domain 5: Security, Compliance & Governance

Q296. What is 'penetration testing' relevant to AI systems?

A. Testing model performance under high load
B. Authorized security testing to find vulnerabilities in AI application infrastructure
C. Testing data penetration into models
D. Stress testing model training jobs

✓ **Correct Answer: B**

Explanation: Penetration testing simulates real attacks on AI systems to discover exploitable security vulnerabilities in APIs, authentication, and infrastructure before malicious actors find them.

Domain 4: Responsible AI

Q297. What is the AWS Responsible AI framework built upon?

A. Cost optimization and performance

B. Fairness, explainability, privacy and security, robustness, governance, and transparency

C. Speed, scalability, and reliability

D. Automation, integration, and compliance

✓ **Correct Answer: B**

Explanation: AWS Responsible AI is built on eight dimensions: fairness, explainability, privacy and security, robustness, verifiability and reproducibility, governance, transparency, and controllability.

Domain 2: Generative AI & Foundation Models

Q298. What is 'Amazon Bedrock's inline agent' capability?

A. Agents running inside SageMaker notebooks

B. Creating lightweight agents programmatically within a single API call without pre-configuration

C. Inline code execution by agents

D. Agents embedded in Lambda functions

✓ **Correct Answer: B**

Explanation: Bedrock Inline Agents allow creating and invoking agents dynamically within a single API call, providing flexibility to define agent behavior programmatically without prior agent resource creation.

Domain 3: AWS AI & ML Services

Q299. What is 'Amazon Transcribe Medical'?

A. Translating medical documents

B. A HIPAA-eligible ASR service for transcribing medical speech with clinical vocabulary

C. Summarizing medical records

D. A medical data extraction service

✓ **Correct Answer: B**

Explanation: Amazon Transcribe Medical is a HIPAA-eligible ASR service that accurately transcribes medical speech including clinical terminology, physician dictation, and patient-doctor conversations.

Domain 1: AI & ML Fundamentals

Q300. What is 'self-supervised learning' and where is it commonly used?

A. A model that supervises its own deployment

B. Learning from unlabeled data by generating labels from the data structure itself, used in LLM pre-training

C. Training without any optimization algorithm

D. A model that trains only on its own outputs

✓ Correct Answer: B

Explanation: Self-supervised learning creates supervisory signals from the data itself (e.g., predicting the next word in a sentence). It is the foundation of pre-training large language models like GPT on massive unlabeled text corpora.

Q301. What does 'gradient clipping' do during neural network training?

A. Removes low-importance gradients

B. Caps gradient magnitudes to prevent exploding gradients during backpropagation

C. Clips outlier features from training data

D. Restricts learning rate changes

✓ Correct Answer: B

Explanation: Gradient clipping sets a maximum threshold for gradient values during backpropagation, preventing the exploding gradient problem that can destabilize deep network training.

Q302. What is the purpose of 'padding' in Convolutional Neural Networks?

A. Adding extra training examples

B. Adding zeros around input edges so feature maps preserve spatial dimensions

C. Padding model weights with zeros

D. Filling missing values in the dataset

✓ Correct Answer: B

Explanation: Padding adds border values (usually zeros) around inputs so that convolution operations can process edge pixels and control output feature map dimensions.

Q303. What is a 'hyperparameter search' strategy called 'grid search'?

A. Searching for patterns in a grid-shaped dataset

B. Exhaustively evaluating every combination of specified hyperparameter values

C. Searching model weights on a grid

D. A visual search technique using grid charts

✓ Correct Answer: B

Explanation: Grid search evaluates model performance for every possible combination of specified hyperparameter values. It is thorough but computationally expensive for large hyperparameter spaces.

Q304. What is 'random search' for hyperparameter tuning?

A. Randomly selecting training data

B. Randomly sampling hyperparameter combinations, often more efficient than grid search

C. Randomizing model weights at each epoch

D. Selecting features at random

✓ **Correct Answer: B**

Explanation: Random search samples random hyperparameter combinations instead of exhaustive grid combinations. Research shows it is often more efficient because not all hyperparameters equally affect performance.

Q305. What is 'max pooling' in CNNs?

A. Averaging values in a pooling window

B. Taking the maximum value from each pooling window to downsample feature maps

C. Pooling resources across multiple GPUs

D. Finding maximum feature importance scores

✓ **Correct Answer: B**

Explanation: Max pooling slides a window over the feature map and takes the maximum value in each window, reducing spatial dimensions while retaining the most prominent features.

Q306. What does 'softmax' activation do in classification models?

A. Converts all values to 0 or 1

B. Converts raw scores into a probability distribution summing to 1 across all classes

C. Applies a soft version of max pooling

D. Reduces output dimensionality

✓ **Correct Answer: B**

Explanation: Softmax converts a vector of raw scores (logits) into probabilities that sum to 1.0, making it the standard output activation for multi-class classification problems.

Q307. What is 'stratified sampling' in dataset splitting?

A. Sampling data from different AWS regions

B. Splitting data while maintaining the same class proportions in each split

C. Sampling the hardest examples for training

D. Splitting data by time period

✓ **Correct Answer: B**

Explanation: Stratified sampling ensures each split (train/val/test) has approximately the same proportion of each class as the full dataset, especially important for imbalanced datasets.

Q308. What is the 'elbow method' used for in K-Means clustering?

A. Removing outliers shaped like elbows

B. Finding the optimal number of clusters by plotting inertia vs. k and finding the 'elbow'

C. A technique for feature selection

D. Identifying elbow-shaped data distributions

✓ **Correct Answer: B**

Explanation: The elbow method plots within-cluster sum of squares (inertia) against the number of clusters k. The optimal k is at the 'elbow' where adding more clusters gives diminishing returns.

Q309. What is 'L1 regularization' (Lasso) known for?

A. Penalizing squared weights to prevent large values

B. Penalizing absolute weight values, tending to produce sparse models with zero weights

C. Adding a layer of neurons to regularize learning

D. Regularizing batch normalization parameters

✓ Correct Answer: B

Explanation: L1 (Lasso) regularization adds the sum of absolute weight values as a penalty, encouraging sparsity by driving some weights to exactly zero, effectively performing feature selection.

Q310. What is 'L2 regularization' (Ridge) known for?

A. Driving model weights to exactly zero

B. Penalizing squared weights, shrinking them toward zero without eliminating them completely

C. Regularizing the second layer only

D. Adding L2 norms to the loss function exclusively for CNNs

✓ Correct Answer: B

Explanation: L2 (Ridge) regularization adds the sum of squared weights as a penalty, discouraging large weights and improving generalization while typically keeping all features with small non-zero weights.

Q311. What is a 'decision boundary' in classification?

A. The accuracy threshold for deploying a model

B. The surface separating regions where the model predicts different class labels

C. The boundary between training and test data

D. The threshold for deciding model hyperparameters

✓ Correct Answer: B

Explanation: A decision boundary is the surface (line in 2D, plane in 3D) that separates the feature space into regions corresponding to different predicted class labels.

Q312. What is 'bootstrapping' in the context of ensemble methods?

A. The startup process for ML infrastructure

B. Random sampling with replacement to create multiple training subsets for ensemble models

C. Bootstrapping a model from random weights

D. A technique for initializing neural networks

✓ Correct Answer: B

Explanation: Bootstrapping creates multiple training sets by sampling the original dataset with replacement. Random Forest uses bootstrapping to train each tree on a different data subset.

Q313. What is 'bagging' in machine learning?

A. Packaging models for deployment

B. Training multiple models on bootstrap samples and averaging their predictions

C. Removing low-quality data bags from training

D. A data compression technique

✓ Correct Answer: B

Explanation: Bagging (Bootstrap Aggregating) trains multiple models independently on different bootstrap samples, then combines predictions by averaging (regression) or voting (classification), reducing variance.

Q314. What is 'boosting' in ensemble learning?

A. Increasing model training speed

B. Sequentially training models where each focuses on correcting errors of the previous one

C. Boosting GPU performance during training

D. Adding more data to improve model performance

✓ Correct Answer: B

Explanation: Boosting trains models sequentially, with each new model placing more weight on the samples the previous model got wrong. Examples include AdaBoost, Gradient Boosting, and XGBoost.

Q315. What is the 'curse of dimensionality'?

A. Too many dimensions on a monitor for data visualization

B. As feature dimensions increase, data becomes sparse making learning harder

C. A curse affecting high-dimensional AWS regions

D. The difficulty of training models with many layers

✓ Correct Answer: B

Explanation: As the number of features grows, the volume of the feature space increases exponentially, making data increasingly sparse, distances less meaningful, and ML algorithms less effective.

Q316. What is 'one-hot encoding'?

A. Encoding the single most important feature

B. Representing categorical variables as binary vectors with one '1' and rest '0's

C. Encoding data with a single hot key

D. A fast encoding technique for large datasets

✓ Correct Answer: B

Explanation: One-hot encoding converts a categorical variable with n categories into n binary columns where exactly one is 1 for each sample. Unlike label encoding, it avoids implying ordinal relationships.

Q317. What is 'precision-recall tradeoff'?

A. The tradeoff between model accuracy and training time

B. Increasing precision typically decreases recall and vice versa by changing classification threshold

C. The tradeoff between model precision and data storage

D. Balancing computing resources between precision and recall calculations

✓ Correct Answer: B

Explanation: By adjusting the classification threshold, you can trade precision for recall. A higher threshold increases precision but decreases recall; lower threshold does the opposite.

Q318. What is 'LSTM' and why was it important for sequential modeling?

A. Large Scale Training Machine for distributed ML

B. Long Short-Term Memory — an RNN variant solving the vanishing gradient problem for long sequences

C. Layered Sequence Transformation Module

D. A linear support tensor machine

✓ Correct Answer: B

Explanation: LSTM networks use gating mechanisms (input, forget, output gates) to selectively retain or discard information over long sequences, solving the vanishing gradient problem of standard RNNs.

Q319. What is 'transfer learning fine-tuning' vs 'feature extraction'?

A. They are identical approaches
B. Fine-tuning updates all weights; feature extraction freezes pre-trained layers and only trains new layers
C. Feature extraction updates all weights; fine-tuning freezes early layers
D. Both approaches require retraining from scratch

✓ Correct Answer: B

Explanation: Feature extraction freezes pre-trained model weights and only trains a new classification head. Fine-tuning updates some or all pre-trained weights, better adapting the model to new tasks.

Q320. What is 'A/B testing' in the context of ML model deployment?

A. Testing two different datasets on the same model
B. Splitting live traffic between two model versions to compare real-world performance
C. A/B testing hyperparameters before training
D. Running A then B phases of model training

✓ Correct Answer: B

Explanation: A/B testing splits production traffic between a current model (A) and a new model (B) to compare real-world performance metrics, enabling data-driven model updates.

Domain 2: Generative AI & Foundation Models

Q321. What is 'prompt chaining'?

A. Linking API keys in a chain for security
B. Breaking complex tasks into sequential prompts where each output feeds the next
C. Chaining multiple LLMs in parallel
D. A technique to extend context window length

✓ Correct Answer: B

Explanation: Prompt chaining decomposes complex tasks into smaller sequential steps, where the output of one prompt becomes the input to the next, enabling more reliable complex reasoning.

Q322. What is 'model distillation' in the context of LLMs?

A. Extracting key concepts from model documentation
B. Training a smaller efficient model to reproduce the behavior of a larger model
C. Distilling training data to its most important examples
D. A technique for removing unsafe content from models

✓ Correct Answer: B

Explanation: LLM distillation trains a smaller 'student' model using the outputs of a larger 'teacher' LLM as soft training targets, producing a compact model that retains much of the teacher's capability.

Q323. What is 'Amazon Bedrock's streaming response' benefit?

A. Streams training data to foundation models
B. Delivers tokens incrementally as generated, improving perceived responsiveness in chat applications

C. Streams model outputs to S3 automatically

D. Enables parallel streaming to multiple users

✓ Correct Answer: B

Explanation: Streaming responses send tokens to the client as they are generated rather than waiting for the complete response, creating a more responsive experience similar to human typing.

Q324. What is 'instruction tuning' for LLMs?

A. Tuning the instruction set of underlying hardware

B. Fine-tuning LLMs on instruction-response pairs to better follow human directions

C. Writing better instructions for prompts

D. Tuning hyperparameters based on instructions

✓ Correct Answer: B

Explanation: Instruction tuning fine-tunes pre-trained LLMs on datasets of instructions paired with desired responses, making models much better at following diverse natural language instructions.

Q325. What is 'semantic chunking' in RAG systems?

A. Breaking data into storage chunks

B. Splitting documents into meaningful semantic units that preserve context for retrieval

C. Chunking API requests for batch processing

D. Dividing model parameters into chunks

✓ Correct Answer: B

Explanation: Semantic chunking splits documents at natural semantic boundaries (sentences, paragraphs, topics) rather than fixed character counts, preserving context and improving retrieval quality in RAG.

Q326. What is 'hybrid search' in RAG applications?

A. Using both SQL and NoSQL databases

B. Combining dense vector similarity search with sparse keyword search for better retrieval

C. Searching across multiple cloud providers

D. Using both fine-tuning and RAG simultaneously

✓ Correct Answer: B

Explanation: Hybrid search combines dense embedding-based semantic search with sparse keyword (BM25) search, capturing both semantic meaning and exact keyword matches for more robust retrieval.

Q327. What is 'Amazon Bedrock Titan Multimodal Embeddings'?

A. A model for processing multi-language text

B. An embedding model that creates unified vector representations for both text and images

C. A model generating multiple types of media

D. A multi-region embedding service

✓ Correct Answer: B

Explanation: Amazon Titan Multimodal Embeddings generates vector representations that can be compared across text and images, enabling cross-modal search like finding images matching a text description.

Q328. What is 'output parsing' in LLM application development?

A. Parsing model source code

B. Extracting structured data from free-form LLM text outputs

C. Parsing input prompts before sending to models

D. Output validation for compliance

✓ Correct Answer: B

Explanation: Output parsing extracts structured information (JSON, lists, entities) from unstructured LLM text responses, enabling downstream processing in application workflows.

Q329. What is 'ReAct' prompting in AI agents?

A. A React.js framework for AI applications

B. A prompting strategy combining reasoning and acting — the model reasons then takes actions iteratively

C. A technique for reactive model retraining

D. A response accuracy testing method

✓ Correct Answer: B

Explanation: ReAct (Reasoning + Acting) is a prompting framework where the LLM alternates between reasoning about a problem and taking concrete actions (tool calls), enabling better multi-step problem solving.

Q330. What is 'Amazon Bedrock's model evaluation automatic metrics' include?

A. CPU and memory usage metrics

B. Accuracy, robustness, toxicity, and factuality scores for comparing foundation models

C. API latency and throughput metrics

D. Cost and billing metrics

✓ Correct Answer: B

Explanation: Bedrock Model Evaluation automatic metrics include built-in scores for accuracy, robustness, toxicity, and factual knowledge, allowing quantitative comparison of models for your use case.

Q331. What is a 'system prompt injection attack' mitigation?

A. Encrypting system prompts

B. Input validation, output filtering, and privilege separation between system and user content

C. Rate limiting API calls

D. Using stronger IAM policies

✓ Correct Answer: B

Explanation: Mitigating prompt injection requires validating and sanitizing user inputs, filtering outputs for policy violations, using Bedrock Guardrails, and maintaining strict separation between trusted system instructions and untrusted user input.

Q332. What is 'Bedrock's cross-account model access'?

A. Using one AWS account's billing for another account's Bedrock usage

B. Sharing foundation model access across AWS accounts using resource-based policies

C. Cross-region model replication

D. Accessing competitor models from one account

✓ Correct Answer: B

Explanation: Bedrock supports cross-account access through AWS resource-based policies, allowing organizations to centralize foundation model access management across multiple AWS accounts.

Q333. What is 'quantization' in the context of LLM optimization?

A. Measuring the quantity of training data

B. Reducing model weight precision (e.g., float32 to int8) to decrease memory and increase speed

C. Quantifying model performance metrics

D. Limiting the number of API calls

✓ Correct Answer: B

Explanation: Quantization reduces the numerical precision of model weights (e.g., from 32-bit floats to 8-bit integers), significantly reducing memory footprint and inference latency with minimal accuracy loss.

Q334. What is 'Amazon Nova Micro' optimized for?

A. Micro-sized image generation

B. Ultra-low latency and cost-efficient text tasks requiring speed over complexity

C. Microscopic scientific data analysis

D. Small-scale model training

✓ Correct Answer: B

Explanation: Amazon Nova Micro is the fastest and most cost-effective Nova model, optimized for high-throughput text tasks where speed and cost matter more than complex reasoning capability.

Q335. What is 'Amazon Nova Pro' designed for?

A. Professional photography editing

B. Highly capable multimodal tasks balancing accuracy, speed, and cost for enterprise workflows

C. Pro-tier AWS account features

D. Professional sports analytics

✓ Correct Answer: B

Explanation: Amazon Nova Pro is a highly capable multimodal model supporting text, image, and video inputs, designed for complex enterprise tasks requiring strong accuracy while remaining cost-effective.

Q336. What is the role of 'embeddings' in a RAG pipeline?

A. Embedding the pipeline in a web application

B. Converting documents and queries into vectors so semantically similar content can be retrieved by similarity

C. Embedding metadata into retrieved documents

D. Embedding the RAG system into a chatbot

✓ Correct Answer: B

Explanation: In a RAG pipeline, embeddings convert both documents (at indexing time) and user queries (at runtime) into vectors. Nearest-neighbor search finds the most semantically relevant documents to include as context.

Q337. What is 'Cohere Embed' available through Amazon Bedrock used for?

A. Embedding Cohere's logo in applications

B. Generating high-quality text embeddings for semantic search and classification

C. Embedding code snippets in documents

D. A video embedding model

✓ **Correct Answer: B**

Explanation: Cohere Embed on Bedrock generates text embeddings optimized for semantic search, clustering, and classification, with multilingual support making it strong for global applications.

Q338. What is 'LangChain' and how does it relate to Amazon Bedrock?

A. A blockchain for LLM data

B. An open-source framework for building LLM applications that integrates with Bedrock

C. Amazon's internal LLM orchestration tool

D. A language learning chain for model training

✓ **Correct Answer: B**

Explanation: LangChain is an open-source framework providing abstractions for building LLM-powered applications. It has native Bedrock integrations enabling developers to use Bedrock models within LangChain chains.

Q339. What is 'Amazon Bedrock's fine-tuning dataset format requirement'?

A. CSV files only

B. JSONL format with prompt-completion pairs stored in S3

C. XML format with labeled examples

D. Plain text files with one example per line

✓ **Correct Answer: B**

Explanation: Bedrock model customization (fine-tuning) requires training data in JSONL format containing prompt-completion pairs, stored in an S3 bucket that Bedrock has permission to access.

Q340. What is 'multimodal RAG'?

A. RAG deployed across multiple regions

B. RAG that retrieves and combines text, images, and other media types as context for generation

C. RAG using multiple LLMs simultaneously

D. A RAG system with multiple knowledge bases

✓ **Correct Answer: B**

Explanation: Multimodal RAG extends traditional text RAG to retrieve and process multiple data types (images, charts, tables alongside text), enabling richer context for answering questions about complex documents.

Domain 3: AWS AI & ML Services

Q341. What is 'Amazon SageMaker Inference Recommender'?

A. A service recommending AI use cases

B. A tool that benchmarks models to recommend optimal instance types and configurations for deployment

C. A recommendation system built on SageMaker

D. A service recommending training algorithms

✓ **Correct Answer: B**

Explanation: SageMaker Inference Recommender automatically benchmarks models across instance types and configurations to recommend the optimal endpoint setup for your performance and cost requirements.

Q342. What is 'Amazon SageMaker Multi-Model Endpoints'?

A. Endpoints accessing multiple AWS regions
B. A single endpoint hosting multiple models, reducing infrastructure costs
C. Multiple endpoints for a single model
D. Endpoints supporting multiple request formats

✓ Correct Answer: B

Explanation: SageMaker Multi-Model Endpoints host multiple models behind a single endpoint, dynamically loading/unloading models as needed, significantly reducing infrastructure costs for serving many models.

Q343. What is 'Amazon SageMaker Asynchronous Inference'?

A. Synchronous inference with async callbacks
B. A deployment option for large payloads and long processing times with queuing
C. Asynchronous model training
D. Inference without waiting for model loading

✓ Correct Answer: B

Explanation: SageMaker Asynchronous Inference queues large or long-running inference requests, returning results to S3 when complete — ideal for video processing or large document analysis.

Q344. What is 'Amazon SageMaker Edge Manager'?

A. Managing network edge connections to SageMaker
B. Deploying, monitoring, and updating ML models on edge devices
C. Managing the SageMaker Studio edge cases
D. An edge computing service for data collection

✓ Correct Answer: B

Explanation: SageMaker Edge Manager optimizes, secures, monitors, and maintains ML models deployed on edge devices (cameras, vehicles, IoT sensors) outside the cloud.

Q345. What is 'Amazon Rekognition Custom Labels'?

A. Adding custom text labels to images
B. Training custom image classification and object detection models with your own labeled images
C. Customizing Rekognition's built-in labels
D. Creating custom label templates for Ground Truth

✓ Correct Answer: B

Explanation: Rekognition Custom Labels lets you train specialized computer vision models for your specific objects and scenes using a small set of your own labeled images, without deep ML expertise.

Q346. What is 'Amazon Comprehend Custom Classification'?

A. AWS's default text classification system
B. Training custom document classifiers for domain-specific text categorization
C. Customizing Comprehend's sentiment categories
D. A classification scheme for AWS service types

✓ Correct Answer: B

Explanation: Comprehend Custom Classification trains models to categorize documents into your own defined categories (e.g., support ticket types, legal document classes) using your labeled examples.

Q347. What is 'Amazon Translate Active Custom Translation'?

A. Manually reviewing all translations
B. Customizing translations using parallel data and terminology for domain-specific language
C. Actively monitoring translation quality
D. Real-time human-assisted translation

✓ Correct Answer: B

Explanation: Amazon Translate supports customization through parallel data (custom training examples) and custom terminology, allowing domain-specific translations that use preferred vocabulary.

Q348. What is 'Amazon Kendra Intelligent Ranking'?

A. Ranking AWS Kendra pricing tiers
B. Re-ranking search results using semantic ML models to improve relevance
C. An internal ranking of Kendra features
D. Ranking customers by their search activity

✓ Correct Answer: B

Explanation: Amazon Kendra Intelligent Ranking uses deep learning to semantically re-rank search results from any search service, improving relevance beyond keyword matching.

Q349. What is 'Amazon SageMaker Training Compiler'?

A. A tool for compiling SageMaker code
B. A built-in compiler that optimizes deep learning models to train faster on GPU instances
C. A compiler for SageMaker pipeline definitions
D. A tool compiling SageMaker documentation

✓ Correct Answer: B

Explanation: SageMaker Training Compiler automatically optimizes deep learning model graphs for GPU training, reducing training time by up to 50% by optimizing memory usage and GPU utilization.

Q350. What is 'AWS Glue' used for in ML data pipelines?

A. Gluing model components together
B. A serverless ETL service for preparing and transforming data for ML training
C. A data visualization service
D. A ML model deployment service

✓ Correct Answer: B

Explanation: AWS Glue is a serverless data integration service providing ETL capabilities to discover, catalog, clean, transform, and move data into the format needed for ML training.

Q351. What is 'Amazon OpenSearch' neural search capability?

A. A neural network training service
B. ML-powered vector search enabling semantic and hybrid search in OpenSearch
C. A search tool for neural science data
D. An open-source neural network library

✓ **Correct Answer: B**

Explanation: Amazon OpenSearch's neural search uses ML models to generate and search vector embeddings alongside traditional keyword search, enabling semantic similarity search for RAG and recommendation systems.

Q352. What is 'Amazon SageMaker Model Cards'?

A. Payment cards for SageMaker billing

B. Documentation framework for recording intended use, risk ratings, and performance details of models

C. Physical cards for configuring ML hardware

D. API keys for accessing deployed models

✓ **Correct Answer: B**

Explanation: SageMaker Model Cards provide a standardized documentation framework for capturing model facts: intended use, risk rating, training details, evaluation results, and ethical considerations.

Q353. What is 'Amazon SageMaker Shadow Testing'?

A. Testing models in dark mode

B. Sending a copy of live traffic to a new model version for comparison without affecting users

C. Running models in shadow AWS accounts

D. Testing models with synthetic shadow data

✓ **Correct Answer: B**

Explanation: SageMaker Shadow Testing routes a copy of live production traffic to a new model candidate in parallel, comparing its performance to the production model without any user impact.

Q354. What is 'Amazon Transcribe Call Analytics'?

A. Analyzing analytics dashboards via voice

B. ML-powered analysis of call center conversations for sentiment, issues, and agent performance

C. Transcribing analytics reports to audio

D. Analyzing transcription job performance

✓ **Correct Answer: B**

Explanation: Amazon Transcribe Call Analytics combines speech transcription with ML-based analysis including sentiment detection, call categorization, issue detection, and agent/customer turn detection for contact centers.

Q355. What is 'Amazon SageMaker Clarify's explainability report'?

A. A report clarifying SageMaker billing

B. A report showing feature importance scores (SHAP values) explaining model predictions

C. Documentation for model clarification

D. A report of training data statistics

✓ **Correct Answer: B**

Explanation: SageMaker Clarify generates explainability reports using SHAP values showing how much each feature contributed to individual predictions, supporting interpretability and regulatory compliance.

Domain 4: Responsible AI

Q356. What is 'equalized odds' in AI fairness?

A. Equal number of samples per class
B. A fairness metric requiring equal true positive and false positive rates across demographic groups
C. Equal model accuracy across regions
D. Equalizing training time across GPUs

✓ Correct Answer: B

Explanation: Equalized odds requires that a model has the same true positive rate AND false positive rate across protected demographic groups, ensuring predictions are equally accurate for all groups.

Q357. What is 'disparate impact' in AI fairness?

A. The impact of deploying AI to disparate regions
B. When an AI system disproportionately harms or benefits a particular demographic group
C. Inconsistent model performance across servers
D. Different API response times for different users

✓ Correct Answer: B

Explanation: Disparate impact occurs when an AI system produces outcomes that have a significantly different effect on one protected group versus another, even without explicit use of protected attributes.

Q358. What is 'transparency in AI' beyond explainability?

A. Making AI code open-source
B. Disclosing that AI is being used, how it works, and what data it was trained on to affected stakeholders
C. Transparent pricing for AI services
D. Transparent hardware specifications for ML

✓ Correct Answer: B

Explanation: AI transparency includes disclosing when AI is being used in decision-making, being open about training data sources, known limitations, and how the system reaches its outputs.

Q359. What is 'AI safety' concerned with?

A. Physical safety of data center hardware
B. Ensuring AI systems behave reliably and do not cause unintended harm to people or society
C. Network safety for AI API communications
D. Data backup safety for ML models

✓ Correct Answer: B

Explanation: AI safety focuses on ensuring AI systems behave as intended, remain aligned with human values, avoid unintended harmful side effects, and can be reliably controlled and corrected by humans.

Q360. What is 'Bedrock Guardrails denied topics' feature?

A. Topics the Bedrock team is not allowed to discuss
B. User-defined topics the LLM application should refuse to engage with
C. Topics removed from foundation model training
D. Topics with restricted API access

✓ Correct Answer: B

Explanation: Bedrock Guardrails denied topics allow organizations to define specific subjects (e.g., competitor products, investment advice, political opinions) that the AI application will decline to discuss.

Q361. What is 'privacy by design' in AI development?

A. Designing private cloud AI infrastructure
B. Embedding privacy protection into AI system architecture from the start rather than adding it later
C. Private design sessions for AI systems
D. Designing AI that can be used privately offline

✓ Correct Answer: B

Explanation: Privacy by design proactively builds privacy protections into AI systems at the design stage, including data minimization, purpose limitation, encryption, and access controls from the outset.

Q362. What is 'debiasing' in ML?

A. Removing all statistical bias from predictions
B. Applying techniques to reduce the effect of unfair bias in models or training data
C. Removing debug logging from production models
D. Biasing the model toward the majority class

✓ Correct Answer: B

Explanation: Debiasing involves preprocessing data (removing biased proxies, rebalancing), in-processing (fairness constraints during training), or post-processing (adjusting predictions) to reduce unfair model bias.

Q363. What is 'AI incident response'?

A. Responding to cloud infrastructure incidents
B. A plan for detecting, investigating, and correcting harmful AI system behaviors after deployment
C. Responding to model training failures
D. Incident reporting for AI billing errors

✓ Correct Answer: B

Explanation: AI incident response establishes processes for detecting when an AI system causes harm, investigating root causes, correcting model or data issues, and communicating with affected stakeholders.

Q364. What is 'Bedrock Guardrails sensitive information filters'?

A. Filtering sensitive AWS account information
B. Automatically detecting and redacting PII and sensitive data in model inputs and outputs
C. Filtering model weights for sensitive parameters
D. Sensitive information storage for compliance

✓ Correct Answer: B

Explanation: Bedrock Guardrails sensitive information filters detect and redact PII types (SSNs, credit card numbers, emails, names) from both prompts sent to the model and responses returned to users.

Q365. What is 'responsible scaling' in AI development?

A. Auto-scaling AI infrastructure responsibly
B. Coupling increases in AI capability with commensurate advances in safety and alignment measures
C. Scaling AI teams at a responsible pace
D. Responsible use of GPU clusters

✓ **Correct Answer: B**

Explanation: Responsible scaling policies (used by companies like Anthropic) establish that increases in model capability must be matched with advances in safety evaluation and risk mitigation before deployment.

Domain 5: Security, Compliance & Governance

Q366. What is 'Amazon SageMaker Studio Domain' security feature?

A. A custom domain name for SageMaker

B. An isolated multi-user environment with IAM-based access control for collaborative ML work

C. A domain registry for ML endpoints

D. A web domain for SageMaker documentation

✓ **Correct Answer: B**

Explanation: A SageMaker Domain creates an isolated environment for teams with individual user profiles, dedicated EFS storage, and IAM-controlled access to SageMaker resources.

Q367. What is 'AWS IAM Identity Center' relevant to enterprise AI?

A. An AI service for identity verification

B. Centralized identity management and SSO access to AWS AI services across multiple accounts

C. An identity recognition service

D. A center for managing IAM documentation

✓ **Correct Answer: B**

Explanation: AWS IAM Identity Center (formerly SSO) provides centralized identity management and single sign-on access to AWS accounts and applications including SageMaker and Bedrock, simplifying enterprise access governance.

Q368. What is 'Amazon SageMaker's VPC-only mode'?

A. A mode limiting SageMaker to virtual processing clusters

B. Restricting all SageMaker traffic to a private VPC, preventing public internet exposure

C. A virtual private cloud mode for cheaper training

D. A mode using only VPC endpoints for billing

✓ **Correct Answer: B**

Explanation: SageMaker VPC-only mode ensures training jobs, notebooks, and endpoints operate entirely within your private VPC without any public internet access, meeting strict security requirements.

Q369. What does 'AWS Certificate Manager (ACM)' provide for ML applications?

A. AWS certification exam management

B. Free SSL/TLS certificates for securing ML API endpoints and web applications

C. ML model certification for deployment

D. Certificate storage for compliance documents

✓ **Correct Answer: B**

Explanation: ACM provides free SSL/TLS certificates for securing communications to ML API endpoints deployed on services like API Gateway, ensuring encrypted connections for AI applications.

Q370. What is 'Amazon Macie's role in ML data governance'?

A. Making ML models faster

B. Continuously discovering and protecting sensitive data (PII, credentials) in S3 training datasets

C. Managing ML experiment metadata

D. A macro execution service for data processing

✓ Correct Answer: B

Explanation: Amazon Macie uses ML to automatically scan S3 buckets (including training data) for sensitive information like PII, financial data, and credentials, alerting teams to data exposure risks.

Q371. What is 'Amazon SageMaker's model artifact encryption'?

A. Encrypting model names

B. Using KMS keys to encrypt trained model files stored in S3 at rest

C. Encrypting model API endpoint URLs

D. Encrypting training log artifacts

✓ Correct Answer: B

Explanation: SageMaker encrypts model artifacts (trained model files saved to S3) using AWS KMS keys, protecting model intellectual property and meeting data security compliance requirements.

Q372. What is 'IP whitelisting' for ML API security?

A. Listing all IP addresses in a model's training data

B. Restricting ML API access to specific allowed IP address ranges

C. Intellectual property whitelisting for model outputs

D. Prioritizing certain IP traffic for lower latency

✓ Correct Answer: B

Explanation: IP whitelisting restricts access to ML APIs (SageMaker endpoints, Bedrock API) to specific trusted IP address ranges using security groups or WAF rules, reducing attack surface.

Q373. What is 'AWS WAF' and its relevance to AI applications?

A. A workflow automation framework

B. A web application firewall protecting AI APIs from common web exploits and bot attacks

C. A write-ahead file for ML data pipelines

D. A warm-up acceleration feature for inference

✓ Correct Answer: B

Explanation: AWS WAF protects web-facing ML APIs from common attacks like SQL injection, XSS, and DDoS, and can be configured to rate-limit AI API calls and filter malicious prompt injection attempts.

Q374. What is 'audit logging' best practice for Bedrock applications?

A. Logging model accuracy metrics only

B. Enabling CloudTrail API logging and Bedrock invocation logging for complete audit trails

C. Auditing model training code

D. Logging only failed API calls for debugging

✓ Correct Answer: B

Explanation: Best practice is combining CloudTrail (API call logging) with Bedrock invocation logging (full prompt/response capture) to create comprehensive audit trails for compliance and security investigation.

Q375. What is 'multi-factor authentication (MFA)' important for in AI governance?

A. Using multiple AI models for authentication
B. Requiring a second verification factor to access sensitive ML systems and training data
C. Multiple authentication factors for model API calls
D. Factor analysis in ML model governance

✓ Correct Answer: B

Explanation: MFA adds a critical security layer for accessing sensitive ML systems — SageMaker Studio, Bedrock console, and S3 training data — preventing unauthorized access even if credentials are compromised.

Q376. What is 'data classification' in ML security?

A. A ML technique for categorizing data
B. Tagging data by sensitivity level to apply appropriate security controls in ML pipelines
C. Classifying training data by volume
D. A compliance classification for AWS services

✓ Correct Answer: B

Explanation: Data classification labels data by sensitivity (public, internal, confidential, restricted) to apply appropriate access controls, encryption, and handling requirements throughout ML pipelines.

Q377. What is 'secrets management' in ML workflows?

A. Managing model secrets (weights) securely
B. Securely storing and rotating API keys, database credentials, and tokens used by ML pipelines
C. Secret governance meetings for AI strategy
D. Managing private model architectures

✓ Correct Answer: B

Explanation: AWS Secrets Manager securely stores and rotates credentials (database passwords, API keys) needed by ML training jobs and inference pipelines, preventing hardcoded credentials in code.

Q378. What is 'Amazon SageMaker Role Manager'?

A. Managing SageMaker Studio user roles
B. A tool simplifying creation of least-privilege IAM roles for SageMaker ML activities
C. A role-playing training environment
D. A manager assignment service for AI teams

✓ Correct Answer: B

Explanation: SageMaker Role Manager provides a guided experience for creating IAM roles with just the right permissions for specific ML activities, reducing the complexity of least-privilege IAM configuration.

Q379. What is 'compliance automation' for AI workloads?

A. Using AI to automate compliance tasks
B. Automatically enforcing and verifying compliance rules using AWS Config, SCPs, and CI/CD checks
C. Automating compliance report generation
D. Automatic pricing compliance for AI services

✓ **Correct Answer: B**

Explanation: Compliance automation uses AWS Config Rules, Service Control Policies, and pipeline checks to automatically detect and remediate non-compliant AI infrastructure configurations in real time.

Q380. What is 'Amazon Bedrock's FIPS 140-2 endpoints' relevance?

A. A fast IP failover system

B. Endpoints using FIPS-validated cryptography required for US federal government AI workloads

C. A financial IP protection standard

D. A framework for inference performance standards

✓ **Correct Answer: B**

Explanation: Amazon Bedrock offers FIPS 140-2 compliant endpoints using validated cryptographic modules, meeting the requirements of US federal government agencies and regulated industries for AI workloads.

Q381. What is 'Amazon SageMaker Governance' as a capability?

A. A government portal for SageMaker users

B. A unified set of tools for ML lifecycle governance including Model Cards, Model Registry, and Model Monitor

C. A governance consulting service

D. SageMaker's pricing governance tools

✓ **Correct Answer: B**

Explanation: SageMaker Governance combines Model Cards (documentation), Model Registry (version management and approval workflows), and Model Monitor (production monitoring) for end-to-end ML lifecycle governance.

Q382. What is 'Amazon Bedrock's cross-region data isolation'?

A. Isolating cross-region latency

B. Ensuring customer data processed in one region does not leave that region without explicit routing

C. Isolating cross-region billing

D. Cross-region model replication isolation

✓ **Correct Answer: B**

Explanation: By default, Bedrock processes requests in the region where the API is called. Cross-Region Inference must be explicitly configured, giving customers control over data residency and regional isolation.

Q383. What is the purpose of 'AWS Service Catalog' in ML governance?

A. A catalog of all AWS services

B. Enabling IT to create approved, pre-configured ML product portfolios that teams can self-service

C. Cataloging ML model versions

D. A pricing catalog for AI services

✓ **Correct Answer: B**

Explanation: AWS Service Catalog lets IT governance teams create pre-approved, compliant ML infrastructure templates (SageMaker environments, Bedrock setups) that data science teams can deploy self-service.

Q384. What is 'AWS Backup' relevance to ML workloads?

A. Backing up ML model predictions

B. Centralized backup of ML training data, model artifacts, and EFS storage for data protection and recovery

C. A backup strategy for GPU instances

D. Backing up SageMaker notebook code

✓ Correct Answer: B

Explanation: AWS Backup provides centralized, automated backup for ML-related storage (S3 training data, EFS notebook storage, EBS model artifacts), ensuring data protection and enabling recovery from accidental deletion.

Q385. What is 'threat modeling' for AI systems?

A. Building ML models to detect threats

B. Systematically identifying potential security threats, attack vectors, and vulnerabilities in AI system design

C. Modeling the impact of AI market threats

D. A threat scoring model for cybersecurity

✓ Correct Answer: B

Explanation: Threat modeling for AI identifies potential attack surfaces (API, training data, model outputs, infrastructure) and adversarial scenarios early in design, guiding security controls and mitigations.

Domain 1: AI & ML Fundamentals

Q386. What is 'cosine similarity' used for in ML?

A. Measuring the angle between two vectors to determine their similarity

B. Measuring the cosine of a model's error curve

C. A trigonometric activation function

D. Measuring Euclidean distance between points

✓ Correct Answer: A

Explanation: Cosine similarity measures the cosine of the angle between two vectors. Values near 1 indicate similar direction (high similarity), used widely in text similarity, recommendation systems, and RAG retrieval.

Q387. What is 'data leakage' in ML and why is it a problem?

A. Training data accidentally deleted from storage

B. When information from the test set leaks into training, causing inflated performance estimates

C. Sensitive data leaking from a model's API

D. Feature data leaking between model layers

✓ Correct Answer: B

Explanation: Data leakage happens when information that would not be available at prediction time is used in training, causing the model to appear better than it is and fail in production.

Q388. What is 'target encoding' for categorical variables?

A. Encoding the target label as text

B. Replacing a categorical value with the mean of the target variable for that category

C. Encoding categories with random target values

D. A label encoding technique for multi-class targets

✓ **Correct Answer: B**

Explanation: Target encoding replaces each category with the mean (or other statistic) of the target variable for that category, creating informative numerical representations for high-cardinality features.

Q389. What does 'model serialization' mean?

A. Running model predictions in series

B. Converting a trained model into a storable format (e.g., pickle, ONNX, SavedModel) for later use

C. Ordering model layers sequentially

D. A serial training process without parallelism

✓ **Correct Answer: B**

Explanation: Model serialization saves a trained model's weights and architecture to a file format so it can be stored, transferred, and reloaded for inference without retraining.

Q390. What is 'ONNX' in the context of ML models?

A. An AWS ML service

B. An open format for representing ML models enabling interoperability across frameworks

C. A neural network optimizer

D. An online annotation tool

✓ **Correct Answer: B**

Explanation: ONNX (Open Neural Network Exchange) is an open standard for ML model representations, allowing models trained in one framework (PyTorch) to be deployed in another (TensorFlow, TensorRT) for optimized inference.

Q391. What is 'SageMaker's built-in XGBoost algorithm' optimized for?

A. Image classification tasks

B. Tabular data classification and regression with high performance and scalability

C. Time series forecasting exclusively

D. Deep learning model training

✓ **Correct Answer: B**

Explanation: SageMaker's built-in XGBoost is an optimized implementation of the gradient boosted decision tree algorithm, highly effective for structured/tabular data tasks and distributed training on large datasets.

Domain 2: Generative AI & Foundation Models

Q392. What is 'tool use' (function calling) in LLM applications?

A. Using developer tools to build LLM apps

B. Allowing LLMs to call external functions/APIs to retrieve data or perform actions

C. A tool for fine-tuning LLMs

D. Using command-line tools to query LLMs

✓ **Correct Answer: B**

Explanation: Tool use (function calling) enables LLMs to decide when to call external functions (weather APIs, databases, calculators) and structure the call parameters, enabling agents to interact with the real world.

Q393. What is 'Amazon Bedrock's Guardrails grounding check'?

A. Checking if models are properly grounded to servers
B. Verifying that model responses are factually grounded in provided source documents
C. A ground truth check for training data
D. Grounding electrical systems for Bedrock hardware

✓ Correct Answer: B

Explanation: Bedrock Guardrails grounding check detects when model responses are not supported by the provided source material, helping identify and filter hallucinations in RAG applications.

Q394. What is 'Amazon Q Apps'?

A. Mobile applications built on Amazon Q
B. A feature letting users build shareable AI-powered apps from natural language descriptions
C. Q's application programming interface
D. Quantum computing applications on AWS

✓ Correct Answer: B

Explanation: Amazon Q Apps allows employees to create lightweight generative AI-powered applications through natural language descriptions within Amazon Q Business, no coding required.

Q395. What is 'model routing' in Amazon Bedrock?

A. Network routing for Bedrock API traffic
B. Automatically selecting the most suitable model for each request based on complexity and cost
C. Routing model outputs to different S3 locations
D. Assigning models to specific users

✓ Correct Answer: B

Explanation: Bedrock model routing (Intelligent Prompt Routing) automatically sends requests to the optimal model balancing capability and cost — routing simple requests to cheaper models and complex ones to more powerful models.

Q396. What is 'Amazon Bedrock's prompt management'?

A. Managing who can send prompts to Bedrock
B. Storing, versioning, and sharing reusable prompts as managed resources in Bedrock
C. Limiting the number of prompts per account
D. A prompt quality scoring service

✓ Correct Answer: B

Explanation: Bedrock Prompt Management allows teams to create, version, test, and share prompt templates as managed resources, enabling prompt reuse, governance, and consistent application behavior.

Domain 3: AWS AI & ML Services

Q397. What is 'Amazon Personalize's real-time recommendations' capability?

A. Recommendations generated weekly in batch

B. Sub-second personalized recommendations served through a REST API endpoint

C. Real-time recommendations from human curators

D. Recommendations updated in real-time during model training

✓ Correct Answer: B

Explanation: Amazon Personalize hosts trained recommendation models behind real-time API endpoints that return personalized recommendations with sub-second latency for live application integration.

Q398. What is 'AWS DeepComposer'?

A. A deep learning compiler

B. An AI-powered musical keyboard for learning generative AI through music composition

C. A service for composing ML pipelines

D. A deep audio processing service

✓ Correct Answer: B

Explanation: AWS DeepComposer is an ML-enabled musical keyboard that generates accompaniment using generative AI models, providing a hands-on way to learn generative AI concepts.

Q399. What is 'Amazon Textract Forms extraction'?

A. Extracting tax forms from documents

B. Automatically identifying and extracting key-value pairs from form fields in documents

C. Formatting documents into standard forms

D. A form builder for data collection

✓ Correct Answer: B

Explanation: Textract Forms extraction identifies form fields and their values (checkboxes, text fields, key-value pairs) in documents like applications and questionnaires, automating form data digitization.

Q400. What is 'Amazon SageMaker Studio Lab'?

A. A physical ML lab at AWS offices

B. A free, no-AWS-account-needed environment for learning ML with Jupyter notebooks

C. A laboratory for testing SageMaker features

D. An advanced Studio tier for enterprise users

✓ Correct Answer: B

Explanation: SageMaker Studio Lab is a free service providing Jupyter notebook environments for ML learning, requiring no AWS account or credit card, designed to lower the barrier to ML education.

Q401. What is 'Amazon Comprehend's dominant language detection'?

A. Detecting the dominant programming language in code

B. Automatically identifying the primary language of a text document

C. Detecting dominant keywords in a document

D. Finding the dominant topic in a corpus

✓ Correct Answer: B

Explanation: Amazon Comprehend automatically detects the dominant language of text from among 100+ supported languages, enabling automatic routing of multilingual content for further processing.

Q402. What is 'Amazon Rekognition Video' used for?

A. Creating video content with AI
B. Analyzing video streams for people tracking, activity detection, and content moderation
C. Transcribing video audio to text
D. Compressing video files using ML

✓ Correct Answer: B

Explanation: Amazon Rekognition Video analyzes stored or streaming video to detect people, track movements, recognize celebrities, detect activities, and moderate inappropriate content at scale.

Q403. What is 'Amazon SageMaker Clarify's bias report types'?

A. Pre-training and post-training bias reports
B. Input bias and output bias reports only
C. Demographic bias and model bias reports
D. Training bias and inference bias reports

✓ Correct Answer: A

Explanation: SageMaker Clarify generates pre-training bias reports (detecting bias in datasets before training) and post-training bias reports (detecting bias in model predictions after training).

Q404. What does 'Amazon SageMaker Automatic Model Tuning' do?

A. Automatically selects the best ML algorithm
B. Automatically searches hyperparameter ranges to find the combination that maximizes a target metric
C. Automatically tunes model weights during inference
D. Automatically scales training infrastructure

✓ Correct Answer: B

Explanation: SageMaker Automatic Model Tuning (hyperparameter optimization) uses Bayesian optimization or random search to find hyperparameter combinations that maximize model performance metrics.

Q405. What is 'Amazon Fraud Detector's event-based model'?

A. Models triggered by calendar events
B. Real-time fraud scoring of individual events (transactions, logins) using ML and business rules
C. Historical event replaying for model training
D. Event-driven model retraining pipelines

✓ Correct Answer: B

Explanation: Amazon Fraud Detector evaluates each event (online transaction, account registration) in real time using ML models trained on your historical data plus Amazon's fraud detection experience.

Domain 4: Responsible AI

Q406. What is 'AI alignment' concerned with?

A. Aligning AI infrastructure in data centers
B. Ensuring AI systems pursue goals that align with human values and intentions
C. Aligning training data distributions

D. Aligning model outputs with business metrics

✓ **Correct Answer: B**

Explanation: AI alignment research focuses on ensuring AI systems behave in ways consistent with human values, intentions, and societal benefit, preventing unintended harmful objectives from emerging.

Q407. What is 'output toxicity' filtering in Bedrock Guardrails?

A. Filtering toxic chemicals in data center outputs

B. Detecting and blocking harmful, offensive, or hateful content generated by LLMs

C. Filtering low-quality model outputs

D. Monitoring model output performance metrics

✓ **Correct Answer: B**

Explanation: Bedrock Guardrails toxicity filtering uses classifiers to detect hate speech, insults, violence, sexual content, and profanity in model responses, blocking them before delivery to users.

Q408. What is 'bias in word embeddings'?

A. Technical bias in embedding calculation

B. Societal stereotypes encoded in embeddings because they were trained on biased text corpora

C. Bias toward certain vocabulary sizes

D. Embedding vectors biased toward zero

✓ **Correct Answer: B**

Explanation: Word embeddings trained on real-world text often encode societal biases (e.g., gender stereotypes, racial associations) present in that text, which can propagate harmful biases into downstream models.

Q409. What is 'informed consent' in AI research and development?

A. Consent forms signed by AI developers

B. Ensuring people understand and agree to how their data will be used in AI systems before collection

C. Obtaining consent from AI models before deployment

D. Legal consent for using third-party AI APIs

✓ **Correct Answer: B**

Explanation: Informed consent requires that individuals receive clear, understandable information about how their data will be collected and used in AI systems, and voluntarily agree before their data is used.

Q410. What is 'model robustness' in responsible AI?

A. A model that never changes after training

B. A model that maintains reliable performance under distributional shifts, noisy inputs, and adversarial attacks

C. Model stability during training convergence

D. Robust hardware for model inference

✓ **Correct Answer: B**

Explanation: Model robustness refers to a model's ability to perform reliably when inputs vary due to noise, distribution shifts, or adversarial perturbations — a key safety property for production AI.

Q411. What is 'disparate treatment' in AI systems?

A. Treating different model versions differently

B. Using protected attributes (race, gender) explicitly in model decisions, causing differential treatment

C. Treating different data types inconsistently

D. Differential pricing for AI services

✓ Correct Answer: B

Explanation: Disparate treatment occurs when a model explicitly uses protected attributes (e.g., race, gender, age) as inputs, directly causing different predictions for different groups.

Q412. What is 'AI auditing' and who can perform it?

A. Financial auditing of AI budgets

B. Independent systematic evaluation of AI systems for compliance, fairness, and safety by internal or third-party auditors

C. Technical auditing of model code only

D. Auditing conducted exclusively by regulators

✓ Correct Answer: B

Explanation: AI auditing involves systematic independent assessment of AI systems against defined criteria (fairness, safety, compliance). It can be conducted by internal teams, third-party firms, or regulatory bodies.

Q413. What does 'human oversight' mean for high-risk AI applications?

A. Humans watching AI training processes

B. Keeping humans involved in reviewing and overriding AI decisions for high-stakes use cases

C. Overhead costs of human-AI teams

D. Humans overseeing AI infrastructure

✓ Correct Answer: B

Explanation: Human oversight ensures that for high-stakes decisions (medical diagnoses, loan approvals, criminal justice), humans review AI recommendations and can override incorrect or unfair decisions.

Q414. What is 'AI ethics' as a field?

A. Ethical hacking of AI systems

B. The study and application of moral principles to AI development, deployment, and governance

C. Ethics training courses for AI engineers

D. Ethical use of AWS credits

✓ Correct Answer: B

Explanation: AI ethics examines moral questions around AI development and use including fairness, privacy, accountability, transparency, safety, and societal impact, guiding responsible AI practices.

Q415. What is 'safety testing' for generative AI before deployment?

A. Testing model infrastructure safety

B. Systematically probing models for harmful outputs, jailbreaks, and policy violations before release

C. Safety testing cloud infrastructure

D. Testing API call safety limits

✓ Correct Answer: B

Explanation: Safety testing (including red teaming) probes generative AI systems for harmful content, jailbreak vulnerabilities, and policy violations before deployment, reducing risk of harmful outputs in production.

Domain 5: Security, Compliance & Governance

Q416. What is 'zero trust security' applied to ML systems?

A. A security model trusting zero users

B. Never trusting any entity by default; always verifying identity and minimizing access for ML resources

C. Trusting zero-cost security solutions

D. A security model with zero configuration

✓ Correct Answer: B

Explanation: Zero trust applies to ML by verifying every access request to SageMaker, Bedrock, and training data regardless of network location, using strong authentication and least-privilege access everywhere.

Q417. What is 'AWS CloudWatch' used for in ML monitoring?

A. Watching cloud weather conditions

B. Monitoring ML training job metrics, endpoint performance, and triggering alarms for anomalies

C. Watching cloud costs only

D. Monitoring developer activity in the console

✓ Correct Answer: B

Explanation: CloudWatch collects metrics from SageMaker training jobs, endpoints, and Bedrock API calls, enabling dashboards, alarms (e.g., endpoint latency exceeded), and log aggregation for ML operations.

Q418. What is 'Amazon Inspector's Lambda scanning' relevant to AI?

A. Inspecting Lambda function source code for bugs

B. Automatically scanning Lambda functions used in ML pipelines for security vulnerabilities

C. Monitoring Lambda invocations during inference

D. Inspecting Lambda billing for ML workflows

✓ Correct Answer: B

Explanation: Amazon Inspector scans AWS Lambda functions (commonly used in Bedrock Agents and ML pipelines) for software vulnerabilities and insecure code, helping secure AI application components.

Q419. What is 'resource policy' for Amazon Bedrock?

A. A policy limiting resource consumption

B. An identity-based policy controlling which principals can invoke Bedrock APIs and access models

C. A resource allocation policy for GPU usage

D. Pricing policy for Bedrock resources

✓ Correct Answer: B

Explanation: Bedrock resource policies (via IAM) control which IAM users, roles, and services can invoke foundation models, create agents, and access model customization, implementing access governance.

Q420. What is 'SageMaker notebook instance lifecycle configuration'?

A. Configuration for the notebook's display lifecycle

B. Shell scripts running on notebook instance start/stop for security hardening and setup automation

C. Lifecycle hooks for model deployment

D. Configuration managing notebook deletion policies

✓ **Correct Answer: B**

Explanation: Lifecycle configurations run shell scripts when SageMaker notebook instances start or stop, used for security hardening (disabling internet, installing security tools, enforcing policies).

Q421. What is 'Amazon Bedrock's service-linked role'?

A. A role linked to Bedrock's service-level agreement

B. A predefined IAM role allowing Bedrock to access required AWS services on your behalf

C. A linked account role for Bedrock billing

D. A role for linking Bedrock to external services

✓ **Correct Answer: B**

Explanation: Bedrock's service-linked role is a pre-configured IAM role automatically created that grants Bedrock the specific permissions needed (S3, KMS, CloudWatch) to perform operations on your behalf.

Q422. What is 'data encryption key rotation' best practice for AI?

A. Rotating training data periodically

B. Periodically replacing encryption keys to limit the blast radius if a key is compromised

C. Rotating model weights for better performance

D. Rotating model deployment keys annually

✓ **Correct Answer: B**

Explanation: Regular KMS key rotation limits exposure — if a key is compromised, only data encrypted with that key version is at risk. AWS KMS supports automatic annual key rotation for customer-managed keys.

Q423. What is 'Amazon VPC Flow Logs' used for in ML security?

A. Logging ML model data flows

B. Capturing network traffic metadata for security analysis of ML workload communications

C. Logging training data flow between S3 and SageMaker

D. Monitoring flow of requests to Bedrock

✓ **Correct Answer: B**

Explanation: VPC Flow Logs capture metadata about network traffic to/from ML resources (SageMaker instances, EKS clusters), enabling security analysis, anomaly detection, and investigation of potential data exfiltration.

Domain 1: AI & ML Fundamentals

Q424. What is 'Gaussian distribution' and its relevance in ML?

A. A distribution only used in Gaussian models

B. A bell-curve probability distribution commonly assumed in many ML algorithms and statistical tests

C. A noise distribution exclusive to image processing

D. A distribution specific to gradient descent

✓ **Correct Answer: B**

Explanation: The Gaussian (normal) distribution is fundamental in ML — many algorithms assume normally distributed data or errors, it describes random noise, and it's used in probabilistic models and uncertainty estimation.

Q425. What is 'model interpretability' vs 'model explainability'?

A. They are completely different concepts with no overlap

B. Interpretability refers to understanding model structure; explainability refers to understanding individual predictions

C. Interpretability applies only to deep learning; explainability to tree models

D. Explainability is a subset of interpretability focusing on regulatory compliance only

✓ Correct Answer: B

Explanation: Interpretability refers to understanding a model's overall structure and decision logic (white-box models). Explainability typically refers to post-hoc methods explaining individual predictions of complex models.

Domain 2: Generative AI & Foundation Models

Q426. What is 'Amazon Bedrock's Automated Reasoning checks'?

A. Automated testing of reasoning benchmark scores

B. A feature that verifies AI responses against logical rules and policies to prevent factual errors

C. Automated reasoning for code generation

D. Benchmarks for measuring model reasoning

✓ Correct Answer: B

Explanation: Bedrock Automated Reasoning checks use formal verification methods to validate that model responses comply with logical rules, policies, and factual constraints, reducing errors in regulated domains.

Q427. What is the 'lost in the middle' problem in LLMs?

A. Training data lost in the middle of a dataset

B. LLMs tend to recall information at the start and end of context better than from the middle

C. Model performance lost during mid-training

D. Tokens lost during tokenization of long texts

✓ Correct Answer: B

Explanation: Research shows LLMs perform worse at retrieving information from the middle of long context windows compared to the start or end, a 'lost in the middle' phenomenon affecting RAG document placement strategy.

Domain 3: AWS AI & ML Services

Q428. What is 'Amazon SageMaker Geospatial ML'?

A. ML models deployed in geographic regions

B. Capabilities for acquiring, processing, and analyzing geospatial data using ML

C. A mapping service using ML

D. Geographic load balancing for ML endpoints

✓ Correct Answer: B

Explanation: SageMaker Geospatial ML provides tools for acquiring satellite and geospatial data, built-in ML models for earth observation tasks like land classification and change detection.

Q429. What does 'Amazon Lex V2' improve over V1?

A. Supports only voice interactions
B. Supports multiple languages per bot, streaming, and improved intent management
C. V2 is exclusively for mobile applications
D. V2 only supports rule-based responses

✓ Correct Answer: B

Explanation: Amazon Lex V2 introduced multi-language support within a single bot, streaming API for low-latency voice, improved intent management UI, and simplified deployment workflows.

Domain 4: Responsible AI

Q430. What is 'Amazon Bedrock's watermark detection' used for?

A. Detecting watermarks in training data
B. Identifying whether images were generated by Amazon Titan Image Generator using embedded signals
C. Watermarking Bedrock API responses
D. Detecting unauthorized use of Bedrock

✓ Correct Answer: B

Explanation: Amazon Titan Image Generator embeds invisible watermarks in generated images. Bedrock's watermark detection API can verify whether an image was created by Titan, supporting AI content attribution.

Domain 5: Security, Compliance & Governance

Q431. What is 'AWS Audit Manager' used for?

A. Managing AWS billing audits
B. Continuously collecting evidence to simplify compliance audits for AI workloads
C. Auditing ML model performance
D. Managing security patch audits

✓ Correct Answer: B

Explanation: AWS Audit Manager continuously collects evidence of AWS resource configurations and API activity, automating the preparation of audit-ready reports for compliance frameworks like SOC 2 and HIPAA.

Domain 1: AI & ML Fundamentals

Q432. What is 'causal inference' in ML?

A. Inferring that correlation implies causation

B. Estimating cause-and-effect relationships rather than just correlations between variables

C. A type of inference used in causal forests only

D. Inferring model parameters from causes

✓ Correct Answer: B

Explanation: Causal inference determines whether a variable X causes changes in Y (not just correlates), using techniques like randomized experiments or causal graphs — important for decision-making and policy evaluation.

Domain 2: Generative AI & Foundation Models

Q433. What is 'speculative decoding' in LLM inference optimization?

A. A technique for generating speculative content

B. Using a small draft model to propose tokens that a larger model quickly verifies, speeding inference

C. Decoding model outputs speculatively without verification

D. A method for reducing speculation in model outputs

✓ Correct Answer: B

Explanation: Speculative decoding uses a small fast draft model to generate candidate tokens in parallel, which a larger verifier model accepts or rejects in batch, achieving significant inference speedup.

Domain 3: AWS AI & ML Services

Q434. What is 'Amazon SageMaker Studio's Code Editor'?

A. A code formatting service

B. A VS Code-based IDE within SageMaker Studio for developing ML scripts and notebooks

C. A compliance code checker

D. An auto-code generator for ML pipelines

✓ Correct Answer: B

Explanation: SageMaker Studio Code Editor brings a VS Code-compatible IDE experience directly into SageMaker Studio, supporting ML development with extensions, terminal access, and seamless SageMaker integration.

Domain 5: Security, Compliance & Governance

Q435. What is the 'AWS Generative AI Security Scoping Matrix'?

A. A matrix comparing prices of Bedrock models

B. A framework categorizing generative AI applications by security risk scope based on customization level

C. A security benchmarking system

D. A compliance scoring matrix for AI audits

✓ **Correct Answer: B**

Explanation: The AWS Generative AI Security Scoping Matrix helps organizations assess security responsibilities based on where they operate in the generative AI stack — from using pre-built AI to training custom models.

Domain 1: AI & ML Fundamentals

Q436. What is 'Shapley value' and its ML significance?

A. A value representing model accuracy

B. A game-theory concept measuring each feature's marginal contribution to a prediction

C. The shape of a model's decision boundary

D. A value representing training data quality

✓ **Correct Answer: B**

Explanation: Shapley values from cooperative game theory measure each feature's average marginal contribution across all possible feature orderings. SHAP (SHapley Additive exPlanations) applies this to explain ML predictions.

Domain 2: Generative AI & Foundation Models

Q437. What is 'Amazon Bedrock's Converse API tool_use feature'?

A. Using tools in the AWS console

B. Enabling models to request execution of defined tools/functions and process their results in conversations

C. A toolkit for Bedrock administrators

D. Tool documentation within Bedrock

✓ **Correct Answer: B**

Explanation: The Converse API tool_use feature provides a standardized way to define tools (functions, APIs) that models can request during a conversation, and to pass back tool results, enabling agentic applications.

Domain 3: AWS AI & ML Services

Q438. What is 'Amazon Translate's real-time translation' useful for?

A. Translating model weights between frameworks

B. Translating customer-facing content instantly as it is created or consumed in applications

C. Real-time language learning assessments

D. Translating code between programming languages

✓ **Correct Answer: B**

Explanation: Amazon Translate's real-time API enables applications to instantly translate text as it is generated (chat messages, support tickets, product listings), enabling multilingual user experiences.

Domain 4: Responsible AI

Q439. What is 'benefit sharing' in responsible AI?

A. Sharing AWS cost benefits across teams

B. Ensuring the benefits of AI are distributed equitably across society rather than concentrated narrowly

C. Sharing model weights with the community

D. Distributing compute resources fairly

✓ Correct Answer: B

Explanation: Benefit sharing in responsible AI calls for ensuring that the productivity gains, economic value, and opportunities from AI are distributed broadly across society, not just to those who build or own AI systems.

Domain 5: Security, Compliance & Governance

Q440. What is 'Amazon SageMaker's support for compliance standards'?

A. SageMaker creates compliance documentation automatically

B. SageMaker is eligible for HIPAA, SOC, PCI DSS, ISO standards enabling regulated industry use

C. SageMaker enforces government regulations on models

D. SageMaker passes all compliance audits automatically

✓ Correct Answer: B

Explanation: Amazon SageMaker is compliant with major standards including HIPAA, SOC 1/2/3, PCI DSS, ISO 27001, enabling healthcare, financial, and other regulated industries to use it for ML workloads.

Domain 1: AI & ML Fundamentals

Q441. What does 'inference latency' mean in ML deployment?

A. Delay in starting model training

B. The time taken for a deployed model to return a prediction after receiving a request

C. Latency in loading training data

D. Network delay in accessing the ML console

✓ Correct Answer: B

Explanation: Inference latency is the end-to-end time from when an input is sent to a deployed model to when the prediction response is received — a critical performance metric for real-time applications.

Domain 2: Generative AI & Foundation Models

Q442. What is 'Amazon Bedrock's model distillation' feature?

A. Distilling complex prompts into simple ones

B. Creating a smaller, efficient custom model by distilling knowledge from a larger Bedrock foundation model

C. A service for distilling training data

D. Filtering distilled information from model outputs

✓ Correct Answer: B

Explanation: Amazon Bedrock Model Distillation trains a smaller, cost-efficient student model using the outputs of a larger teacher foundation model as synthetic training data, delivering similar quality at lower cost.

Domain 3: AWS AI & ML Services

Q443. What is 'Amazon SageMaker's distributed training strategies'?

A. Training models in distributed geographic locations

B. Data parallelism and model parallelism for training large models across multiple GPU instances

C. Distributing training jobs to different customers

D. Distributing training datasets across S3 buckets

✓ Correct Answer: B

Explanation: SageMaker supports data parallelism (splitting data across GPUs) and model parallelism (splitting model layers across GPUs) for efficiently training very large models that exceed single GPU memory.

Domain 4: Responsible AI

Q444. What is 'dataset documentation' (Datasheets for Datasets) in responsible AI?

A. Spreadsheets containing the dataset

B. Standardized documentation of dataset creation, composition, uses, and limitations for responsible reuse

C. Data quality reports from ETL pipelines

D. Documentation of data storage costs

✓ Correct Answer: B

Explanation: Datasheets for Datasets are structured documents recording how a dataset was collected, its composition, recommended and discouraged uses, potential biases, and maintenance information to enable informed reuse.

Domain 5: Security, Compliance & Governance

Q445. What is 'Amazon Bedrock's model invocation logging to CloudWatch'?

A. Watching model invocation dashboards

B. Sending model input/output logs to CloudWatch for analysis, monitoring, and compliance retention

C. Invoking CloudWatch from within models

D. A CloudWatch integration for billing logs

✓ Correct Answer: B

Explanation: Bedrock invocation logging can send full request/response logs to CloudWatch Logs, enabling log analysis, alerting on unusual patterns, and meeting compliance requirements for conversation retention.

Domain 1: AI & ML Fundamentals

Q446. What is 'stacking' in ensemble machine learning?

A. Stacking layers of neural networks

B. Training a meta-model on the predictions of several base models to produce a final output

C. Stacking training datasets together

D. A method for stacking AWS EC2 instances

✓ Correct Answer: B

Explanation: Stacking trains multiple diverse base models and feeds their predictions as features into a meta-learner, often producing better results than any single base model alone.

Q447. What is 'label noise' in training data?

A. Noise added to labels for data augmentation

B. Incorrectly assigned labels in the training dataset that degrade model quality

C. Missing labels in training data

D. Extra labels added to features

✓ Correct Answer: B

Explanation: Label noise refers to incorrectly assigned class labels in training data (e.g., misannotated images), which degrades model performance by training on wrong examples.

Q448. What is a 'learning curve' used for in ML?

A. Analyzing how fast students learn ML

B. Plotting model performance vs. training data size to diagnose bias or variance problems

C. Analyzing the model's learning rate schedule

D. Tracking business value of ML over time

✓ Correct Answer: B

Explanation: Learning curves plot training and validation performance against training set size, revealing whether a model suffers from high bias (more data won't help) or high variance (more data will help).

Q449. What is 'online learning' in machine learning?

A. Learning ML through online courses

B. Continuously updating model weights as new data arrives in a streaming fashion

C. Training models over the internet

D. Learning from online databases

✓ **Correct Answer: B**

Explanation: Online learning updates model parameters incrementally as each new data sample arrives, enabling models to adapt to changing data distributions in real-time without full retraining.

Q450. What is 'cosine similarity' used for in ML?

A. Measuring the angle between two vectors to determine their similarity

B. Measuring the cosine of a model's error curve

C. A trigonometric activation function

D. Measuring Euclidean distance between data points

✓ **Correct Answer: A**

Explanation: Cosine similarity measures the cosine of the angle between two vectors. Values near 1 indicate high similarity, used widely in text similarity, recommendation systems, and RAG retrieval.

Q451. What is 'data leakage' in ML and why is it a problem?

A. Training data accidentally deleted from storage

B. When information from the test set leaks into training, causing inflated performance estimates

C. Sensitive data leaking from a model API

D. Feature data leaking between model layers

✓ **Correct Answer: B**

Explanation: Data leakage occurs when information unavailable at prediction time is used in training, causing the model to appear better than it is and fail in production.

Q452. What is 'model serialization' in ML deployment?

A. Running model predictions in series

B. Converting a trained model into a storable file format (pickle, ONNX, SavedModel) for later use

C. Ordering model layers sequentially

D. A serial training process without parallelism

✓ **Correct Answer: B**

Explanation: Model serialization saves a trained model's weights and architecture to a file format so it can be stored, transferred, and reloaded for inference without retraining.

Q453. What is 'class weight' adjustment in ML training?

A. Adjusting model weight for each class layer

B. Increasing the loss contribution of minority class samples to address class imbalance

C. Weighting classes by prediction confidence

D. A weight initialization technique

✓ **Correct Answer: B**

Explanation: Class weighting assigns higher loss penalties to misclassifying minority class samples, forcing the model to pay more attention to underrepresented classes without resampling the data.

Q454. What does 'cold start problem' mean in recommendation systems?

A. A model that is slow to start training

B. Difficulty making recommendations for new users or items with no historical interaction data

C. A caching problem in recommendation APIs

D. Starting a model in a cold data center

✓ **Correct Answer: B**

Explanation: The cold start problem occurs when a recommendation system cannot generate accurate recommendations for new users or new items due to lack of historical interaction data.

Q455. What is 'feature store' and why is it valuable?

A. A store selling ML features and plugins

B. A centralized repository for storing, sharing, and serving ML features consistently across teams and models

C. A database of feature flags for ML pipelines

D. A store for model architecture templates

✓ **Correct Answer: B**

Explanation: A feature store centralizes ML feature computation, storage, and serving, enabling teams to reuse features across models, ensure training-serving consistency, and retrieve point-in-time features.

Q456. What is 'model complexity' and how does it relate to bias-variance tradeoff?

A. The number of files in a model repository

B. Higher model complexity reduces bias but increases variance, requiring regularization to balance the tradeoff

C. Complexity measured by training time

D. The complexity of the model's API interface

✓ **Correct Answer: B**

Explanation: Simple models have high bias (underfit) while complex models have high variance (overfit). The bias-variance tradeoff requires selecting complexity that minimizes total error.

Domain 2: Generative AI & Foundation Models

Q457. What is 'negative prompting' in image generation?

A. Writing negative reviews as prompts

B. Specifying what should NOT appear in a generated image to guide the output

C. A technique for generating negative sentiment text

D. Prompts that produce lower quality outputs

✓ **Correct Answer: B**

Explanation: Negative prompting in image generation (Stable Diffusion) tells the model what to avoid (e.g., 'blurry, low quality'), steering generation away from unwanted elements.

Q458. What is 'model collapse' in generative AI?

A. A model crashing during inference

B. Degradation in AI output quality when models are repeatedly trained on AI-generated data

C. Model weights collapsing to zero

D. Collapse of a multi-model ensemble

✓ **Correct Answer: B**

Explanation: Model collapse occurs when AI models trained on synthetic data from prior AI generations progressively lose diversity and accuracy, producing low-quality homogeneous outputs.

Q459. What is 'tool use' (function calling) in LLM applications?

A. Using developer tools to build LLM apps
B. Allowing LLMs to call external functions or APIs to retrieve data or take actions
C. A tool for fine-tuning LLMs
D. Using command-line tools to query LLMs

✓ Correct Answer: B

Explanation: Tool use (function calling) enables LLMs to decide when to call external functions (APIs, calculators, databases) and structure the parameters, enabling agentic real-world interactions.

Q460. What is 'hallucination mitigation' best practice in production LLM systems?

A. Using higher temperature settings
B. Combining RAG, grounding checks, output validation, and human review for high-stakes responses
C. Reducing context window size
D. Disabling model creativity features

✓ Correct Answer: B

Explanation: Production hallucination mitigation combines RAG (factual grounding), Bedrock Guardrails grounding checks, output validation, and human-in-the-loop review for critical decisions.

Q461. What is 'structured output' generation in LLMs?

A. Outputting model architecture diagrams
B. Constraining LLM outputs to produce valid JSON, XML, or other structured formats reliably
C. Structured training output logs
D. Outputs formatted for structured databases

✓ Correct Answer: B

Explanation: Structured output generation uses JSON mode, function calling, or constrained decoding to force LLMs to produce consistently formatted structured data for programmatic downstream processing.

Q462. What is 'Amazon Bedrock's support for multi-agent collaboration'?

A. Multiple AWS accounts using Bedrock together
B. Enabling multiple specialized AI agents to collaborate on complex tasks by delegating sub-tasks
C. Multi-agent reinforcement learning on Bedrock
D. Collaboration features between Bedrock and SageMaker

✓ Correct Answer: B

Explanation: Bedrock supports multi-agent architectures where a supervisor agent breaks complex tasks into sub-tasks and delegates to specialized sub-agents, enabling sophisticated workflows beyond single-agent capability.

Q463. What is 'chain-of-thought few-shot prompting'?

A. Chaining few models together
B. Providing step-by-step reasoning examples in the few-shot prompt to guide complex problem-solving
C. Chaining few-shot models in a pipeline

D. Combining chains of data with few-shot examples

✓ **Correct Answer: B**

Explanation: Chain-of-thought few-shot prompting includes examples showing intermediate reasoning steps, teaching the model to reason through complex problems step-by-step before giving a final answer.

Q464. What is 'Amazon Bedrock's RetrieveAndGenerate API'?

A. An API for retrieving Bedrock service documentation

B. A unified RAG API that retrieves from a Knowledge Base and generates a grounded response in one call

C. An API for retrieving training data and generating model checkpoints

D. An API combining model retrieval with fine-tuning

✓ **Correct Answer: B**

Explanation: The Bedrock RetrieveAndGenerate API performs the complete RAG workflow in a single call — retrieving relevant documents from a Knowledge Base and generating a grounded response with citations.

Q465. What are 'inference profiles' in Amazon Bedrock?

A. User profile pages for Bedrock users

B. Pre-configured sets of inference parameters that can be saved and reused across applications

C. AWS profiles for accessing Bedrock

D. Performance profiles benchmarking model speed

✓ **Correct Answer: B**

Explanation: Bedrock inference profiles (system-defined or customer-defined) package foundation model and inference settings for consistent reuse across applications, simplifying management.

Domain 3: AWS AI & ML Services

Q466. What is 'Amazon SageMaker's Clarify online explainability'?

A. Explanations delivered via an online portal

B. Real-time feature attribution for individual predictions at inference time on SageMaker endpoints

C. Online documentation for SageMaker Clarify

D. Explanations computed overnight in batch

✓ **Correct Answer: B**

Explanation: SageMaker Clarify online explainability computes SHAP-based feature attributions for individual predictions in real-time at inference, enabling per-prediction explanations in production applications.

Q467. What is 'Amazon SageMaker's Managed MLflow' capability?

A. Managing fluid flows in industrial ML

B. A fully managed MLflow server for tracking experiments, models, and deployments within SageMaker

C. Flow management for ML data pipelines

D. A managed service for ML financial flows

✓ **Correct Answer: B**

Explanation: SageMaker Managed MLflow provides a fully managed MLflow tracking server integrated with SageMaker, enabling experiment tracking and model registry without managing MLflow infrastructure.

Q468. What is 'Amazon SageMaker's Bring Your Own Container (BYOC)' feature?

A. Bringing physical containers to AWS data centers

B. Using custom Docker containers for training and inference when built-in containers don't meet requirements

C. A container rental service for SageMaker

D. Bringing containers from other cloud providers

✓ Correct Answer: B

Explanation: SageMaker BYOC lets you package any ML framework or algorithm into a Docker container and run it on SageMaker infrastructure, providing maximum flexibility beyond built-in container options.

Q469. What is 'Amazon Forecast's AutoPredictor'?

A. A self-predicting model

B. An automated feature selecting the best forecasting algorithm and ensembling models for optimal accuracy

C. A predictor that automatically retrains on schedule

D. An auto-scaling predictor for high traffic

✓ Correct Answer: B

Explanation: Amazon Forecast AutoPredictor automatically selects, combines, and ensembles the best algorithms for your specific time series data, typically outperforming any individual forecasting algorithm.

Q470. What is 'Amazon SageMaker's real-time inference auto-scaling'?

A. Scaling inference capacity manually

B. Automatically adjusting endpoint instances based on traffic to balance cost and performance

C. Auto-scaling model weights during inference

D. Automatically retraining models based on traffic patterns

✓ Correct Answer: B

Explanation: SageMaker endpoint auto-scaling uses Application Auto Scaling to automatically add or remove instances behind a real-time endpoint based on metrics like invocations per instance.

Q471. What is 'Amazon Rekognition's PPE detection' capability?

A. Processing pipeline errors detection

B. Detecting personal protective equipment (hard hats, masks, vests) in images for workplace safety

C. Privacy-preserving encryption detection

D. Detecting payment processing errors

✓ Correct Answer: B

Explanation: Rekognition PPE Detection identifies whether workers are wearing required personal protective equipment in images, automating workplace safety compliance monitoring.

Q472. What is 'Amazon Comprehend's events detection'?

A. Detecting AWS CloudTrail events

B. Extracting business events and their participants from news and financial documents

C. Detecting calendar events in text

D. Event-driven Comprehend API triggers

✓ **Correct Answer: B**

Explanation: Comprehend Events detection identifies specific business event types (mergers, IPOs, bankruptcies) along with their participants, dates, and attributes from financial and news text.

Q473. What is 'Amazon SageMaker Studio Lab'?

A. A physical ML lab at AWS offices

B. A free, no-AWS-account-needed environment for learning ML with Jupyter notebooks

C. A laboratory for testing SageMaker features

D. An advanced Studio tier for enterprise users

✓ **Correct Answer: B**

Explanation: SageMaker Studio Lab is a free service providing Jupyter notebook environments for ML learning requiring no AWS account or credit card, designed to lower the barrier to ML education.

Domain 4: Responsible AI

Q474. What is 'AI alignment' concerned with?

A. Aligning AI infrastructure in data centers

B. Ensuring AI systems pursue goals that align with human values and intentions

C. Aligning training data distributions

D. Aligning model outputs with business metrics

✓ **Correct Answer: B**

Explanation: AI alignment research focuses on ensuring AI systems behave in ways consistent with human values, intentions, and societal benefit, preventing unintended harmful objectives from emerging.

Q475. What is 'disparate treatment' in AI systems?

A. Treating different model versions differently

B. Using protected attributes explicitly in model decisions, causing differential treatment of groups

C. Treating different data types inconsistently

D. Differential pricing for AI services

✓ **Correct Answer: B**

Explanation: Disparate treatment occurs when a model explicitly uses protected attributes (race, gender, age) as inputs, directly causing different predictions for different demographic groups.

Q476. What is 'model robustness' in responsible AI?

A. A model that never changes after training

B. A model that maintains reliable performance under distributional shifts, noisy inputs, and adversarial attacks

C. Model stability during training convergence

D. Robust hardware for model inference

✓ **Correct Answer: B**

Explanation: Model robustness refers to a model's ability to perform reliably when inputs vary due to noise, distribution shifts, or adversarial perturbations — a key safety property for production AI.

Q477. What is 'societal impact assessment' for AI?

A. Assessing how AI affects social media platforms

B. Evaluating broader societal effects including impacts on employment, equity, and communities

C. A social media impact tracking tool

D. Measuring user satisfaction with AI products

✓ Correct Answer: B

Explanation: Societal impact assessment examines how AI deployment affects society broadly — job displacement, inequality, community impacts, and long-term systemic effects beyond direct users.

Q478. What is 'value alignment' in generative AI specifically?

A. Aligning model values with business KPIs

B. Ensuring LLM outputs reflect human values such as honesty, helpfulness, and avoiding harm

C. Aligning embedding values across models

D. Value alignment in financial pricing models

✓ Correct Answer: B

Explanation: Value alignment in generative AI ensures models produce outputs reflecting human values — being truthful, helpful, and respectful, avoiding harmful or deceptive content — through RLHF and Constitutional AI.

Q479. What is 'stakeholder engagement' in responsible AI development?

A. Engaging with AWS account holders

B. Involving affected communities, domain experts, and ethicists in AI design to surface concerns early

C. Engaging shareholders in AI investment decisions

D. Stakeholder meetings for sprint planning

✓ Correct Answer: B

Explanation: Responsible AI development involves engaging diverse stakeholders — communities affected, domain experts, ethicists, and regulators — to identify risks, biases, and concerns before deployment.

Q480. What is 'AI literacy' and why is it important?

A. The ability to read AI-generated content

B. Understanding AI capabilities, limitations, and societal impacts to make informed decisions about AI use

C. Literacy programs delivered by AI tutors

D. The number of AI-related books available

✓ Correct Answer: B

Explanation: AI literacy enables individuals and organizations to understand what AI can and cannot do, recognize its risks, and make informed decisions — essential for responsible adoption and governance.

Domain 5: Security, Compliance & Governance

Q481. What is 'zero trust security' applied to ML systems?

A. A security model trusting zero users

B. Never trusting any entity by default; always verifying identity and minimizing access for all ML resources

C. Trusting zero-cost security solutions

D. A security model with zero configuration

✓ Correct Answer: B

Explanation: Zero trust applies to ML by verifying every access request to SageMaker, Bedrock, and training data regardless of network location, using strong authentication and least-privilege access everywhere.

Q482. What is 'Amazon CloudWatch' used for in ML monitoring?

A. Watching cloud weather conditions

B. Monitoring ML training metrics, endpoint performance, and triggering alarms for anomalies

C. Watching cloud costs only

D. Monitoring developer activity in the console

✓ Correct Answer: B

Explanation: CloudWatch collects metrics from SageMaker training jobs, endpoints, and Bedrock API calls, enabling dashboards, alarms (e.g., endpoint latency spikes), and log aggregation for ML operations.

Q483. What is 'data encryption key rotation' best practice for AI?

A. Rotating training data periodically

B. Periodically replacing encryption keys to limit risk if a key is compromised

C. Rotating model weights for better performance

D. Rotating model deployment keys annually

✓ Correct Answer: B

Explanation: Regular KMS key rotation limits exposure — if a key is compromised only data encrypted with that key version is at risk. AWS KMS supports automatic annual rotation for customer-managed keys.

Q484. What is 'secrets management' in ML workflows?

A. Managing model secrets (weights) securely

B. Securely storing and rotating API keys, database credentials, and tokens used by ML pipelines

C. Secret governance meetings for AI strategy

D. Managing private model architectures

✓ Correct Answer: B

Explanation: AWS Secrets Manager securely stores and rotates credentials (database passwords, API keys) needed by ML training jobs and inference pipelines, preventing hardcoded credentials in code.

Q485. What is 'Amazon Bedrock's private deployment using VPC endpoints'?

A. Deploying models on private hardware

B. Accessing Bedrock APIs entirely within a VPC using interface endpoints without public internet exposure

C. A private beta deployment program

D. Deploying models in a private AWS account only

✓ Correct Answer: B

Explanation: Amazon Bedrock supports AWS PrivateLink interface endpoints allowing your VPC to access Bedrock APIs without traffic traversing the public internet, meeting strict data isolation requirements.

Q486. What is 'compliance automation' for AI workloads?

A. Using AI to automate compliance tasks

B. Automatically enforcing and verifying compliance rules using AWS Config, SCPs, and CI/CD checks
C. Automating compliance report generation only
D. Automatic pricing compliance for AI services

✓ Correct Answer: B

Explanation: Compliance automation uses AWS Config Rules, Service Control Policies, and pipeline checks to automatically detect and remediate non-compliant AI infrastructure configurations in real time.

Q487. What is 'AWS Audit Manager' used for?

A. Managing AWS billing audits
B. Continuously collecting evidence to simplify compliance audits for AI and cloud workloads
C. Auditing ML model performance
D. Managing security patch audit cycles

✓ Correct Answer: B

Explanation: AWS Audit Manager continuously collects evidence of AWS resource configurations and API activity, automating the preparation of audit-ready reports for compliance frameworks like SOC 2 and HIPAA.

Q488. What is 'Amazon SageMaker's Model Approval workflow'?

A. AWS legal approval for model deployment
B. A governance workflow requiring human sign-off before a model version can be deployed to production
C. Automatic model approval based on accuracy thresholds
D. An approval process for model architecture changes

✓ Correct Answer: B

Explanation: SageMaker Model Registry's approval workflow requires designated approvers to review model metrics and documentation before a model version is approved for production deployment, ensuring governance.

Q489. What is 'Amazon SageMaker's compliance with HIPAA'?

A. A medical diagnosis feature in SageMaker
B. SageMaker's eligibility to be used for workloads processing protected health information under HIPAA
C. Healthcare AI compliance enforcement
D. A compliance program for medical ML researchers

✓ Correct Answer: B

Explanation: Amazon SageMaker is HIPAA-eligible, meaning healthcare organizations can sign a Business Associate Agreement with AWS and use SageMaker for ML workloads involving protected health information.

Q490. What is 'Amazon Bedrock's FIPS 140-2 compliant endpoints' relevance?

A. A fast IP failover system
B. Endpoints using FIPS-validated cryptography required for US federal government AI workloads
C. A financial IP protection standard
D. A framework for inference performance standards

✓ Correct Answer: B

Explanation: Amazon Bedrock offers FIPS 140-2 compliant endpoints using validated cryptographic modules, meeting the requirements of US federal government agencies and regulated industries for AI workloads.

Q491. What is 'responsible disclosure' in AI security?

A. Disclosing AI capabilities in marketing materials
B. Reporting discovered AI vulnerabilities to the vendor privately before public disclosure to allow fixes
C. Disclosing model training data sources publicly
D. Publicly disclosing all AI security incidents immediately

✓ Correct Answer: B

Explanation: Responsible disclosure is the security practice of privately reporting AI system vulnerabilities to the vendor with a reasonable fix timeline before any public announcement, protecting users while enabling remediation.

Domain 1: AI & ML Fundamentals

Q492. What is a ROC curve used to evaluate?

A. Model training speed
B. A classifier ability to separate classes by plotting true positive rate vs false positive rate across thresholds
C. Regression model accuracy
D. Feature importance ranking

✓ Correct Answer: B

Explanation: The ROC curve plots True Positive Rate vs False Positive Rate at every classification threshold. The AUC score summarises overall discriminative power, with 1.0 being perfect and 0.5 being random.

Q493. What does Mean Squared Error (MSE) penalise more heavily than Mean Absolute Error (MAE)?

A. Small prediction errors
B. Large prediction errors, because errors are squared before averaging
C. Missing values in predictions
D. Categorical mislabelling

✓ Correct Answer: B

Explanation: MSE squares each error before averaging, giving disproportionately higher weight to large mistakes. This makes MSE preferable when large errors are especially costly to the business.

Domain 2: Generative AI & Foundation Models

Q494. What is the lost-in-the-middle problem for LLMs?

A. Training data randomly deleted mid-run
B. Models tend to recall content from the start and end of a long context more reliably than from the middle
C. Gradient values disappearing in mid-network layers
D. Tokens being dropped during tokenisation

✓ Correct Answer: B

Explanation: Research demonstrates that LLMs retrieve information placed in the middle of a long context window less reliably than information at the edges, which influences how documents should be ordered in RAG pipelines.

Domain 3: AWS AI & ML Services

Q495. What framework containers does Amazon SageMaker provide out of the box for deep learning?

A. Only its own proprietary containers

B. AWS Deep Learning Containers with optimised builds of PyTorch and TensorFlow for training and inference

C. TensorFlow containers only

D. Third-party containers sourced from the AWS Marketplace

✓ Correct Answer: B

Explanation: SageMaker ships AWS Deep Learning Containers, which are pre-built and security-hardened Docker images for PyTorch and TensorFlow. They are optimised for AWS GPU and Inferentia hardware.

Q496. What key improvement did Amazon Lex V2 introduce compared to V1?

A. Voice-only interaction support

B. Support for multiple languages within a single bot, a streaming API, and an improved visual intent editor

C. Exclusive support for mobile channels

D. Rule-based responses replacing ML intents

✓ Correct Answer: B

Explanation: Lex V2 added per-bot multi-language support, a streaming conversational API for lower latency, a redesigned intent management console, and simplified versioning compared to Lex V1.

Domain 4: Responsible AI

Q497. What does intersectionality mean in an AI fairness context?

A. The point where two ML algorithms intersect on a performance chart

B. Examining whether bias emerges at the intersection of multiple protected attributes such as race combined with gender

C. Combining several training datasets at their overlapping records

D. The overlap between supervised and unsupervised learning techniques

✓ Correct Answer: B

Explanation: Intersectional fairness analysis checks that a model does not produce disparate outcomes for people who belong to multiple protected groups simultaneously, capturing harms that per-attribute analysis alone can miss.

Q498. What should a responsible model deprecation plan include?

A. Immediate deletion of model artefacts with no notice

B. Advance stakeholder notice, a migration path to a replacement model, impact assessment, and documentation of the retirement rationale

C. Keeping the old model running indefinitely as a fallback

D. Simply removing API access without further action

✓ Correct Answer: B

Explanation: Responsible deprecation gives downstream teams time to migrate, documents why the model is being retired, assesses impact on users and systems, and ensures continuity through a replacement or migration guide.

Domain 5: Security, Compliance & Governance

Q499. What security information do Amazon VPC Flow Logs capture for ML workloads?

A. Model prediction payloads in full

B. Network traffic metadata including source and destination IPs, ports, and accept or reject status for flows to and from ML resources

C. Training data transferred between S3 and SageMaker

D. Bedrock API request and response bodies

✓ Correct Answer: B

Explanation: VPC Flow Logs record metadata about IP traffic flowing to and from network interfaces. Security teams use them to detect unexpected outbound connections, potential data exfiltration, or unusual access patterns around ML infrastructure.

Q500. How do SageMaker IAM condition keys strengthen least-privilege access control?

A. They encrypt SageMaker data at rest

B. They let administrators write policies that restrict actions based on attributes like resource tags, instance type, or VPC configuration

C. They generate temporary credentials for notebook users

D. They activate optional SageMaker features conditionally

✓ Correct Answer: B

Explanation: IAM condition keys for SageMaker allow fine-grained rules such as requiring a specific VPC, enforcing encrypted storage volumes, or limiting training to approved instance families, tightening least-privilege beyond simple Allow or Deny statements.

Domain 1: AI & ML Fundamentals

Q501. What is 'feature scaling' and why is it important?

A. Scaling the number of features in a dataset

B. Normalising feature magnitudes so algorithms like k-NN and gradient descent are not dominated by large-valued features

C. Scaling compute resources during feature extraction

D. Increasing the feature count to improve model accuracy

✓ Correct Answer: B

Explanation: Distance-based and gradient-based algorithms are sensitive to feature scale. Normalising features prevents large-valued ones from dominating, leading to faster convergence and better model performance.

Q502. What is 'min-max normalisation'?

A. Subtracting the mean and dividing by standard deviation

B. Rescaling each feature to a fixed range, typically 0 to 1, using the observed minimum and maximum

C. Removing minimum and maximum outliers before training

D. A normalisation technique only applicable to image pixel values

✓ Correct Answer: B

Explanation: Min-max normalisation maps values to [0,1] using (x - min)/(max - min), preserving distribution shape while ensuring all features are on the same scale.

Q503. What is 'Z-score standardisation'?

A. Rescaling all values to the range 0 to 1

B. Subtracting the mean and dividing by standard deviation so each feature has zero mean and unit variance

C. A standardised way to split data into train and test sets

D. Normalising output labels rather than input features

✓ Correct Answer: B

Explanation: Z-score standardisation produces features with mean 0 and standard deviation 1. It is preferred when outliers are present because it is less sensitive to extreme values than min-max scaling.

Q504. What is a 'sparse feature vector' in ML?

A. A vector with very few total dimensions

B. A high-dimensional vector where most values are zero, common in text and recommendation system data

C. A vector stored in a sparse cloud storage tier

D. A compressed feature vector produced after dimensionality reduction

✓ Correct Answer: B

Explanation: Sparse vectors arise in bag-of-words NLP representations and user-item recommendation matrices where most entries are zero because users interact with only a tiny fraction of all items.

Q505. What is 'TF-IDF' in natural language processing?

A. A deep learning technique for text generation

B. A term-weighting scheme that increases weight for rare informative words and decreases weight for common words across a corpus

C. A transformer-based feature extraction method

D. A text formatting and document indexing standard

✓ Correct Answer: B

Explanation: TF-IDF (Term Frequency-Inverse Document Frequency) highlights words that are frequent in a specific document but rare across the corpus, making them more discriminative for classification and retrieval.

Q506. What is 'bag-of-words' text representation?

A. A model that stores words in a Python bag (set) data structure

B. Representing a document as word occurrence counts while ignoring word order

C. A language model generating words one bag at a time

D. A document clustering method using word similarity

✓ Correct Answer: B

Explanation: Bag-of-words creates a fixed-length vector of word counts per document, discarding order. It is simple and effective for tasks like spam detection, topic classification, and sentiment analysis.

Q507. What is 'beam search' in sequence generation models?

A. A linear scan through model weight space

B. A decoding strategy keeping the k most probable partial sequences at each step to find a higher quality overall output

C. A tree search algorithm used in reinforcement learning

D. A search algorithm for hyperparameter optimisation

✓ Correct Answer: B

Explanation: Beam search maintains k candidate sequences simultaneously, expanding and pruning at each step to balance quality and computational cost, outperforming greedy decoding for most sequence tasks.

Q508. What is 'perplexity' as a language model metric?

A. A measure of how confused human annotators are by model outputs

B. The exponentiated average negative log-likelihood per token — lower perplexity means the model better predicts held-out text

C. The total number of parameters in a language model

D. A metric for evaluating image generation fidelity

✓ Correct Answer: B

Explanation: Perplexity quantifies how surprised a language model is by held-out text. Lower values indicate the model assigns higher probability to the test corpus, reflecting better language understanding.

Q509. What is a 'false positive' in a confusion matrix?

A. A positive sample the model correctly identified

B. A negative sample the model incorrectly predicted as positive, also called a Type I error

C. A positive prediction that matched the true label

D. A negative prediction that matched the true label

✓ Correct Answer: B

Explanation: A false positive (FP) occurs when the true label is negative but the model predicts positive. In spam detection this means classifying a legitimate email as spam.

Q510. What is 'k-nearest neighbours' (k-NN) classification?

A. A neural network with k hidden layers

B. A non-parametric algorithm classifying a new point based on majority vote among its k closest training examples

C. A clustering algorithm finding k centroids

D. A feature selection method keeping the top k features

✓ **Correct Answer: B**

Explanation: k-NN stores all training examples and classifies new points by majority vote among the k nearest neighbours by distance. It requires no explicit training but inference scales with dataset size.

Q511. What is 'support vector machine' (SVM) key property?

A. Provides computational GPU support for deep learning

B. Finds the maximum-margin hyperplane that best separates classes, maximising generalisation to unseen data

C. A reinforcement learning algorithm using support rewards

D. An unsupervised clustering method based on support vectors

✓ **Correct Answer: B**

Explanation: SVM maximises the margin between classes by finding the decision boundary whose distance to the nearest points from each class (support vectors) is as large as possible.

Q512. What is 'early stopping' in neural network training?

A. Stopping training after a fixed predefined number of epochs

B. Halting training when validation loss stops improving, preventing overfitting without manual epoch tuning

C. Stopping purely to reduce cloud compute cost

D. Preventing gradient flow in the earliest network layers

✓ **Correct Answer: B**

Explanation: Early stopping monitors validation loss and terminates training when it plateaus within a patience window, acting as a regulariser that prevents the model from memorising training examples.

Q513. What is 'batch normalisation' in deep learning?

A. Setting an optimal mini-batch size before training

B. Normalising layer activations within each mini-batch to stabilise training, enabling higher learning rates and faster convergence

C. Normalising the loss function value

D. Normalising gradient magnitudes during backpropagation

✓ **Correct Answer: B**

Explanation: Batch normalisation standardises inputs to each layer within mini-batches, reducing internal covariate shift and making training more stable, faster, and less sensitive to weight initialisation.

Q514. What is 'multi-head attention' in Transformer models?

A. Attention computed by multiple independent models that vote on output

B. Running several attention computations in parallel on different linear projections then concatenating results to jointly attend to multiple aspects

C. Multiple sequential attention layers stacked one after another

D. Attention that processes multiple separate input datasets in parallel

✓ **Correct Answer: B**

Explanation: Multi-head attention projects queries, keys, and values into h subspaces, computes attention independently in each, then concatenates, allowing the model to focus on different relationship types simultaneously.

Q515. What is 'positional encoding' in Transformers?

A. Encoding the AWS region where the model is deployed

B. Adding sequence position information to token embeddings since Transformers process all tokens in parallel with no built-in order

C. Encoding token positions for relational database storage

D. A positional argument passed in the model API request

✓ Correct Answer: B

Explanation: Without recurrence or convolution, Transformers need positional encodings to distinguish token order. These are added to embeddings before the first layer so the model can leverage sequence position.

Q516. What is a 'residual connection' (skip connection) used for?

A. Carrying residual prediction errors back to the input

B. Adding a layer's input directly to its output so gradients can flow through very deep networks unimpeded

C. A connection for residual compute capacity overflow

D. A shortcut that bypasses the activation function in a layer

✓ Correct Answer: B

Explanation: Residual connections (introduced by ResNet) allow gradients to bypass blocks of layers during backpropagation, solving the vanishing gradient problem and enabling successful training of hundreds of layers.

Q517. What is 'model pruning' used for?

A. Removing outlier samples from training data before fitting

B. Removing low-importance weights or neurons to create a smaller, faster model with minimal accuracy loss

C. Pruning the hyperparameter search space before tuning

D. Removing redundant features from the input feature set

✓ Correct Answer: B

Explanation: Pruning eliminates parameters whose removal has the smallest impact on model output, reducing memory footprint and inference latency while retaining most performance — useful for edge deployment.

Q518. What is 'curriculum learning' in model training?

A. Following a formal AWS course curriculum to learn ML

B. Presenting easy examples first and progressively harder ones, mimicking how humans learn to improve convergence

C. A scheduled learning rate curriculum during training

D. Organising training data into batches following a fixed curriculum

✓ Correct Answer: B

Explanation: Curriculum learning structures the training order from simple to complex examples. This can lead to faster convergence and better final performance compared to randomly shuffled presentation.

Q519. What is 'out-of-distribution' (OOD) detection in deployed models?

A. Detecting data stored outside the primary storage distribution tier

B. Identifying test-time inputs that differ significantly from training data where model predictions may be unreliable

C. Detecting outliers in model weight value distributions

D. Flagging predictions that fall outside acceptable business value ranges

✓ **Correct Answer: B**

Explanation: OOD detection allows a deployed model to recognise when inputs are unlike anything it was trained on, enabling it to abstain, flag for human review, or return a calibrated low-confidence response.

Q520. What is 'neural architecture search' (NAS)?

A. Manually searching research papers for useful neural architectures

B. Using automated search algorithms or reinforcement learning to discover optimal neural network architectures for a task

C. Searching a registry for the best pre-trained model to deploy

D. Searching model documentation for architecture diagrams

✓ **Correct Answer: B**

Explanation: NAS automates the design of neural network architectures (layer types, connections, widths), finding configurations that outperform hand-designed ones while targeting specific hardware constraints.

Q521. What is a 'proxy metric' risk in ML optimisation?

A. A backup metric computed only during proxy model evaluation

B. A measurable surrogate for the true goal that can diverge from it once optimised, leading to unintended model behaviour

C. A secondary metric recorded only at inference time

D. A metric computed through a third-party benchmarking API

✓ **Correct Answer: B**

Explanation: Goodhart's Law warns that once a measure becomes a target it ceases to be a good measure. Optimising a proxy (e.g., engagement rate) can produce models that game the metric rather than serving the true goal.

Q522. What is 'imitation learning' used for?

A. Training one model architecture to imitate another model's architecture

B. Learning a policy by imitating expert demonstrations rather than defining and optimising a reward function

C. Generating synthetic training data that imitates real data statistics

D. Knowledge distillation where a student model imitates a teacher

✓ **Correct Answer: B**

Explanation: Imitation learning is practical when reward functions are hard to define. For example, autonomous driving agents can be trained by imitating logged human driving trajectories rather than engineering reward signals.

Q523. What is 'conformal prediction' used for?

A. Ensuring model predictions conform to predefined business rules

B. Producing prediction sets or intervals with a guaranteed probability of containing the true label, providing rigorous uncertainty quantification

C. Conforming model output formats to a standard schema

D. Verifying that model outputs conform to compliance regulations

✓ **Correct Answer: B**

Explanation: Conformal prediction wraps any trained model and produces prediction sets whose coverage is guaranteed to contain the true label at a user-specified probability (e.g., 90%), regardless of model type or data distribution.

Q524. What is 'model compression ratio' in ML?

A. The ratio of training compute to inference compute

B. How much smaller a compressed model is relative to the original, combining savings from pruning, quantisation, or distillation

C. Data compression applied to training inputs before ingestion

D. The ratio of model accuracy before and after compression

✓ Correct Answer: B

Explanation: Compression ratio summarises size reduction efficiency. A model with 100M original parameters compressed to 10M has a 10x compression ratio, greatly reducing memory and inference cost.

Q525. What is 'layer normalisation' and when is it preferred over batch normalisation?

A. Normalising the total number of layers in a network

B. Normalising activations across the feature dimension for each individual sample, preferred for sequences and Transformers where batch statistics are unstable

C. A technique used only for image models

D. Normalising weight matrices between adjacent layers

✓ Correct Answer: B

Explanation: Layer normalisation computes statistics over features for each sample independently, making it unaffected by batch size — essential for variable-length sequences in Transformers and recurrent networks.

Domain 2: Generative AI & Foundation Models

Q526. What is a 'prompt template' in LLM applications?

A. A CloudFormation template for LLM infrastructure provisioning

B. A reusable text skeleton with placeholder variables filled at runtime before sending to the model

C. A template defining the model's neural network architecture

D. A prompt stored in S3 and loaded for batch processing

✓ Correct Answer: B

Explanation: Prompt templates separate static structure from dynamic runtime values (user query, retrieved context, user name), enabling consistent and maintainable prompts across an application's multiple call sites.

Q527. What is 'parent-child chunking' in Bedrock Knowledge Bases?

A. An account hierarchy within Amazon Bedrock

B. Indexing small child chunks for precise retrieval while supplying larger parent context to the LLM, balancing precision with completeness

C. Using a parent foundation model to oversee a child model during fine-tuning

D. A chunking strategy exclusive to hierarchical book-length documents

✓ Correct Answer: B

Explanation: Parent-child chunking retrieves precise small child chunks by similarity, then expands to the surrounding parent chunk before generation, so the LLM has sufficient context to produce a complete answer.

Q528. What is 'metadata filtering' in Bedrock Knowledge Bases retrieval?

A. Stripping metadata from model API responses

B. Pre-filtering the vector store by document attributes such as date, category, or department before semantic similarity search

C. Filtering metadata fields out of fine-tuning training data

D. Removing embedding metadata to reduce storage cost

✓ Correct Answer: B

Explanation: Metadata filtering narrows the search space before vector similarity comparison, improving retrieval precision — for example, limiting retrieval to policy documents published after a specific date.

Q529. What is 'semantic reranking' in Amazon Bedrock?

A. Re-ranking foundation model leaderboard scores

B. Applying a cross-encoder model as a second stage to re-order initially retrieved chunks by semantic relevance before LLM generation

C. Re-prioritising user queries before retrieval

D. Sorting API responses by response time latency

✓ Correct Answer: B

Explanation: A cross-encoder reranker scores each (query, chunk) pair jointly rather than independently, capturing finer-grained relevance signals that first-stage approximate nearest-neighbour retrieval misses.

Q530. What is 'Amazon Bedrock Data Automation'?

A. Automated cost management for Bedrock storage

B. Using foundation models to extract structured data from unstructured documents, images, audio, and video at scale

C. Automated CI/CD pipelines for Bedrock model deployment

D. Synthetic training data generation for Bedrock fine-tuning

✓ Correct Answer: B

Explanation: Bedrock Data Automation intelligently processes unstructured content using foundation models, outputting structured data (JSON, tables, key-value pairs) without requiring custom extraction model development.

Q531. What is 'Amazon Nova Canvas'?

A. A visual workflow design canvas for Bedrock agents

B. Amazon's image generation and editing foundation model in the Nova family available on Bedrock

C. A prompt design canvas used in Bedrock Playground

D. A no-code interface for deploying Bedrock agents

✓ Correct Answer: B

Explanation: Nova Canvas generates and edits images from text descriptions, supporting background removal, inpainting, style transfer, and image conditioning within the Amazon Bedrock Nova model family.

Q532. What is 'Amazon Nova Reel'?

A. A showcase reel feature for Bedrock capabilities

B. Amazon's text-to-video and image-to-video generation foundation model in the Nova family on Bedrock

C. A short-form content generator for social media platforms

D. An audio-to-video synchronisation model

✓ Correct Answer: B

Explanation: Nova Reel is Amazon's video generation foundation model on Bedrock, producing short video clips from text prompts or seed images with smooth motion and cinematic quality.

Q533. What is 'Amazon Nova Sonic'?

A. An audio effects generation and mixing model

B. Amazon's bidirectional speech model enabling real-time low-latency conversational voice AI on Bedrock

C. A sonic branding and jingle creation tool

D. A sensor data transcription model

✓ Correct Answer: B

Explanation: Nova Sonic supports end-to-end speech-to-speech conversations with natural prosody, interruption handling, and low latency, enabling builders to create real-time voice AI applications on Bedrock.

Q534. What is 'grounded generation' and why does it matter?

A. Generation using only ground truth training labels

B. Producing responses with citations to specific source passages making each claim verifiable rather than potentially hallucinated

C. Setting temperature to zero to ground outputs in determinism

D. Grounding LLM inference to a geographically proximate AWS region

✓ Correct Answer: B

Explanation: Grounded generation anchors each response claim to a retrieved source passage, enabling users to verify statements and reducing the risk of acting on hallucinated information in high-stakes applications.

Q535. What is 'Amazon Bedrock session management' for agents?

A. AWS session token management for Bedrock console access

B. Persisting conversation context across turns using session IDs so the agent can reference earlier exchanges in follow-up questions

C. Session-level billing granularity for Bedrock API calls

D. Timeout management for long-running model inference tasks

✓ Correct Answer: B

Explanation: Bedrock Agents sessions maintain a memory buffer of prior turns identified by a session ID, allowing natural multi-turn conversations where the agent references context from earlier in the session.

Q536. What is 'custom orchestration' for Amazon Bedrock Agents?

A. Orchestrating workloads across multiple AWS accounts

B. Replacing the built-in ReAct reasoning loop with a custom Lambda function implementing your own agent orchestration logic

C. Scheduling custom Bedrock inference jobs at specific times

D. Orchestrating multiple fine-tuning jobs concurrently

✓ Correct Answer: B

Explanation: Custom orchestration lets developers implement proprietary planning, tool-calling sequences, and memory management in a Lambda function rather than relying on Bedrock's default ReAct orchestrator.

Q537. How does fine-tuning differ from continued pre-training in Bedrock?

A. The two terms are completely interchangeable

B. Fine-tuning uses labelled prompt-completion pairs to teach new behaviours; continued pre-training uses unlabelled domain text to deepen knowledge

C. Continued pre-training requires more labelled examples than fine-tuning

D. Fine-tuning modifies the tokeniser while continued pre-training only adjusts transformer weights

✓ Correct Answer: B

Explanation: Supervised fine-tuning (SFT) shapes model behaviour with labelled examples. Continued pre-training adapts the model's knowledge base from raw domain text, making them complementary and often used together.

Q538. What is 'Mistral Mixtral' architecture advantage?

A. A mixture of multiple Mistral API versions behind a router

B. Sparse mixture-of-experts activating only a subset of parameters per token, achieving high performance at lower inference cost

C. A generative music model from Mistral AI

D. A multi-task transfer learning framework from Mistral

✓ Correct Answer: B

Explanation: Mixtral's MoE design activates 2 of 8 expert networks per token, matching the quality of much larger dense models while using the compute of a smaller model during inference.

Q539. What is 'Cohere Command R+' optimised for?

A. Large-scale image generation and editing tasks

B. Enterprise RAG and agentic workflows with 128K context, multilingual support, and built-in multi-step tool use

C. Real-time speech transcription and voice synthesis

D. Competitive code generation and security vulnerability scanning

✓ Correct Answer: B

Explanation: Command R+ is Cohere's model purpose-engineered for retrieval-augmented generation and agentic workflows, excelling in long-document analysis, multi-hop reasoning, and tool-use across many languages.

Q540. What is 'retrieval precision vs recall' in RAG evaluation?

A. They are the same metric expressed differently

B. Precision measures the fraction of retrieved chunks that are relevant; recall measures the fraction of relevant chunks that were retrieved

C. Precision scores the LLM answer; recall scores the retrieval index quality

D. Both metrics measure only the final end-to-end answer accuracy

✓ Correct Answer: B

Explanation: High retrieval precision means few irrelevant chunks clutter the LLM context. High recall ensures relevant information is not missed. Both affect final answer quality and should be optimised together.

Q541. What is 'Bedrock Guardrails word filter'?

A. A filter removing AWS service terms from model responses

B. A configurable blocklist of specific words and phrases that Bedrock detects and blocks in both prompts and model responses

C. An autocorrect filter fixing spelling in model outputs

D. A frequency-based filter removing common words from retrieved documents

✓ Correct Answer: B

Explanation: Word filter lets organisations define specific terms they never want to appear in conversations — competitor names, profanity, regulatory-sensitive phrases — enforced consistently across all model interactions.

Q542. What is 'Bedrock Playground compare mode'?

A. A gaming leaderboard comparing foundation model benchmark scores

B. Sending the same prompt to multiple foundation models simultaneously to view and compare responses side-by-side before choosing one

C. Comparing Bedrock API costs across different AWS regions

D. A side-by-side comparison of fine-tuned versus base model outputs

✓ Correct Answer: B

Explanation: Compare mode in Bedrock Playground accelerates model selection by letting you test a representative prompt against multiple models at once, visually comparing quality, style, and accuracy before committing.

Q543. What is 'model merging' for foundation models?

A. Merging billing from two separate Bedrock AWS accounts

B. Arithmetically combining weights from separately trained models to create one model inheriting capabilities from both parents

C. Batching multiple inference calls into a merged single request

D. Consolidating multiple model versions into a single registry entry

✓ Correct Answer: B

Explanation: Weight merging techniques like SLERP and DARE-TIES linearly combine or interpolate weights from models fine-tuned for different tasks, producing a blended model with mixed capabilities without retraining.

Q544. What is 'speculative decoding' for LLM inference?

A. Generating more speculative or creative outputs using high temperature

B. Using a small fast draft model to propose k tokens then batch-verifying them with the large model, significantly accelerating throughput

C. Reducing hallucinations by having a second model verify outputs

D. Estimating token cost before committing to a full generation

✓ Correct Answer: B

Explanation: Speculative decoding achieves 2-3x speedup: the draft model generates a token sequence cheaply, then the oracle model verifies all k tokens in one forward pass, accepting correct ones and restarting from the first error.

Q545. What is 'AI21 Jamba' on Amazon Bedrock?

A. A jazz music composition model

B. AI21 Labs hybrid SSM-Transformer model offering efficient long-context processing with lower memory than pure Transformer architectures

C. An AWS-native model optimised for JVM-based Java workloads

D. A model specialised in structured JSON output generation

✓ Correct Answer: B

Explanation: Jamba blends Mamba state-space model layers with standard Transformer attention, processing long contexts more memory-efficiently than pure Transformer models while maintaining strong task performance.

Q546. What is 'Amazon Bedrock cost per token' pricing consideration?

A. A flat monthly subscription fee per model regardless of usage

B. Input and output tokens are priced separately per thousand tokens, output typically costing more, making prompt design and max_tokens management important

C. Tokens are only charged for custom fine-tuned models not base models

D. Cost is based on inference duration in seconds not token count

✓ Correct Answer: B

Explanation: Understanding token pricing is critical for cost governance. Output tokens typically cost more than input tokens. Concise prompts and appropriate max_tokens limits directly control per-call costs.

Domain 3: AWS AI & ML Services

Q547. What is 'Amazon SageMaker Unified Studio'?

A. A rebrand of the legacy SageMaker Classic notebook experience

B. A unified workspace integrating data engineering, SQL analytics, ML development, and generative AI with centralised governance

C. A video studio for creating AI product demonstration content

D. A consolidated billing dashboard covering all SageMaker services

✓ Correct Answer: B

Explanation: SageMaker Unified Studio (formerly incorporating DataZone) brings together data cataloguing, SQL querying, feature engineering, model training, and generative AI application development in a single governed environment.

Q548. What is 'AWS HealthScribe'?

A. A health monitoring service for AWS cloud infrastructure

B. A service using ASR and generative AI to automatically generate structured clinical notes from patient-clinician conversations

C. A prescription and pharmacy data management service

D. A healthcare data migration tool for moving records to AWS

✓ Correct Answer: B

Explanation: AWS HealthScribe transcribes patient-clinician conversations and uses generative AI to extract SOAP notes, referral summaries, and structured data, reducing administrative burden on healthcare professionals.

Q549. What is 'Amazon Bedrock integration with Aurora PostgreSQL pgvector'?

A. Aurora hosting Bedrock API endpoints on PostgreSQL

B. Using Aurora PostgreSQL with the pgvector extension as a managed relational vector store for Bedrock Knowledge Bases

C. Aurora automatically backing up Bedrock model artefacts

D. Aurora providing GPU compute for Bedrock model inference

✓ Correct Answer: B

Explanation: pgvector in Aurora PostgreSQL provides ACID-compliant vector storage alongside relational data, making it a natural choice for teams wanting to consolidate structured metadata and embeddings in one managed database.

Q550. What is 'OpenSearch Serverless vector engine'?

A. A serverless keyword-only search platform

B. Auto-scaling serverless vector storage and k-NN search for embeddings with no cluster management required

C. A serverless dashboard for the OpenSearch console

D. A serverless inference engine for LLMs hosted on OpenSearch hardware

✓ Correct Answer: B

Explanation: OpenSearch Serverless vector engine spins up capacity on demand for embedding storage and similarity search, eliminating the need to pre-provision or manage OpenSearch cluster resources for RAG applications.

Q551. What is 'Amazon DataZone' and its ML governance role?

A. A DNS zone service for ML application APIs

B. A data governance service for cataloguing, discovering, sharing, and governing data across an organisation for analytics and ML workloads

C. A data centre physical zone management tool

D. A data partitioning service for SageMaker training input channels

✓ Correct Answer: B

Explanation: DataZone provides business-friendly data cataloguing with fine-grained access policies, subscription workflows, and data lineage, enabling ML teams to find and access governed training data without manual data sharing.

Q552. What is a 'Personalize recipe' in Amazon Personalize?

A. A cooking and food recommendation template

B. A pre-defined algorithm configuration aligned to a recommendation use case such as user personalisation, similar items, or re-ranking

C. A billing alert configuration template for Personalize

D. A data preprocessing workflow before Personalize model training

✓ Correct Answer: B

Explanation: Personalize recipes abstract algorithm selection: USER_PERSONALIZATION generates ranked item lists for a user, RELATED_ITEMS finds similar items, and PERSONALIZED_RANKING reorders a candidate set by user preference.

Q553. What is 'Amazon Comprehend targeted sentiment'?

A. Sentiment analysis aimed at specific AWS service reviews

B. Identifying the sentiment expressed toward each specific entity in a text rather than a single document-level score

C. Sentiment tracking for targeted marketing campaigns

D. Sentiment analysis restricted to social media and Twitter content

✓ Correct Answer: B

Explanation: Targeted sentiment associates a sentiment label with each mentioned entity, returning the fine-grained opinion expressed about staff, food, location, or other named targets within the same document.

Q554. What is 'Amazon Transcribe custom vocabulary'?

A. A vocabulary quiz built into Transcribe

B. A user-supplied list of domain-specific terms with phonetic hints that improves transcription accuracy for specialised language

C. A custom language model the user trains on their data

D. A vocabulary size setting controlling transcription job output length

✓ Correct Answer: B

Explanation: Custom vocabulary tables let customers supply medical terms, product names, unusual proper nouns, and technical acronyms with display forms and phonetic pronunciations, reducing transcription errors for domain language.

Q555. What is 'Amazon Polly SSML support'?

A. A serverless scripting markup language for Polly workflow automation

B. Speech Synthesis Markup Language tags controlling rate, pitch, volume, pauses, pronunciation, and emphasis in synthesised speech

C. A SQL-like query language for selecting Polly voice options

D. A security markup language for authenticating Polly API requests

✓ Correct Answer: B

Explanation: SSML gives fine-grained control over speech prosody and pronunciation in Polly, enabling natural pauses, emphasis, custom phoneme pronunciation, and whispering effects for high-quality voice experiences.

Q556. What is 'Amazon Textract Expense Analysis'?

A. An analysis of how much customers spend on Textract API calls

B. A purpose-built mode that extracts vendor, date, line items, and totals from receipts and invoices as structured data

C. An accounts payable automation application layer

D. Infrastructure cost analysis for document processing workloads

✓ Correct Answer: B

Explanation: Textract Expense Analysis is purpose-built for financial documents, automatically identifying and returning structured expense fields without needing to train a custom model for each invoice or receipt format.

Q557. What is 'Amazon SageMaker Model Monitor data quality baseline'?

A. An AWS-set minimum data quality standard

B. Statistical reference properties computed from a representative dataset used to detect feature distribution drift in production traffic

C. A cost baseline estimate for running Model Monitor

D. Minimum accuracy the model must achieve before deployment approval

✓ Correct Answer: B

Explanation: The baseline captures feature statistics and constraint schemas. Model Monitor compares each production request batch against the baseline, raising CloudWatch alarms when feature distributions drift significantly.

Q558. What is 'Amazon Forecast what-if analysis'?

A. An AWS pricing what-if calculator

B. Running counterfactual forecast scenarios by varying time series inputs to see how changes affect business metric predictions

C. Post-hoc analysis of forecast accuracy without Forecast

D. Comparing accuracy across different Forecast algorithm choices

✓ Correct Answer: B

Explanation: What-if analysis generates multiple future forecasts under different input assumptions (promotion spend up 20%, supply disrupted), enabling scenario planning and sensitivity analysis for business decisions.

Q559. What is 'Amazon SageMaker distributed data parallelism (SMDDP)'?

A. A library distributing data preprocessing tasks across CPU cores

B. An optimised AllReduce gradient aggregation library for multi-GPU distributed training that reduces inter-instance communication overhead

C. A parallel data loader for SageMaker notebook environments

D. A distributed database parallel query library for ML feature retrieval

✓ Correct Answer: B

Explanation: SMDDP implements fused AllReduce operations optimised for AWS network topology, achieving faster gradient synchronisation than generic PyTorch DDP for distributed training on multiple GPU instances.

Q560. What is 'Amazon SageMaker model parallelism library (SMP)'?

A. A library enabling a single endpoint to serve multiple models

B. An AWS library partitioning model layers and tensors across multiple GPUs to train models too large for a single GPU's memory

C. A parallel processing library for notebook cell execution

D. A parallel hyperparameter search library for SageMaker

✓ Correct Answer: B

Explanation: SMP automatically partitions large model graphs across GPU instances, enabling training of models with tens or hundreds of billions of parameters that would exhaust single-GPU memory.

Q561. What is 'Amazon SageMaker Debugger built-in rules'?

A. Code style linting rules applied to SageMaker notebook scripts

B. Pre-built tensor analysis rules automatically detecting common training problems such as vanishing gradients, overfitting, and poor initialisation

C. Security policy rules applied to SageMaker IAM configurations

D. Data validation rules for SageMaker Processing job inputs

✓ **Correct Answer: B**

Explanation: Debugger ships with built-in rules (VanishingGradient, Overfit, WeightUpdateRatio, etc.) that continuously monitor captured training tensors and raise CloudWatch alarms when training health issues are detected.

Q562. What is 'Amazon SageMaker Training Compiler'?

A. A compiler for SageMaker Pipeline step YAML definitions

B. A built-in graph compiler optimising deep learning model computation for GPU execution, reducing training time up to 50 percent

C. A source code compiler for SageMaker Python SDK scripts

D. A documentation compiler aggregating SageMaker API references

✓ **Correct Answer: B**

Explanation: Training Compiler applies XLA-based graph compilation to optimise GPU kernel fusion and memory layout, improving GPU utilisation and reducing training time without requiring code changes from the developer.

Q563. What is 'Amazon SageMaker Inference Recommender'?

A. A service recommending which AI application to build next

B. A tool that benchmarks a model across instance types and configurations to recommend the optimal endpoint setup for performance and cost

C. A built-in recommendation system trained and served on SageMaker

D. A service recommending the best training algorithm for a given dataset

✓ **Correct Answer: B**

Explanation: Inference Recommender runs load tests across multiple instance types with your model, returning a ranked leaderboard of configurations with p50/p99 latency, throughput, and cost-per-inference metrics.

Q564. What is 'Amazon SageMaker Shadow Testing'?

A. Evaluating models using synthetic shadow datasets

B. Routing a copy of live production traffic to a new challenger model simultaneously, comparing performance without affecting real users

C. Running model inference in a shadow non-production AWS region

D. Testing models privately before making them visible in the registry

✓ **Correct Answer: B**

Explanation: Shadow Testing mirrors real traffic to a new model candidate in parallel with the production model, providing unbiased real-world performance comparison before deciding whether to promote the new model.

Q565. What is 'Amazon Comprehend async analysis' used for?

A. Running analysis without configuring any parameters

B. Processing large batches of documents stored in S3 asynchronously in the background, with results written back to S3

C. Asynchronous API calls that fire and forget with no response

D. Parallel simultaneous analysis across multiple language models

✓ **Correct Answer: B**

Explanation: Comprehend async batch jobs process tens of thousands of documents in the background for entity recognition, sentiment, key phrases, and other analyses at scale without consuming synchronous API quota.

Q566. What is 'Amazon SageMaker JumpStart' model hub?

A. A quick start guide for new SageMaker users

B. A repository of hundreds of pre-trained open-source models, foundation models, and ML solution templates deployable with one click

C. A jump server providing access to SageMaker Studio

D. A starter kit for deploying SageMaker in a new AWS account

✓ Correct Answer: B

Explanation: JumpStart provides a model hub with state-of-the-art open models (Llama, Stable Diffusion, BERT, etc.), one-click fine-tuning and deployment, and pre-built ML solution templates for common use cases.

Domain 4: Responsible AI

Q567. What is 'AI truthfulness' as a design principle?

A. The model should agree with everything the user says

B. The AI makes only sincere assertions it believes to be true and avoids deliberate deception even when that might be convenient

C. Guaranteeing 100 percent factual accuracy in every output

D. Telling users the exact books and websites used for training

✓ Correct Answer: B

Explanation: Truthfulness means the model does not knowingly assert falsehoods. A model can be wrong and still truthful — but it should not manufacture false information or frame inaccuracies as certain facts.

Q568. What is 'model calibration' in probabilistic AI?

A. Physically calibrating the hardware running the model

B. Ensuring predicted confidence scores reflect actual empirical accuracy — a model that says 80 percent confident should be right 80 percent of the time

C. Calibrating training hyperparameters before a training run

D. Adjusting model outputs to meet a specific performance threshold

✓ Correct Answer: B

Explanation: Well-calibrated models produce reliable uncertainty estimates. Overconfident models lead users to accept wrong answers as certain; underconfident models lead them to unnecessarily disregard correct ones.

Q569. What is 'output refusal' in responsible generative AI?

A. A model failure mode producing empty outputs

B. The AI declining to comply with requests that violate safety guidelines instead of producing harmful content

C. An HTTP 400 error returned for malformed API requests

D. Refusing to generate when temperature is set to zero

✓ Correct Answer: B

Explanation: Output refusal is an intentional safety behaviour. Rather than generating instructions for harm or prohibited content, the model explains that it cannot help with that request, protecting users and society.

Q570. What does the EU AI Act define as 'high-risk AI'?

A. Any model with more than one billion parameters

B. AI used in critical infrastructure, employment decisions, education, law enforcement, biometrics, and access to essential services

C. All AI systems deployed on public cloud providers

D. Only generative AI above a specific benchmark performance threshold

✓ Correct Answer: B

Explanation: High-risk AI under the EU AI Act faces the strictest requirements: conformity assessment, registration, human oversight, data governance, and transparency obligations — proportionate to the potential impact on people's rights.

Q571. What is 'reversibility' in responsible AI design?

A. The ability to mathematically reverse-engineer model weights

B. Designing AI actions to be undoable where possible, limiting permanent harm from incorrect automated decisions

C. Reversing training data presentation order for regularisation

D. Rolling back a model version in the SageMaker Model Registry

✓ Correct Answer: B

Explanation: Designing for reversibility means an AI system's actions can be undone when errors are found, limiting harm. Irreversible high-stakes actions — financial, medical, physical — warrant mandatory human confirmation.

Q572. What is 'purpose limitation' in GDPR and AI data use?

A. Limiting the number of ML use cases per model deployment

B. Personal data may only be used for the specific purpose it was collected for and cannot be repurposed for AI training without a new legal basis

C. Limiting data retention to the minimum storage duration

D. Restricting data processing to the geographic region of collection

✓ Correct Answer: B

Explanation: Purpose limitation prevents 'function creep' where data collected for customer support gets repurposed for model training without users' knowledge or consent, a common compliance risk in AI development.

Q573. What are 'data subject rights' in AI context?

A. Rights of data scientists who access training datasets

B. Legal rights individuals hold over their personal data including access, correction, deletion, and explanation of automated decisions affecting them

C. AWS account owner rights regarding data storage

D. Dataset creator rights to restrict commercial use of their data

✓ Correct Answer: B

Explanation: GDPR grants individuals the right to access their data, correct errors, request erasure, port their data, and receive a meaningful explanation of automated decisions — all affecting how AI systems must be designed.

Q574. What is 'model over-reliance' risk in AI deployment?

A. A model depending on too many input features at inference

B. Users uncritically accepting AI outputs without applying judgement, missing errors especially in edge cases and high-stakes decisions

C. Over-dependence on a single AI vendor for all machine learning needs

D. An ML model consuming more compute than its business value justifies

✓ Correct Answer: B

Explanation: Over-reliance leads to serious harm when a confident but incorrect model output is accepted without scrutiny, particularly in medical, legal, or financial decisions where human oversight remains essential.

Q575. What is 'tiered risk' in AI governance frameworks?

A. Different pricing tiers for different risk levels of AI API calls

B. Classifying AI applications by potential harm and applying proportionate governance — minimal oversight for low-risk, stringent controls for high-risk

C. A three-tier cloud architecture pattern applied to AI infrastructure

D. Insurance premium tiers for AI liability coverage

✓ Correct Answer: B

Explanation: Tiered risk governance avoids both under-governing dangerous systems and over-burdening harmless ones by calibrating requirements (audits, documentation, human oversight) to the severity of potential harms.

Q576. What is 'AI auditability' in governance?

A. Tracking API call counts for billing audit reconciliation

B. Maintaining records of training data, model logic, decisions, and outcomes that can be reviewed and verified by auditors

C. Automatic source code auditing of model implementation files

D. Logging all Bedrock invocations to S3 for archiving

✓ Correct Answer: B

Explanation: Auditability requires complete traceability from training data provenance through model decisions to outcomes, enabling regulators, internal teams, and independent auditors to verify compliant and ethical behaviour.

Q577. What is 'societal benefit' framing in AI development?

A. Calculating financial ROI for society from AI projects

B. Explicitly considering how AI affects not only direct users and the business but also broader communities and society at large

C. A philanthropic AI research grant programme

D. Social media engagement metrics for AI-generated content

✓ Correct Answer: B

Explanation: Societal benefit framing asks designers to consider second-order effects such as job displacement, inequality widening, community surveillance risks, and democratic integrity when making AI product decisions.

Q578. What is 'participatory design' in responsible AI?

A. An open competition inviting all practitioners to submit AI designs

B. Involving end users and communities affected by an AI system in the design process to surface real needs and identify potential harms early

C. Requiring all team members to vote on design decisions by majority

D. An agile methodology specifically adapted for AI product development sprints

✓ Correct Answer: B

Explanation: Participatory design brings diverse stakeholder voices — especially those from marginalised communities — into development, reducing the risk of harm by surface real-world impacts that developers might overlook.

Q579. What is 'predictive policing' AI concern?

A. Using AI to predict and throttle API traffic in policing applications

B. Applying ML to forecast crime or flag high-risk individuals, perpetuating historical bias and raising civil liberties concerns

C. Policing AI models themselves for regulatory compliance

D. Predicting when law enforcement IT systems need software patches

✓ Correct Answer: B

Explanation: Predictive policing trained on historically biased law enforcement data systematically over-targets minority communities, amplifying existing inequities and creating feedback loops that entrench discriminatory outcomes.

Q580. What is 'AI documentation' requirement in governance?

A. Writing README files for every model code repository

B. Maintaining comprehensive records of model purpose, training data, performance, limitations, and testing to support accountability and regulatory review

C. Automatically generated API reference documentation for model endpoints

D. Cloud infrastructure documentation for the servers running the model

✓ Correct Answer: B

Explanation: Governance frameworks require documentation beyond code comments — covering intended use cases, training data provenance, bias testing results, known failure modes, and deployment conditions for accountability.

Q581. What is 'facial recognition' primary ethical concern?

A. The high cost relative to other identity verification methods

B. Higher error rates for minority demographic groups, potential for mass surveillance, and use without subject awareness or consent

C. Operational limitations in low-light or outdoor environments

D. Integration complexity with legacy identity management systems

✓ Correct Answer: B

Explanation: Independent audits have documented significantly higher false-match and false-non-match rates for darker-skinned and female subjects, and surveillance deployments raise fundamental privacy and civil liberties concerns.

Domain 5: Security, Compliance & Governance

Q582. What is 'Amazon SageMaker SOC 2 report' relevant to?

A. A social responsibility report for AWS staff

B. A third-party audit report verifying SageMaker's security, availability, and confidentiality controls available via AWS Artifact

C. An internal SageMaker performance scorecard

D. A compliance report customers must generate for their own auditors

✓ Correct Answer: B

Explanation: SOC 2 Type II reports provide independent auditor evidence that SageMaker's controls were operating effectively over a period, helping enterprise customers satisfy vendor security assessments.

Q583. What is 'SageMaker IAM execution role' least-privilege best practice?

A. Granting the execution role AdministratorAccess for simplicity

B. Granting only read access to specific input S3 buckets, write access to output buckets, access to the specific KMS key, and CloudWatch publish rights

C. Using a shared execution role across all training and inference jobs

D. Granting the execution role read-only access to all AWS services

✓ Correct Answer: B

Explanation: Least-privilege execution roles minimise blast radius if credentials are compromised. Scoping S3 access to specific bucket ARNs and KMS access to specific key IDs follows the principle of least privilege.

Q584. What is 'AWS SCP' primary limitation in AI governance?

A. SCPs can override root account permissions

B. SCPs only restrict the maximum permissions available — they cannot grant permissions beyond what IAM identity policies already allow

C. SCPs apply only to the root management account

D. SCPs are limited to S3 and IAM services only

✓ Correct Answer: B

Explanation: SCPs set the permission ceiling for an account or OU. Even if an SCP allows an action, the IAM identity policy must also permit it. SCPs alone cannot grant access; they can only further restrict it.

Q585. What is 'data sovereignty' for AI on AWS?

A. The legal right to commercially sell your training datasets

B. The requirement that data and its processing remain under the legal jurisdiction of the originating country or organisation

C. AWS claiming sovereignty over data processed by its services

D. A concept relevant only to government-classified data

✓ **Correct Answer: B**

Explanation: Data sovereignty drives region selection and PrivateLink configuration. AWS region, data residency guarantees, and cross-region inference controls all help customers maintain sovereignty compliance.

Q586. What is 'S3 Block Public Access' protecting in ML workflows?

A. Blocking public access to ML documentation in S3

B. Preventing training datasets and model artefacts stored in S3 from being accidentally exposed to the internet via misconfigured ACLs or policies

C. Blocking known malicious public IP addresses from S3

D. Preventing public users from calling SageMaker inference endpoints

✓ **Correct Answer: B**

Explanation: S3 Block Public Access provides a safety net — even if a bucket policy or ACL is inadvertently set to public, the account-level block prevents the data from actually becoming publicly accessible.

Q587. What is 'ML supply chain security' risk mitigation?

A. Physically securing the GPU hardware supply chain

B. Scanning dependencies, verifying artefact checksums, using private model registries, and auditing pre-trained model sources to prevent backdoors

C. Using ML models to secure physical supply chain logistics

D. Applying SOC 2 compliance to manufacturing AI applications

✓ **Correct Answer: B**

Explanation: ML supply chain attacks (malicious packages, backdoored public models, poisoned datasets) require defence-in-depth: dependency scanning, provenance verification, private registries, and model integrity checks.

Q588. What is 'model signing' in ML security?

A. Having a manager digitally sign the model deployment approval form

B. Applying a cryptographic signature to model artefacts enabling detection of any tampering during storage or transfer

C. Signing inference API requests with HMAC for authentication

D. Attaching a digital signature to model performance documentation

✓ **Correct Answer: B**

Explanation: Model signing enables integrity verification — the SHA-256 or asymmetric signature on the artefact confirms it has not been modified after creation, protecting against malicious substitution in the deployment pipeline.

Q589. What is 'model inventory' in enterprise AI governance?

A. The list of Python packages installed in a model's runtime environment

B. A centralised registry cataloguing every AI model in production with owner, purpose, risk rating, deployment date, and review schedule

C. Counting learnable parameters across all deployed model versions

D. AWS inventory tracking of SageMaker training instance usage

✓ **Correct Answer: B**

Explanation: A model inventory provides the organisational visibility needed for risk management, regulatory reporting, decommissioning planning, and incident response across all AI systems deployed enterprise-wide.

Q590. What is 'continuous compliance monitoring' for AI infrastructure?

A. Manually completing compliance checklists every quarter

B. Automatically and continuously evaluating AI resource configurations against compliance rules using AWS Config and Security Hub with auto-remediation

C. A managed compliance consulting engagement from AWS

D. Continuously retraining models to maintain regulatory accuracy thresholds

✓ Correct Answer: B

Explanation: AWS Config custom and managed rules continuously evaluate resource configurations (encryption enabled, VPC-only, tags present) and Security Hub aggregates findings, enabling real-time compliance posture management.

Q591. What is 'Amazon SageMaker Model Dashboard' used for?

A. A financial reporting dashboard for SageMaker spending

B. A centralised governance view aggregating all deployed endpoints with their Model Monitor alarm states for data quality, bias, and model drift

C. A user management dashboard for SageMaker Studio domains

D. A real-time request traffic dashboard for inference endpoints

✓ Correct Answer: B

Explanation: Model Dashboard gives ML governance teams a single-pane view of every deployed model's health — flagging endpoints with active data quality, bias drift, feature attribution drift, or model quality alarms.

Q592. What is 'Amazon Bedrock PCI DSS eligibility'?

A. Bedrock automatically achieves PCI DSS compliance for customers

B. Bedrock is covered under AWS PCI DSS attestation enabling use in cardholder data environments when customers implement their controls

C. PCI DSS requires avoiding all AI services in payment systems

D. PCI DSS compliance for Bedrock requires a separate AWS contract

✓ Correct Answer: B

Explanation: Under the shared responsibility model Bedrock customers can scope it into PCI environments, but must implement their own required controls: encryption, access control, logging, and network segmentation.

Q593. What is 'SageMaker multi-account MLOps architecture'?

A. SageMaker functionality available only in accounts with multiple regions

B. Using separate AWS accounts for dev, staging, and production with cross-account IAM roles sharing model packages and pipelines across boundaries

C. Multi-tenancy within a single SageMaker account

D. An automatic account replication feature for SageMaker resources

✓ Correct Answer: B

Explanation: Enterprise MLOps uses account-per-environment isolation for blast radius containment and compliance. AWS RAM and cross-account IAM roles allow governed promotion of model packages from dev to prod accounts.

Q594. What is 'Bedrock invocation logging' compliance best practice?

A. Logging only failed API invocations to reduce storage cost

B. Logging all inputs and outputs to S3 with Object Lock and to CloudWatch with retention aligned to regulatory requirements

C. Allowing Bedrock to select optimal log verbosity automatically

D. Logging only model inputs and never responses to protect user privacy

✓ Correct Answer: B

Explanation: Compliance best practice captures both prompts and responses in S3 (immutability via Object Lock) and CloudWatch (real-time alerting), with retention periods meeting regulatory minimums (e.g., 7 years for financial services).

Q595. What is 'SageMaker VPC-only endpoint configuration'?

A. Restricting endpoints to virtual PC access only

B. Deploying inference endpoints inside a customer VPC with subnets and security groups so no traffic leaves the private network

C. Allowing only VPN-connected users to call SageMaker endpoints

D. A configuration available only for batch transform jobs not real-time endpoints

✓ Correct Answer: B

Explanation: VPC-only endpoints prevent public internet exposure of model inference traffic. Combined with VPC endpoints (PrivateLink) for the SageMaker API itself, the entire workflow stays within the private network perimeter.

Q596. What is 'NIST AI RMF' and how does AWS support it?

A. National Institute for SaaS Technology AI Reference Model Framework

B. NIST's voluntary AI Risk Management Framework with four functions — Map, Measure, Manage, Govern — supported by SageMaker Clarify, Model Cards, Monitor, and Pipelines

C. A binding federal AI regulation for all US companies

D. A financial risk management framework adapted for AI investment decisions

✓ Correct Answer: B

Explanation: NIST AI RMF provides structured guidance for AI risk management. AWS maps SageMaker features to each function: Pipeline lineage (Map), Clarify bias/explainability (Measure), Model Monitor (Manage), Model Cards (Govern).

Q597. What is 'Bedrock PrivateLink interface endpoint' key benefit?

A. Providing faster responses via a content delivery network

B. Keeping all traffic between a customer VPC and Bedrock API inside the AWS network without traversing the public internet

C. Generating private API keys unique to each Bedrock account

D. Hiding Bedrock usage from appearing in the AWS billing console

✓ Correct Answer: B

Explanation: PrivateLink interface endpoints ensure Bedrock API calls remain on AWS-managed private networking, satisfying strict network isolation, data residency, and regulatory requirements for sensitive enterprise workloads.